

DEVELOPMENTS OF MATCHED-FIELD PROCESSING FOR PARAMETER INVERSION IN PRESSURIZED PIPELINES

by

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ABSTRACT

This thesis develops an innovative inverse analysis framework for accurate determination of defect characteristics and material properties of pressurized water supply systems (WSSs) based on the matched-field processing (MFP) methodology—a signal processing approach widely and successfully used in other domains such as underwater acoustics for source localization and environmental characterization. The MFP methodology involves (i) an experimental step where transient waves, propagating at high speed (km/s), are introduced into the pipe system and the response is sensed; and (ii) a theoretical step where the measured response is “matched” with the replica response (derived from the mathematical model of waves in pipes). The proposed MFP framework is found to be (i) super-resolving; thus, beating the diffraction limit and enabling each defect in a cluster of

densely (defect-to-defect distance is less than half of the minimum probing wavelength) spaced defects to be individually and clearly detected; (ii) able to significantly suppress unwanted side-lobes; thus, addressing the issue of ambiguous main-lobe identification; (iii) robust and stable against various noise types, including white, colored, and impulse noise, even under low signal-to-noise ratio conditions; and (iv) accurate. These substantial advantages are due to the following three attributes. First, MFP fully leverages both amplitude and phase information of the broadband transient signals. Second, MFP is statistically optimal under Gaussian noise; thus, ideally suited for WSSs since laboratory and field data show that the noise in these systems is well described by the Gaussian probability distribution. Third, a novel understanding of the mechanism responsible for the development of side-lobes is acquired in this research, resulting in approaches to deal with these unwanted phenomena in parameter estimation in WSSs. The proposed MFP methodology is validated through comprehensive simulation and experimental studies in three specific applications: determination of location and size of a single leak; determination of location and size of clustered leaks where the leak-to-leak distance is smaller than half a wavelength; and determination of the viscoelastic parameters of our testing facilities. The MFP method is generalizable across pipeline systems, applicable to all types of unknown parameters, rapid to perform, non-intrusive, non-destructive, and supports real-time health condition assessment of WSSs.

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This is to certify that I have examined the above Ph.D. thesis
and have found that it is complete and satisfactory in all respects,
and that any and all revisions required by
the thesis examination committee have been made.

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NOMENCLATURE

κ	bulk modulus of water, page 23
μ	the propagation function, page 27
ν	Poisson's ratio, page 23
ω	the angular frequency, page 26
ω_{th}	the fundamental frequency
ω	the utilized frequency range, page 38
ψ	non-dimensional parameter that depends on the elastic properties of pipelines, page 22
ρ	water density , page 23
σ	the standard deviation of the Gaussian white noise, page 39
$\tau_{n_{kv}}$	the retardation time coefficient in K-V model, page 24
θ	leak location, page 50
θ	the vector of concerned parameters, page 45
θ^{IP}	the intact-pipeline parameters, page 26
θ^L	the leak properties, page 31
ε	total strain response, page 24
ε_e	instantaneous elastic strain response, page 24
$\varepsilon_r(t)$	retarded strain response, page 24

A	cross-sectional area, page 23
a	frequency-dependent wave speed , page 26
a_0	the elastic component of wave speed, page 24
B	the single-frequency MFP output function, page 45
B_{MB}	the output function of MB-MFP, page 48
B_{MB}^1	the output function of MB-MFP ¹ , page 50
B_{MB}^2	the output function of MB-MFP ² , page 51
C	Creep compliance function, page 24
C_0	the instantaneous creep compliance, page 24
$C_{n_{kv}}$	the creep compliance coefficient in K-V model, page 24
D	pipe internal diameter, page 23
E	Young's modulus of elasticity of pipe wall, page 23
$\mathbb{E}$	the expectation operator used to calculate the expected value, page 39
e	pipe wall thickness, page 23
f_{DW}	Darcy-Weisbach friction factor, page 23
f_{th}	the system's fundamental frequency, page 58
G, G_{ω}^M	the Green's function of leak scattering at frequency ω and measurement location x^M , page 60
g	gravitational acceleration, page 23
H	this superscript signifies the operation of taking the conjugate transpose of a complex vector, page 45

H_0^L	the steady-state pressure head at the leak , page 29
h	the transient-induced fluid head oscillation, page 22
h_ω^L	the pressure head oscillations at the leak, page 29
h_ω^M	the head h at frequency ω and measurement location x^M , page 26
$\mathbf{h}_\omega^M$	the measured pressure head vector for all N_f frequencies and N_m measurement locations, page 38
$h_\omega^{U'}$	the measured pressure head at $x^{U'}$ near x^U , page 28
h_ω^U	the head h at frequency ω and upstream end x^U , page 26
L	pipe length, page 25
M_a	Mach number, page 25
N_f	the number of frequencies, page 37
N_{kv}	the total number of K-V model elements, page 24
N_L	the number of leaks, page 34
N_m	the number of measurement locations, page 37
n_L	the leak number, page 38
$\mathbf{n}_\omega^M$	noise vector for all frequencies and measurement locations, page 39
n_ω^M	noise included in the real measured head signal, page 39
Q_0	steady-state flow rate, page 23
Q_0^L	the steady-state discharge extracted through leak , page 29
q	the transient-induced flow rate oscillation, page 22

q^L	the discharge perturbation escaping through the leak, page 22
q_ω^L	the discharge oscillations at the leak, page 29
q_ω^M	the discharge q at frequency ω and measurement location x^M , page 26
q_ω^U	the discharge q at frequency ω and upstream end x^U , page 26
R	the linearized steady-state friction term, page 24
R_{error}	the ratio of the Euclidean norm (or modulus) of the difference between $\tilde{\mathbf{h}}_\omega^{\mathbf{M}}$ and $\mathbf{h}_\omega^{\mathbf{M}*}$ to the norm of $\mathbf{h}_\omega^{\mathbf{M}*}$, page 132
T_m	the measurement time, page 88
T_{th}	the system's fundamental period, page 58
t	time, page 22
U^O	the overall transfer matrix, page 31
U^P	the point transfer matrix, page 30
U^{IP}	the field transfer matrix for an intact pipe (IP) , page 27
x	spatial coordinate along pipeline, page 24
x^D	the downstream end of pipe, page 58
x^{L^-} or L^-	the left point close to the leak, page 29
x^{L^+} or L^+	the right point close to the leak, page 29
x^M	measurement location, page 26
$\mathbf{x}^{\mathbf{M}}$ or $\mathbf{M}$	the measurement locations, page 38
x^U	the upstream end of pipe, page 26

$x^{U'}$ the location near x^U , page 28

Z the characteristic impedance of the pipe, page 27

Z^L the hydraulic impedance of leak, page 29

$\mathbb{Z}$ this symbol represents the domain of positive integers, page 73

K-V Kelvin-Voigt model, page 24

MB-MFP modified broadband matched-field processing, page 14

MB-MFP¹ the first improved version of MB-MFP which uses a side-lobe canceling algorithm, page 50

MB-MFP² the second improved version of MB-MFP which uses a peak extraction algorithm, page 51

MFP matched-field processing, page 9

MLE maximum likelihood estimation, page 9

MV manual ball valve, page 86

SNR the signal-to-noise ratio, page 39

WSSs urban water supply systems, page 3

*

this “*” symbol is used to indicate “real” or “true”. For example, $\mathbf{h}_{\omega}^{\mathbf{M}*}$ denotes the real measured signal, page 39

~

this symbol indicates that the values are esimted/prediced. For example, $\tilde{\mathbf{h}}_{\omega}^{\mathbf{M}}$ denotes the reproduced head signal, see equation (4.29), page 77

CHAPTER 1

INTRODUCTION AND LITERATURE REVIEW

1.1 Motivation: Crucial Parameter Estimation in WSSs

Water supply systems (WSSs) are lifelines of cities and towns, relying on the stable and safe operations of extensive pressurized pipelines to meet the fluctuating demand while maintaining the water quality and pressure [1]. However, these systems now encounter significant challenges, including rising water demand, aging infrastructure, and increasingly complex network topology. Studies report that these factors can reduce operational efficiencies [2, 3] and result in pipeline defects such as leakage [4, 5], corrosion [6], and blockage [7]. Among these defects, leakage is particularly problematic as it is the ultimate result of many issues like cracks and can lead to substantial resource and energy wastage, decreased supply pressure, environmental damage, and other adverse impacts [8, 9]. Leakage can be caused by multiple factors, including material degradation, soil or ground movement, vibrations from traffic, extreme weather conditions, and operational stresses [10–13]. In recent years, water leakage has significantly intensified across global WSSs, particularly in rapidly urbanizing Asian regions. A 2010 report [14] stated that WSSs in Asia lost about 29 billion cubic meters of non-revenue water (NRW, defined as treated water that is produced but not delivered to customers), primarily due to leaks. This volume of water was valued at around 8.7 billion US dollars. A more recent report [15] in 2018 indicated that over an eight-year period, the volume of lost NRW in Asian cities nearly doubled to 52 billion cubic meters, valued at 15 billion US dollars. Such NRW losses can account for up to 35% of the total water supply. Although leak-

age often receives the most attention due to its significant immediate impact, other pipeline defects also significantly compromise system performance (e.g., a corroded pipe is more prone to bursts and incurs higher frictional losses) [16]. Combined, these defects create compounding risks to the stability and safety of WSSs. This reality highlights the need for early, targeted defect detection.

While the defect determination is a critical step for system reliability, another important aspect of WSSs maintenance is understanding the mechanical behavior of the pipeline materials themselves. Over the past few decades, the adoption of polymeric pipes made from materials such as polyvinyl chloride (PVC), polyethylene (PE), and high-density polyethylene (HDPE) has grown. These materials offer additional advantages over traditional metal materials, including resistance to corrosion, lower weight, ease of installation, and cost-effectiveness [17]. Unlike metal pipes which exhibit predominantly elastic behavior, plastic pipes display viscoelastic properties – meaning their deformation under stress has both an instantaneous elastic component and a time-dependent viscous component [18]. This viscoelasticity is typically characterized by a creep function often derived from mechanical analog models, such as the Kelvin-Voigt model which represents the pipe wall as a spring and dashpot in parallel. The creep function details the evolution of material strain under stress over time, incorporating parameters such as creep compliance (a measure of the material’s deformability) and retardation time (the timescale of the viscoelastic response) [19]. Engineers need to quantify these viscoelastic parameters to accurately predict pipe behavior under pressure surges and even under normal operation [20, 21]. In addition, the viscoelastic properties of plastic pipe walls can “mask” or distort transient wave reflections caused by leaks or other anomalies [22–24]. If the viscoelastic parameters are not accounted for, anomaly detection via transient wave-based methods becomes significantly more complex [25]. Moreover, calibrating viscoelastic parameters also helps in longevity studies (e.g., viscoelastic creep over many years might lead to pipe sagging

or need for support) [26]. In summary, accurately estimating material properties like viscoelasticity in polymeric pipes is crucial in WSSs.

In addition to understanding defect characteristics (e.g. defect size and location) and material properties (e.g. viscoelastic parameters), knowing other pipeline parameters — such as wave speed, friction coefficients, and network topology metrics — is also important to facilitate pipeline condition assessment and informed decisions regarding maintenance and upgrades in WSSs [27–29]. Each of the parameters uniquely alters hydraulic characteristics, such as pressure head and flow rate, under both steady-state and transient operating conditions within WSSs. This functional relationship suggests that accurate pipeline parameter estimation can be achieved through inverse analysis to the measurements of these hydraulic characteristics.

1.2 Background

This section first highlights the inherently ill-posed nature of the inverse problem of using hydraulic transient waves for pipeline parameter estimation, then reviews the existing applications of matched-field processing (MFP) as an inversion technique for obtaining parameter estimates, setting the stage for discussing our research objectives.

1.2.1 Inverse Problem Framework for Parameter Estimation Using Hydraulic Transient Waves

Accurate estimation of the unknown pipeline parameters can be achieved by measuring and analyzing physical signals that are influenced by these parameters. For example, pressure head and flow discharge within pipelines are known to be affected by pipe dimensions, pipe defects and pipe wall material properties. In recent years, hydraulic transient waves (also known as water hammer waves or hydraulic transients), historically associated with destructive water hammer phenomena, are increasingly recognized in the literature as a promising tool for parameter estimation in pipelines. These transient-based methods offer advantages such as scalability to large systems, cost-effectiveness, and non-disruptive operations [30]. Transient waves can be generated in pipelines through intentional (active) or unintentional (passive) flow perturbations or disturbances, such as starting/stopping a pump, adjusting valves [31, 32], or using specialized devices like the Portable Pressure Wave Maker in large-diameter pipes [33]. These generated waves propagate inside the pipelines with a fast wave speed (e.g., 400 m/s in polymeric pipes and 1300 m/s in steel pipes), and encode information about the unknown system parameters by their interaction with pipelines. Through measuring and analyzing the transient signals, it is possible to extract out the information of these unknown pipeline parameters like the defect characteristics, the material properties, and so on [34–38]. For instance, the signals shown in Fig. 1.1 clearly indicate that due to the existence of leak, the transient signal of intact pipe differs from the one of leaky pipe. This pressure difference means that the transient signal can capture the leak information. In essence, the methods using transient waves operate similarly to radar or sonar systems: an injected transient wave acts as a probing signal, while the reflected waveform reveals system properties.

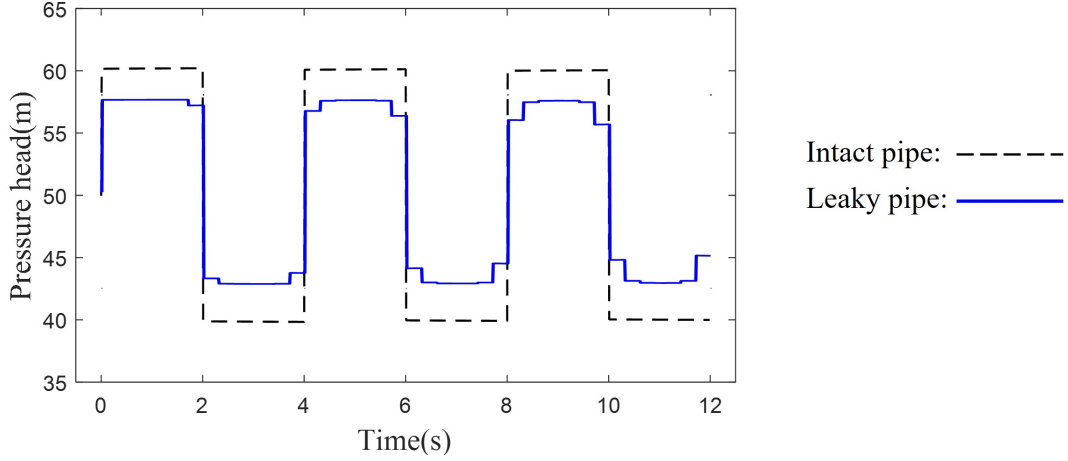


Figure 1.1. The calculated transient signals corresponding to intact pipe and leaky pipe from numerical simulations.

Given the established role of transient waves in encoding pipeline parameters, our next step is to extract/infer these parameters from transient signals. Such task of determining the unknown parameters from observed data is typically, in many scientific and engineering disciplines, formulated as an inverse problem. Specifically, our goal is to infer unknown parameters using the system's transient response and a transient wave propagation model. However, this inverse problem is inherently ill-posed as the exact solution cannot be directly obtained owing to the complexity of transient wave propagation (thus introducing model inaccuracies) and the inevitable presence of measurement noise. Consequently, advanced inversion techniques (or optimization methods) are required to obtain an estimated solution. To address this challenge, a promising way is the adoption of matched-field processing (MFP), a powerful inversion approach widely used in fields such as underwater acoustics for source localization and environmental characterization [39]. In the context of pipeline parameter estimation, MFP can be employed to match transient wave signals measured along the pipeline with those computed from a mathematical model, thereby inferring the unknown parameters. In the next subsections, we introduce MFP's concept, terminology, exist-

ing applications in pipeline parameter estimation along with its advantages and key limitations to contextualize our contributions.

1.2.2 Inversion Technique of Matched-Field Processing

Terminology

Before we explore the applications of matched-field processing (MFP) method in pipeline parameter estimation, it is important to define several commonly used terminologies. These definitions enhance our understanding of the method and prepare us for the subsequent discussions on MFP's advantages and research gaps. The relevant terms are outlined below:

1. ***field***: The term “field” typically refers to the state of physical quantities at points in the environment. In our pipeline scenario, “field” is the measured or modeled pressure information at points along pipeline during a hydraulic transient event. The field can also be called the wavefield/data/signal.
2. ***ambiguity space***: The output space of MFP is commonly called “ambiguity space” in literature, due to the fact that there are often multiple candidate model parameter sets producing high match, leading to ambiguous determination of the optimal parameter set. While the term "match space" or "correlation space" might more precisely reflect the match/correlations process between fields, the high-resolution MFP proposed in this thesis involves more complex steps instead of just simple correlation comparisons, making "ambiguity space" the more accurate term.
3. ***peaks***: These very distinct, high-intensity MFP outputs, which represent good matches be-

tween the measured and modeled fields, are naturally referred to as “peaks” in ambiguity space. They are easily identifiable.

4. ***main-lobe***: The highest peak showing in ambiguity space is called the “main-lobe”. The region (or the point) where this main-lobe locating at in ambiguity space indicates the predicted optimal parameter set.
5. ***side-lobes***: Peaks that are not close to the main-lobe but still show high MFP outputs are called “side-lobes.” These side-lobes spread out with the increasing range from main-lobe on ambiguity space. When side-lobes become comparable in peak levels to main-lobe, it bring ambiguity to identify the main-lobe, thus questioning the credibility of the main-lobe identification.
6. ***mismatch***: Mismatch refers to any errors that occur in the modeled or replica fields. These errors can arise due to uncertainties in known parameters, inaccuracies in the propagation or scattering models (e.g., we often simplify real-world processes), and approximate nature of numerical solutions. Mismatch can impact MFP performance in varying degrees, depending on the error type.
7. ***degradation***: Degradation indicates a severe loss or mislocation of the main-lobe. This can occur due to mismatch, low signal-to-noise ratios, or any other factors, and represents a performance failure of the MFP method.
8. ***super-resolution***: Super-resolution refers to MFP’s super-resolving capability to distinguish clustered defects which are spaced closer than half of the minimum probing wavelength (the wave diffraction limit). In this thesis, to avoid confusion, “super-resolution” or “super-resolving” are specially used to describe MFP’s spatial resolution, which signifies the smallest distance capable of distinguishing between different defects when using MFP. For describing the main-lobe width, we use the term “high-resolution” when the main-lobe width is

smaller than half of the minimum probing wavelength.

9. **noise**: Broadly speaking, noise can be defined as any unwanted signals in the measurements.

In this thesis, we consider three types of noise: white noise, colored noise (e.g., blue noise), and impulse noise, which primarily differ in how their power spectra are distributed across frequencies. Specifically, (i) white noise has a uniform power spectrum, meaning it maintains equal energy across all frequencies, (ii) blue noise exhibits greater energy at higher frequencies, with power increasing as frequency increases, and (iii) impulse noise is characterized by high-amplitude disturbances that overlay the baseline white noise spectrum.

Existing Applications of MFP

Matched-field processing (MFP) is a powerful inversion technique that has enjoyed widespread use in various domains such as ocean acoustics and earthquake research [40–42], wherein it is the premier choice to solve a variety of inverse problems, including source localization, environmental parameter estimation, bearing of passive sources, and model accuracy evaluation. Its underlying principle is straightforward: (i) collecting the measured field $\mathbf{b}$, (ii) calculating the replica field $A(\mathbf{x})$ for each candidate model parameter set $\mathbf{x}$, (iii) Selecting the parameter set yielding the main-lobe in ambiguity space (i.e., showing the highest match/cross-correlation between $\mathbf{b}$ and $A(\mathbf{x})$) to be the predicted/estimated result. In essence, the name “matched-field processing” explicitly reflects this process of field matching.

MFP holds significant promise in pipeline parameter inversion using transient wavefields, due to the following technical merits: (i) MFP is particularly recommended for low-frequency applications (e.g., a few Hz to several tens of Hz, which coincides with the frequency range of transient waves),

because low-frequency waves have long wavelengths which yield fewer and more stable acoustic modes, enabling more reliable wavefield matching; (ii) MFP can use both amplitude and phase information of the whole frequency band; i.e., it uses all available data components rather than just phase (as is done in most time-domain theoretic modeling) or just specific frequencies. This indicates that MFP, as a full-waveform inversion method [36], can achieve more accurate and robust results; (iii) MFP is statistically equivalent to maximum likelihood estimation (MLE) and optimal when the measurement noise follows a Gaussian distribution [43], thus, ideally suited for WSSs since laboratory and field data show that the noise in these systems is well described by the Gaussian probability distribution [44].

Recently, MFP has been successfully applied in water pipeline systems as a frequency-domain inversion method, although its applications have been primarily limited to defect detection. The following provides a chronological overview of MFP's applications and performance in pipeline engineering: Specifically, Wang et al. [43] have proposed a transient-based leak detection methodology using the conventional broadband MFP (CB-MFP, also known as the Bartlett processor) method, which is the most widely used and best understood MFP technique. CB-MFP solves the inverse problem of leak detection by inferring the leak location and size in model given measured transient wavefield dependent on such information, and it is proved to be robust against white noise. Additionally, as CB-MFP uses both amplitude and phase information of signals over a broadband low-frequency band, it exhibits superior robustness and precision relative to techniques that solely rely on the resonant frequencies of the system (such as [45]) and those that are only based on phase information (such as [4]). Moreover, the CB-MFP method reduces the two-dimensional search space (i.e., the leak size and leak location) to one-dimensional searches where the leak location is first found, and the result is substituted back into the defect detection functional to determine

the leak size. This dimensionality reduction improves computational efficiency. This method was later adapted for leak localization in a single viscoelastic pipe, where the pipe wall's viscoelastic properties were included in the wave propagation model by defining a frequency-dependent wave speed [46]. Subsequently, the CB-MFP method was further developed to account for uncertainties arising from incomplete knowledge, by incorporating prior uncertainty estimates derived from leak-free transient experiments. This improvement was validated using experimental data from quasi-field tests on a 135-meter-long HDPE pipe at Beacon Hill, Hong Kong SAR [47]. Furthermore, the application of MFP for leak detection has been extended from single pipe to more complex pipe networks such as tree-type and loop-type, wherein a derivation to factorized wave propagation model is given [48–50]. Despite these extensions of MFP, it still has notable research gaps (detailed in the next section).

Current Research Gaps in MFP

Matched-field processing (MFP) is still in the early stages of application in pipeline systems and encounters significant challenges. One major difficulty is the presence of side-lobes in ambiguity space, which are undesirable and cause ambiguity in determining the main-lobe associated with parameter estimates [51–54]. This phenomenon is illustrated in Fig. 1.2(a), where the side-lobes bring ambiguity for correct main-lobe identification. Although MFP has been applied and extended in several studies for leak detection [43, 46–50, 55–59], the physical and mathematical origins of side-lobe generation remain unexamined. Empirical studies suggest that the side-lobe resolution can be increased (i.e., the side-lobe width can be narrowed) by using more sensors [43, 60], but their amplitudes maintain as the sensor number increases. Additionally, in real-world systems, spatial constraints (e.g., accessibility, cost) can often restrict measurement point density. Therefore, there

is a need of inversion techniques that can produce accurate parameter estimations with suppressed side-lobes, by using as few measurement points as possible. Notably, side lobe interference plagues inverse problems across disciplines, including ocean acoustics [41], geophysics [61], and pipeline diagnostics [43, 62], underscoring its universality. Systematically addressing side-lobe suppression challenges is therefore a cross-disciplinary imperative to unlock MFP’s full potential in pipeline parameter estimation and beyond.

Another critical challenge involves potential degradation of MFP performance due to nonwhite noise interference. As shown in Fig. 1.2(b), the CB-MFP method cannot localize the leak correctly under nonwhite noise conditions. This limitation is especially critical in practical pipeline environments, where nonwhite-noise sources include systematic errors (e.g., model inaccuracies) and environmental disturbances (e.g., flow turbulence, traffic-induced vibrations, and sensor measurement errors) [43, 62, 63]. The proposed CB-MFP approach [43] directly sums the complex information across all frequencies via the inner product operation. During this summation process, white noise can be canceled out with each other as white noise has uniform spectral density over frequencies. As a result, white noise does not introduce any frequency-dependent bias into the inner product, which consequently ensures the robustness of CB-MFP against white noise. However, this robustness diminishes for colored or impulse noise, where spectral correlation introduces persistent interference through frequency summation and may result in performance failures, as shown in Fig. 4.3. Incorporating prior information about the probabilistic distribution of noise could theoretically enhance MFP’s robustness. Yet, obtaining complete noise statistics is hindered by non-stationary noise profiles and limited spatial-temporal sampling in real-world operational pipelines.

Moreover, MFP’s spatial resolution is constrained by wave diffraction limitations. That is, the MFP

methods based on a single-defect model are unable to detect clustered defects, defined as defects spaced less than half the minimum probing wavelength apart [60, 64]. For instance, as depicted in Fig. 1.2(c), the MFP output results has only one peak at the clustered defects, i.e., the MFP method fails to distinguish the closely-spaced leaks. While the MFP methods based on a multi-defect model can be developed to address this issue, the challenge of an unknown number of defects remains unresolved. Last but not the least, to date, MFP applications remain largely restricted to pipeline defect detection, with few efforts to extend the methodology to broader parameter estimation tasks such as material property estimation.

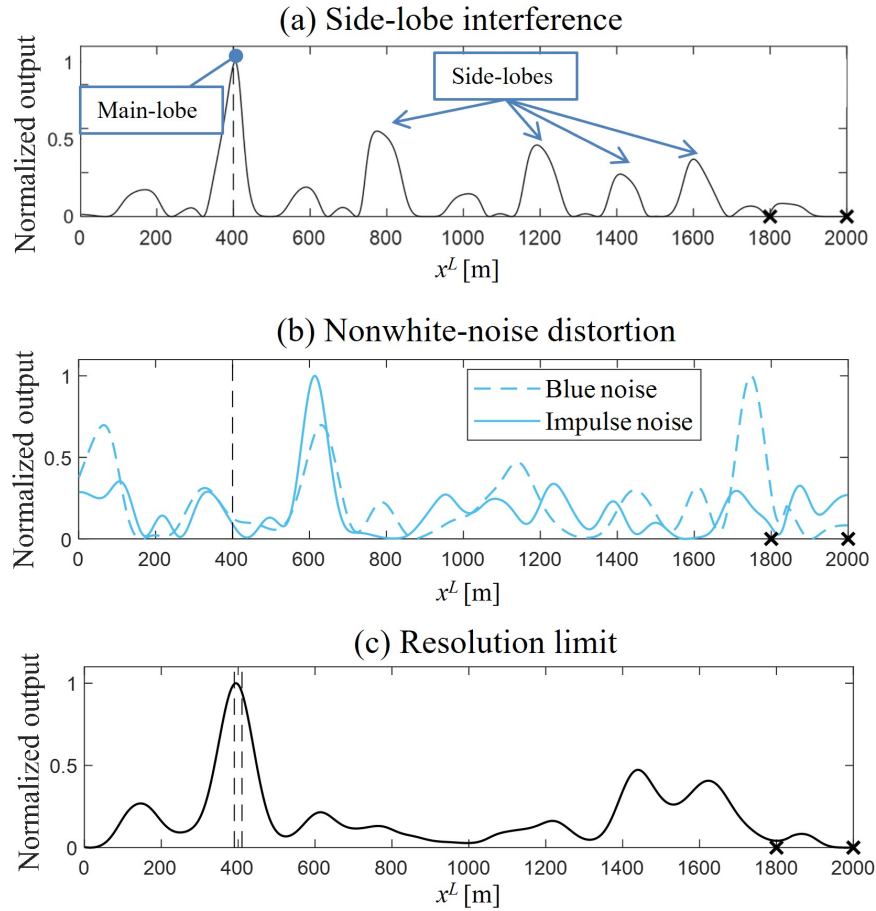


Figure 1.2. Current research gaps of using CB-MFP for leak localization.

In conclusion, while the existing studies validate MFP’s potential for leak detection, critical gaps persist. These gaps include: 1) the unclear mechanism of side-lobe generation, 2) the ways to suppress side-lobes, 3) the spatial resolution limit, 4) the performance degradation under nonwhite noise, and 5) the currently limited application scope. In the next section, we outline our research objectives with targeted solutions, aiming at overcoming these limitations.

1.3 Thesis Objectives

Building upon the detailed introduction of the existing applications and limitations of MFP methods in pipeline parameter estimation, this section outlines research objectives with proposed solutions through theoretical enhancements and algorithmic innovations. Ultimately, our goal is to propose a novel transient-based MFP methodology (serving as a generalized inverse analysis framework) capable of yielding accurate, robust, and super-resolution parameter estimations, with greatly suppressed side-lobes and enhanced main-lobe resolution.

- **Objective 1: Develop a generalized MFP framework for pipeline parameter inversion**

Proposed solution: To develop a versatile MFP framework capable of estimating various pipeline parameters, an important option is the decomposition of transient signals. Specifically, we first formulate transient wave propagation models to analytically derive frequency-domain pressure head responses, which encode the pipeline parameter information for subsequent MFP analysis. This expressed pressure head is then decomposed into distinct components, with each component predominantly associated with a subset of parameters exhibiting distinct spectral signatures. For example, the total pressure response in a leaky pipe can

be separated into one component representing the intact pipeline’s transient response (capturing material properties) and individual components representing the transient responses from each leak (capturing leak-specific information). This decomposition enables selective isolation of parameter-specific signal components, suppressing interference from irrelevant signals during MFP analysis. The next critical step involves establishing a generalized MFP mathematical formulation in the form of an objective function dependent on the general unknown parameters, with the optimal parameter values being those that maximize the function’s output. To further enhance versatility, we utilize broadband excitations rather than single-frequency inputs, as the latter can provide insufficient information for determining parameters [65].

- **Objective 2: Enhance the robustness for parameter estimation in nonwhite noise environments**

Proposed solution: We propose modified broadband MFP (MB-MFP) techniques that enhance robustness under nonwhite noise conditions. MB-MFP initiates with per-frequency normalization of measured signals across the broadband spectrum, followed by the average of single-frequency MFP outputs. This normalization scales both the signal and noise components so that their combined magnitude equals unity, thereby effectively mitigating frequency-localized noise amplification (e.g., impulse noise with peaked spectra). As a result, MB-MFP methods not only preserve white noise robustness inherent to CB-MFP while significantly improving tolerance to non-stationary, spectrally correlated noise including colored and impulse noise.

- **Objective 3: Investigate the generation mechanism of side-lobes**

Proposed solution: To suppress side-lobes which cause ambiguity to accurate parameter estimation, it is crucial to first understand how these side-lobes are generated and then develop advanced optimization algorithms to reduce them. Thode et al. [66] demonstrated that in underwater acoustics, the side-lobe patterns observed in the spatial range-frequency plane tend to converge toward the true source location. Motivated by this insight, we leverage the hidden information embedded in side-lobes of single-frequency MFP results. We begin with a thorough theoretical and numerical investigation into the mechanisms behind side-lobe formation. Our analysis reveals that the side-lobes in broadband MFP ambiguity space result from the cumulative superposition of individual side-lobes generated in single-frequency MFP results.

- **Objective 4: Achieve side-lobe suppression and increase main-lobe resolution**

Capitalizing on the insight of single-frequency MFP output space, we propose two enhanced MB-MFP techniques: (1) MB-MFP¹, which incorporates a side-lobe cancellation algorithm, and (2) MB-MFP², which utilizes a peak extraction algorithm. They can significantly improve the main-lobe-to-side-lobe ratios in the final ambiguity spaces. Both approaches use the gradient information in single-frequency MFP output to preserve the main-lobes while selectively reduce or eliminate side-lobes of each single-frequency MFP output, before integrating the single-frequency MFP outputs to define the broadband MFP methods. Therefore, MB-MFP¹ and MB-MFP² can achieve side-lobe suppression. In addition, MB-MFP² can provide a improved high-resolution main-lobe identification.

- **Objective 5: Achieve super-resolution in clustered-defect detection**

Proposed solution: To achieve super-resolution for clustered-defect scenarios, we propose

MB-MFP methods based on a theoretical model with $N_L \geq N_{L*}$, where N_L and N_{L*} respectively represent the number of defects given in the model and in the real pipeline system. The multi-defect model expands the solution space dimensionality, enabling MB-MFP to resolve up to N_L defects. Thus, when $N_L \geq N_{L*}$, the real defects are guaranteed to lie within this expanded subspace. This conclusion draws from the concept of “overcomplete representation,” which is commonly used in signal processing and machine learning to enhance the correctness and robustness of results [67]. An overcomplete representation, where the number of latent features exceeds the signal dimensions, is particularly useful when the signal consists of multiple sets of complete linear features. In pipeline systems, by assuming weak defect interactions (valid for the usual defect detection cases [68]), pressure responses can be approximated as linear superpositions of individual defect contributions, and thus this over-completeness holds true. More importantly, it is found that the estimated leak locations (N_L) tend to cluster around the actual leak locations (N_{L*}) as N_L increases and eventually surpasses N_{L*} . This clustering effect makes MB-MFP method able to simultaneously estimate the number of leaks and achieve precise leak localization, and thus addresses the challenge of unknown defect number.

In summary, we aim to develop an generalized inverse analysis framework for pipeline parameter estimation by synergistically combining wave physics principles and MFP methodology. We addresses several critical challenges, including the unclear generation mechanism of MFP side-lobes, the ambiguity caused by side-lobes, the spatial resolution limit imposed by wave diffraction effect, and the performance degradation under nonwhite noise. Beyond pipelines, the framework’s adaptability positions it as a promising versatile inverse analysis toolbox for other applications such as geophysical imaging, and acoustic sensing.

1.4 Thesis Organization

1.4.1 Thesis Outline

The thesis progresses from fundamental theories (Chapters 2: governing equations, and Chapter 3: MFP methodologies) to targeted applications (Chapters 4: single-leak detection, Chapter 5: multi-leak detection, and Chapter 6: viscoelastic parameter estimation), culminating in conclusions and future directions (Chapter 7). These chapters are organized as follows:

- **Chapter 2** presents the governing equations for transient wave propagation in pressurized pipelines. Based on the mass and momentum conservation laws, we formulate the governing partial differential equations (PDEs) in the time domain. Using Fourier transform analysis, these PDEs are systematically converted to their frequency-domain representations. By applying the appropriate boundary conditions, we analytically solve the equations governing the hydraulic quantities (pressure head and flow rate) at a single frequency and a designated measurement location. These solutions are represented using the transfer matrix method, where the transfer matrix intuitively describes the transient wave propagation process along the pipeline and explicitly relates the hydraulic quantities at the boundary to those at the measurement location. This transfer matrix contains key system parameters, including defect characteristics and material properties, and can be decomposed into distinct parameter-specific submatrices (e.g., the overall transfer matrix for a leaky pipe can be decomposed into a field transfer matrix representing wave propagation in an intact pipe, and point transfer matrices characterizing localized defects). This decomposition then enables the mathematical isolation of parameter-specific pressure head components, which subsequently serve as

inputs for MFP-based parameter inversion.

- **Chapter 3** introduces the generalized MFP methodology for parameter inversion by matching measured and modeled wavefields. We first formalize the single-frequency MFP, whose output function is defined as a function of the angular frequency and the model parameter vector in the candidate search space. Through analytical derivations and numerical simulations, we characterize the spatial-spectral mechanisms governing main-lobe and side-lobe generation in single-frequency MFP outputs. By leveraging the gradient information inherent in the single-frequency MFP outputs, we develop modified broadband MFP (MB-MFP) methods that are robust against various noise types while providing high-resolution parameter estimation and significant side-lobe suppression.
- **Chapter 4** focuses on the application of MFP in determining the location and size of a single leak in pipelines. We first investigate the mechanism of side-lobes generation in the single-frequency MFP outputs for leak detection. Systematic analysis reveals a frequency-dependent convergence pattern: the side-lobe trajectories on the MFP output surface rendered in candidate location-frequency plane asymptotically converge into the real leak position as frequency increases. This convergence property implies the concealed leak information behind the side-lobes. Such information is then explored and utilized through adopting the proposed MB-MFP techniques. These techniques are robust and provide high-resolution main-lobe for leak detection. Both theoretical and experimental results validate the effectiveness of these novel methods in noisy environments.
- **Chapter 5** tackles the challenge of detecting clustered leaks in pipelines, wherein the leaks are closely spaced with a distance less than half minimum wavelength. The proposed MB-MFP techniques achieve super-resolution localization and are robust to various noise types. Additionally, a key finding of our research is that when the number of leaks assumed in the

model equals or exceeds the actual number of leaks, the estimated leak locations reliably cluster around the real leak sites. This clustering effect facilitates not only the precise localization of leaks but also the accurate enumeration of them. The proposed methods are analyzed through numerical simulations and validated with experimental data from laboratory facilities.

- **Chapter 6** presents the application of MFP in estimating pipeline material properties. The methodology involves an inversion on the measured transient data to the parameter estimates through a systematic process of data collection, model calculation, and the cross-correlation between the measured and modeled data. Due to the fact that the initial range of candidate parameters may not include the optimal estimates, iterative implementations are suggested. The MFP framework is applied to a real-world HDPE pipeline system to estimate viscoelastic parameters, and achieves accurate estimates. Key advantages include the fast computational speed, noise resilience, scalability, and real-time monitoring capability.
- **Chapter 7** concludes our findings and lists the future directions which are worthy further investigation.

1.4.2 Journal Paper Publications

1. **Xu, C.**, Waqar, M., Louati, M., and Ghidaoui, M. S. (2024). High-resolution pipeline leak localization using the side-lobes of the defect detection functional of matched-field processing. *Mechanical Systems and Signal Processing*, 217:111529.
DOI: <https://doi.org/10.1016/j.ymssp.2024.111529>
2. **Xu, C.**, Waqar, M., Louati, M., and Ghidaoui, M. S. (2025). Enhanced Multi-Leak Detection in Pressurized Pipelines using Super-Resolution Matched-Field Processing. *Water Research*

(under review).

3. **Xu, C.**, Waqar, M., and Ghidaoui, M. S. (2025). Parameter Inversion in Pressurized Pipelines via Hydraulic Transient Waves and Matched-Field Processing. *Mechanical Systems and Signal Processing* (under review).
4. Cui, Y., **Xu, C.**, Waqar, M., Louati, M., and Ghidaoui, M. S. (2025). Pipe-wall thickness estimation using high-frequency waves and the matched-field processing technique. *The Journal of the Acoustical Society of America* (To be submitted).
5. Wang, X., **Xu, C.**, Waqar, M., Zhang, J., and Ghidaoui, M. S. (2025). Physics-informed neural network for hydraulic transient wave in the pipeline. *Mechanical Systems and Signal Processing* (To be submitted).

CHAPTER 2

GOVERNING EQUATIONS

In this chapter, we present the governing partial differential equations of hydraulic transient waves, focusing on solving the mass and momentum conservation equations in the frequency domain via the transfer-matrix method. We first introduce the time-domain governing equations for transient wave propagation. These equations include the parameters corresponding to defect characteristics (e.g., leakage as a mass source) and material properties (e.g, viscoelasticity). Subsequently, we employ Fourier transformation to the time-domain equations to obtain their frequency-domain counterparts, and then derive solutions for pressure head and flow rate in the frequency domain. We express these solutions via a transfer-matrix method, where the matrix characterizes transient wave propagation along the pipeline. Finally, we extend the expression of pressure head from single frequency and measurement location to multiple frequencies and measurement locations. The resulting pressure head expression yields the computed modeled wavefield, which is incorporated into the MFP methodology for parameter estimation, as detailed in the following chapter.

2.1 One-dimensional Governing Equations in Time Domain

Since the foundational theories of transient waves in the time domain have been comprehensively addressed in prior studies [19, 31, 69, 70], we do not repeat these well-established derivations here

to ensure clarity and brevity. Instead, we refer these works and focus on explaining the equations that describe the frequency-domain pressure head relevant to our research. Consider the simplified pipeline configuration as depicted in Fig. 2.1. Let q and h respectively denote the transient-induced flow rate and pressure head oscillations which are relative to the steady-state condition. They follow the linearized unsteady-oscillatory mass and momentum equations along time t and spatial coordinate x as:

$$\frac{\partial q}{\partial x} + \frac{gA}{a_0^2} \frac{\partial h}{\partial t} + 2A\psi \frac{\partial \varepsilon_r}{\partial t} = -q^L \delta(x - x^L) \quad (2.1)$$

and

$$\frac{1}{gA} \frac{\partial q}{\partial t} + \frac{\partial h}{\partial x} + Rq = 0 \quad (2.2)$$

whose systematical derivation can be referred to the Chapter on Governing Equations in the theses [19, 69] or the Chapter on Transient-Flow Equations in the books [31, 32]. The linearization is based on the facts: 1) the small Mach number during transient event (i.e., the ratio of fluid velocity and wave speed is much smaller than one), and 2) the ratio of flow rate oscillation and steady-state flow rate is much smaller than one. Definitions for most variables in Eqs. 2.1 and 2.2 and in some subsequent equations, are summarized in Table 2.1. q^L denotes the discharge perturbation that escape through the leak, which acts as a mass point source locating at a point of negligible extent relative to the system's other geometric scales, such as length and diameter. $\delta(x - x^L)$ is a one-dimensional Dirac delta function with dimension $[L]^{-1}$. Its inclusion arises from modeling the leak mass flux as a point source, which facilitates the integration of the mass conservation equation with respect to the spatial variable x . The term R is the linearized steady-state friction term, calculated as $R = (f_{DW}Q_0)/(gDA^2)$. ψ is a non-dimensional parameter that depends on the elastic properties of pipelines, and $\psi = (1 - \nu^2)$ for the thin-walled pipe anchored throughout its length against axial movement, which we consider in this thesis.

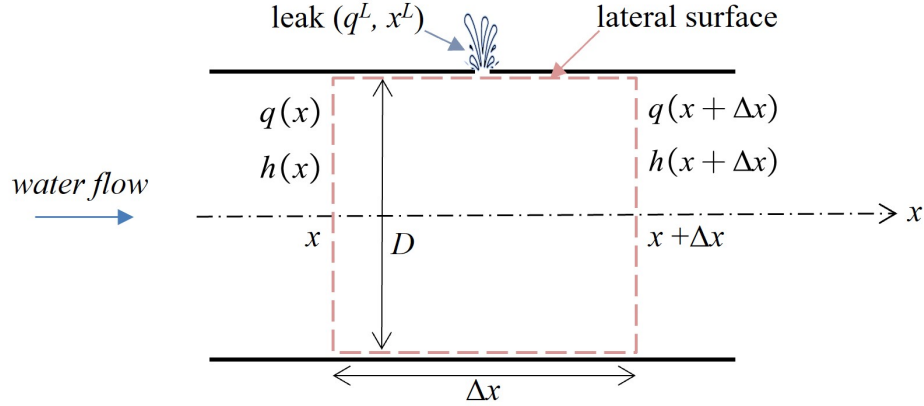


Figure 2.1. The general control volume for pipe flow with a leak, from which q and h can be derived as functions of time t and spatial coordinate x during transient wave propagation.

Table 2.1. The variable definitions for a pipe system

Gravitational acceleration g	Cross-sectional area A
Poisson's ratio ν	Darcy-Weisbach friction factor f_{DW}
Steady-state flow rate Q_0	Pipe internal diameter D
Water density ρ	Bulk modulus of water κ
Pipe wall thickness e	Young's modulus of elasticity of pipe wall E

Analyzing hydraulic dynamics in polymeric pipelines is inherently more complex than in metallic pipelines. This complexity arises from the viscoelastic behavior, which includes both instantaneous and retarded (time varying) strain responses, adding a layer of difficulty in understanding their dynamic responses [18, 71]. That is, the total strain response ε , in a viscoelastic pipe under transient stress consists of two parts: an instantaneous elastic response ε_e and retarded response $\varepsilon_r(t)$. In Eq. 2.1, the term a_0 represents the elastic component of wave speed, incorporating the instantaneous

elastic modulus and accounting for elastic strain ε_e [19], defined as

$$a_0 = \left(\frac{\rho}{\kappa} + (1 - \nu^2) \frac{\rho D}{eE} \right)^{-1/2} \quad (2.3)$$

The retarded strain, $\varepsilon_r(t)$, is given by

$$\varepsilon_r(t) = \frac{\rho g D}{e} \int_0^t h(t-s) \frac{\partial C(s)}{\partial s} ds \quad (2.4)$$

in which $C(t)$ is the creep compliance function and s is a dummy variable used in integration. Note that in a purely elastic case, $C(t)$ is time-independent, resulting in $\varepsilon_r = 0$.

Viscoelasticity is often modeled using the Generalized Kelvin-Voigt (K-V) model, which is reported to accurately capture the creep effects of viscoelastic materials [25, 72]. This model has been applied both in time domain, as seen in the works of Covas et al. [73], Soares et al. [74], and Tjuatja et al. [75], and in frequency domain, as developed by Wang et al. [46]. For the Generalized K-V model, $C(t)$ is defined as [19]

$$C(t) = C_0 + \sum_{n_{kv}=1}^{N_{kv}} C_{n_{kv}} \left(1 - \exp\left(\frac{-t}{\tau_{n_{kv}}}\right) \right) \quad (2.5)$$

wherein C_0 is the instantaneous creep compliance and $C_0 = 1/E$; $C_{n_{kv}}$ and $\tau_{n_{kv}}$ are respectively the creep compliance coefficient and retardation time coefficient in the n_{kv} -th K-V model element; N_{kv} is the total number of K-V model elements. In this formulation, the viscoelasticity in polymeric pipes is defined by introducing additional viscoelastic coefficients into the transient wave model.

The unsteady friction effect is not included in Eq. 2.2 because it is reasonably neglected, as justified in [71]. Specifically, in our study of transient wave propagation in pipelines, the dimensionless

scales L/D and $f \cdot M_a$ are sufficiently large (e.g., the length scale $f_{DW}M_aL/D$ is on the order of 0.1 or greater); therefore, incorporating unsteady friction into the model offers little additional benefit in terms of accurately predicting pressure envelope damping [76]. Here, L and D denote the pipe length and diameter, f_{DW} is the Darcy-Weisbach friction factor, and M_a represents the Mach number.

2.2 Transformed Governing Equations in Frequency Domain

We have represented the creep compliance function $C(t)$ using the Generalized K-V model (Eq. 2.5). We substitute this expression into the retarded strain ε_r (Eq. 2.4), and then insert the result into the time-domain governing Eqs. 2.1 and 2.2. After applying the Fourier transform to these equations (refer to Appendix A.5 in Waqar’s Ph.D. thesis [19] or the paper [46]) and rearranging the terms, we obtain the following transformed governing equations in frequency domain:

$$\frac{\partial q}{\partial x} + \frac{gA}{a^2}i\omega h = -q^L \delta(x - x^L) \quad (2.6)$$

and

$$\frac{\partial h}{\partial x} + \left(\frac{i\omega}{gA} + R \right) q = 0 \quad (2.7)$$

wherein

$$a(\omega; \boldsymbol{\theta}^{IP}) = \left[\frac{1}{a_0^2} + \frac{\rho D}{e} (1 - \nu^2) \sum_{n_{kv}=1}^{N_{kv}} \frac{C_{n_{kv}}}{1 + i\omega\tau_{n_{kv}}} \right]^{-1/2} \quad (2.8)$$

is the formulated wave speed that depends on the frequency ω . It incorporates the viscoelastic effect through K-V model elements with a coefficient set $\boldsymbol{\theta}^{IP} = [\tau_1, C_1, \dots, \tau_{N_{kv}}, C_{N_{kv}}]^\top$. The superscript IP stands for “intact pipe”, indicating that the K-V coefficients correspond to an intact pipe. This

superscript distinguishes the pipe-related parameters from the leak-related parameters introduced later. The superscript $^\top$ indicates vector transpose.

2.3 Solving the Frequency-Domain Equations with Transfer Matrix Method

2.3.1 Solution for An Intact Pipe Expressed by Using Field Transfer Matrix

For the intact pipe shown in Fig. 2.2, the term related to the leak on the right-hand side of Eq. 2.6 is neglected. Then, solving the frequency-domain governing Eqs. 2.6 and 2.7, and introducing boundary conditions q_ω^U and h_ω^U (i.e., the discharge q and head h at frequency ω and upstream end x^U), we can derive the flow rate response q_ω^M and the pressure head response h_ω^M (i.e., q and h at frequency ω and measurement location x^M). This derivation is detailed in Appendix A.1 and the results are

$$q_\omega^M = \cosh\left(\mu(x^M - x^U)\right) q_\omega^U - \left(\frac{1}{Z}\right) \sinh\left(\mu(x^M - x^U)\right) h_\omega^U \quad (2.9)$$

and

$$h_\omega^M = -Z \sinh\left(\mu(x^M - x^U)\right) q_\omega^U + \cosh\left(\mu(x^M - x^U)\right) h_\omega^U \quad (2.10)$$

where Z is the characteristic impedance of the pipe and $Z = (\mu a^2) / (i\omega g A)$; μ is the propagation function and $\mu = a^{-1} (-\omega^2 + i\omega g A R)^{1/2}$.

The solutions to q_ω^M and h_ω^M (Eqs. 2.9 and 2.10) can be rearranged as a matrix form:

$$\begin{bmatrix} q_\omega^M \\ h_\omega^M \end{bmatrix} = U^{IP} \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP} \right) \begin{bmatrix} q_\omega^U \\ h_\omega^U \end{bmatrix} \quad (2.11)$$

wherein U^{IP} is the field transfer matrix for an intact pipe and

$$U^{IP} \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP} \right) = \begin{bmatrix} \cosh(\mu(x^M - x^U)) & -\left(\frac{1}{Z}\right) \sinh(\mu(x^M - x^U)) \\ -Z \sinh(\mu(x^M - x^U)) & \cosh(\mu(x^M - x^U)) \end{bmatrix} \quad (2.12)$$

where the semicolon “;” is used to separate the given parameters (i.e., the predetermined known parameters like the utilized frequency and measurement location) on the left side from the concerned but unknown parameters on the right side. As expressed in Eq. 2.11, U^{IP} relates q and h between locations x^M and x^U at frequency ω , providing intuitive insights into how the wave propagates from one point to another point through the pipeline system, which is depicted in Fig. 2.2. $\boldsymbol{\theta}^{IP}$ represents the vector of interested parameters corresponding to the intact pipe. In our study, we concern the K-V coefficients, i.e., $\boldsymbol{\theta}^{IP} = [\tau_1, C_1, \dots, \tau_{N_{kv}}, C_{N_{kv}}]^\top$.

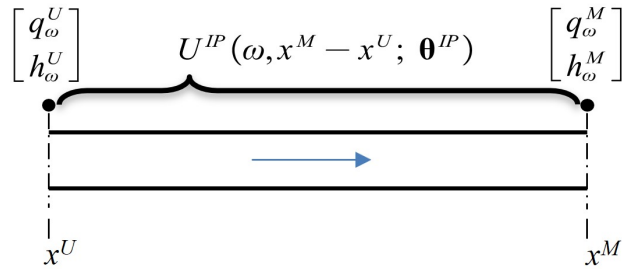


Figure 2.2. Field transfer matrix for an intact pipe segment.

Note that in Eq. 2.11, we assume that q_ω^U and h_ω^U are known to solve for q_ω^M and h_ω^M , which is

practical. For instance, if there is a reservoir at x^U , it is reasonable to assume $h_\omega^U = 0$ as the pressure head at the reservoir is assumed to be constant [31, 32, 46]. Then, we can measure the data $h_\omega^{U'}$ at $x^{U'}$ which is near x^U , and combine it with h_ω^U and Eq. 2.11 to calculate q_ω^U . Furthermore, if the pipeline setup permits to measure h at two points upstream, we can directly calculate an upstream-end boundary condition by combining the measurements with Eq. 2.11. This approach helps to avoid the complexities of modeling unknown or difficult boundary conditions and is employed in the experimental studies of this thesis.

2.3.2 Mass and Momentum Conservation Equations across A Leak and Their Solutions Expressed by Using Point Transfer Matrix

Let Q_0^L represent the steady-state discharge extracted through leak and H_0^L denote the steady-state pressure head at the leak location. Their relation can be expressed by the orifice equation as

$$Q_0^L = s^L \sqrt{2gH_0^L} \quad (2.13)$$

where s^L is the effective cross-sectional area of the leak. Often the leak size change during unsteady state is small and assumed negligible, thus for the unsteady state during a transient, the discharge escaped through a leak can be modeled by using a linearized orifice equation as [19, 69, 77–79]:

$$q_\omega^L = \frac{1}{Z^L} h_\omega^L \quad (2.14)$$

whose derivation can be referred to Appendix A.2 and wherein q_ω^L and h_ω^L are respectively the discharge and pressure head oscillations at the leak; and

$$Z^L = 2H_0^L / Q_0^L = \left(s^L \sqrt{\frac{g}{2H_0^L}} \right)^{-1} \quad (2.15)$$

is called the hydraulic impedance of leak.

Utilizing the mass conservation across the leak yields

$$q_\omega^{L^-} = q_\omega^L + q_\omega^{L^+} \quad (2.16)$$

where L^- and L^+ respectively denote the left and right points close to the leak, i.e., $x^{L^-} \approx x^L \approx x^{L^+}$, as shown in Fig. 2.3. By assuming the head loss across the leak is negligible, the momentum conservation derives the head continuity:

$$h_\omega^{L^-} = h_\omega^L = h_\omega^{L^+} \quad (2.17)$$

Substituting the linearized orifice Eq. 2.14 into the mass conservation Eq. 2.16, and then combining the results with Eq. 2.17 derives the relation for q and h across the leak as

$$\begin{cases} q_\omega^{L^+} = q_\omega^{L^-} - \frac{1}{Z^L} h_\omega^{L^-} \\ h_\omega^{L^+} = h_\omega^{L^-} \end{cases} \quad (2.18)$$

which can be rearranged as

$$\begin{bmatrix} q_\omega^{L^+} \\ h_\omega^{L^+} \end{bmatrix} = U^P \left(s^L \right) \times \begin{bmatrix} q_\omega^{L^-} \\ h_\omega^{L^-} \end{bmatrix} \quad (2.19)$$

wherein U^P is the point transfer matrix and

$$U^P(s^L) = \begin{bmatrix} 1 & -\frac{1}{Z^L} \\ 0 & 1 \end{bmatrix} \quad (2.20)$$

U^P includes the information of leak size s^L through the hydraulic impedance Z^L (Eq. 2.15).

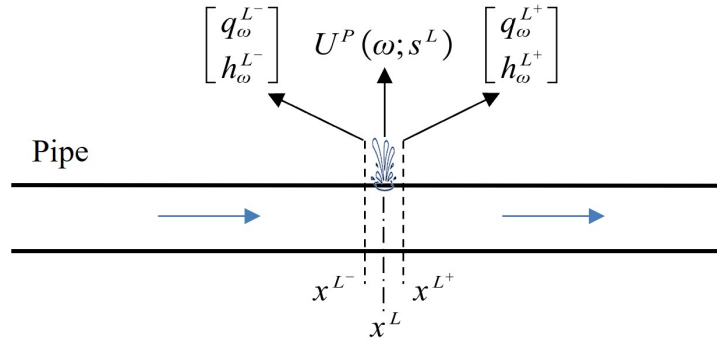


Figure 2.3. Point transfer matrix for a leak in pipe.

2.3.3 Overall Transfer Matrix for A Single-leak Pipe

A single-leaky pipeline, shown in Fig. 2.4, helps to intuitively explain the form of an overall transfer matrix U^O for a system combining both intact pipe and leak. Assuming the two pipe segments separated by the leak have identical properties except the segment length. Then, the combination of Eq. 2.11) connecting q and h for a pipe segment and Eq. 2.19 connecting q and h across a leak can link q and h at different nodes along the pipe as follows:

$$\begin{aligned}
\begin{bmatrix} q_{\omega}^M \\ h_{\omega}^M \end{bmatrix} &= U^{IP} \left(\omega, x^M - x^{L^+}; \boldsymbol{\theta}^{IP} \right) \begin{bmatrix} q_{\omega}^{L^+} \\ h_{\omega}^{L^+} \end{bmatrix}, \\
\begin{bmatrix} q_{\omega}^{L^+} \\ h_{\omega}^{L^+} \end{bmatrix} &= U^P \left(s^L \right) \begin{bmatrix} q_{\omega}^{L^-} \\ h_{\omega}^{L^-} \end{bmatrix}, \\
\begin{bmatrix} q_{\omega}^{L^-} \\ h_{\omega}^{L^-} \end{bmatrix} &= U^{IP} \left(\omega, x^{L^-} - x^U; \boldsymbol{\theta}^{IP} \right) \begin{bmatrix} q_{\omega}^U \\ h_{\omega}^U \end{bmatrix} \quad (2.21)
\end{aligned}$$

which derives an overall transfer matrix U^O that can relate q and h from one end x^U to another end x^M as:

$$\begin{bmatrix} q_{\omega}^M \\ h_{\omega}^M \end{bmatrix} = U^O \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}^L \right) \begin{bmatrix} q_{\omega}^U \\ h_{\omega}^U \end{bmatrix} \quad (2.22)$$

where

$$U^O \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}^L \right) = U^{IP} \left(\omega, x^M - x^{L^+}; \boldsymbol{\theta}^{IP} \right) \times U^P \left(s^L \right) \times U^{IP} \left(\omega, x^{L^-} - x^U; \boldsymbol{\theta}^{IP} \right) \quad (2.23)$$

which is obtained by multiplying the field transfer matrices U^{IP} (Eq. 2.11) for the two pipe segments and the point transfer matrix U^P (Eq. 2.19) for the leak. The two vectors $\boldsymbol{\theta}^{IP}$ and $\boldsymbol{\theta}^L$ respectively represent the parameters corresponding to the intact pipe (for example, viscoelastic parameters, i.e., $\boldsymbol{\theta}^{IP} = [\tau_1, C_1, \dots, \tau_{N_{kv}}, C_{N_{kv}}]^T$) and the parameters associated with the leak (such as leak size and location, i.e., $\boldsymbol{\theta}^L = [s^L, x^L]$).

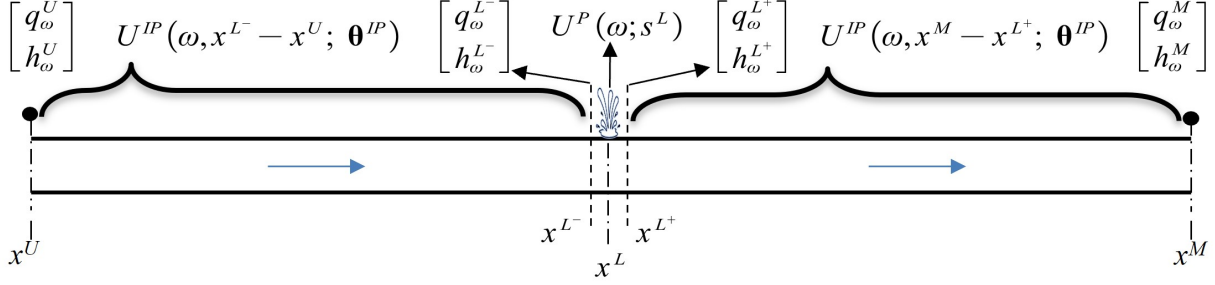


Figure 2.4. Overall transfer matrix for a single-leak pipe by multiplying the field and point transfer matrices.

By substituting $x^{L-} \approx x^L \approx x^{L+}$ and the expressions of U^{IP} (Eq. 2.12) and U^P (Eq. 2.20) into Eq. 2.23, and then expanding and rearranging the result, we obtain

$$U^O(\omega, x^M - x^U; \theta^{IP}, \theta^L) = U^{IP}(\omega, x^M - x^U; \theta^{IP}) + U^L(\omega, x^M - x^U; \theta^L) \quad (2.24)$$

where

$$U^{IP}(\omega, x^M - x^U; \theta^{IP}) = \begin{bmatrix} \cosh(\mu(x^M - x^U)) & -\left(\frac{1}{Z}\right) \sinh(\mu(x^M - x^U)) \\ -Z \sinh(\mu(x^M - x^U)) & \cosh(\mu(x^M - x^U)) \end{bmatrix} \quad (2.25)$$

is the field transfer matrix for the whole intact pipe from x^U to x^M and same to Eq. 2.12, and

$$U^L(\omega, x^M - x^U; \theta^L) = s^L \sqrt{\frac{g}{2H_0^L}} \times \begin{bmatrix} Z \cosh(\mu(x^M - x^L)) \sinh(\mu(x^L - x^U)) & -\cosh(\mu(x^M - x^L)) \cosh(\mu(x^L - x^U)) \\ -Z^2 \sinh(\mu(x^M - x^L)) \sinh(\mu(x^L - x^U)) & Z \sinh(\mu(x^M - x^L)) \cosh(\mu(x^L - x^U)) \end{bmatrix} \quad (2.26)$$

is the leak-induced scattering matrix. It is important to note that U^L here truly encompasses all the information related to the leak (i.e., both leak size s^L and location x^L in θ^L), which is different from

$U^P(s^L)$ (Eq. 2.20) mentioned above, as the latter only contains information about the leak size s^L . The presented overall transfer matrix in Eq. 2.24 effectively separates the leak information $\boldsymbol{\theta}^L$ and the intact pipe information $\boldsymbol{\theta}^{IP}$, transforming them into a linear additive form, thereby achieving a “modular” representation.

The comparison between the two different definitions (Eq. 2.23 and Eq. 2.24) indicates that the overall transfer matrix for the leaky pipe can be understood in two ways: (i) as the multiplication of three matrices corresponding respectively to the left segment of the pipe ($x^M - x^L$), the leak (x^L), and the right segment of the pipe ($x^L - x^U$), as shown in Fig. 2.4; (ii) as the sum of the field transfer matrix for the intact pipe and the leak-induced scattering matrix, as illustrated in Fig. 2.5. Both perspectives provide intuitive and accurate interpretations.

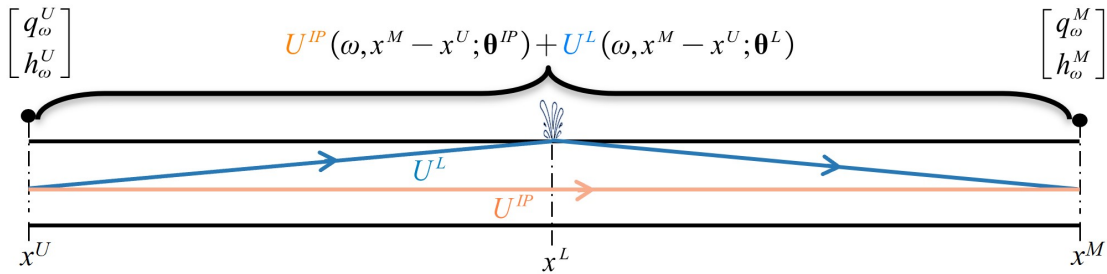


Figure 2.5. Overall transfer matrix for a single-leak pipe by summing the field transfer matrix of intact pipe and the leak-induced scattering matrix.

2.3.4 Overall Transfer Matrix for A Multi-leak Pipe

Next, we extend our discussion to the case of multiple leaks, with a particular emphasis on how to modularize the overall transfer matrix in such scenarios, expressing it in a linear additive form. We

first introduce the overall transfer matrix results for the cases of two leaks, and then expand this framework to scenarios where the number of leaks $N_L \geq 2$.

For a leaky pipeline with $N_L = 2$ leaks, the overall transfer matrix can be intuitively represented as the product of field transfer matrices (U^{IP} , Eq. 2.12) for pipe segments and point transfer matrices (U^P , Eq. 2.20) for leaks in sequence, as shown in Fig. 2.6. Then, q and h at the two ends (x^M and x^U) can be related as follows:

$$\begin{bmatrix} q_\omega^M \\ h_\omega^M \end{bmatrix} = U^O \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}_1^L, \boldsymbol{\theta}_2^L \right) \begin{bmatrix} q_\omega^U \\ h_\omega^U \end{bmatrix} \quad (2.27)$$

where the overall transfer matrix

$$\begin{aligned} U^O \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}_1^L, \boldsymbol{\theta}_2^L \right) \\ = U^{IP} \left(\omega, x^M - x_2^L; \boldsymbol{\theta}^{IP} \right) \times U^P \left(s_2^L \right) \times U^{IP} \left(\omega, x_2^L - x_1^L; \boldsymbol{\theta}^{IP} \right) \\ \times U^P \left(s_1^L \right) \times U^{IP} \left(\omega, x_1^L - x^U; \boldsymbol{\theta}^{IP} \right) \end{aligned} \quad (2.28)$$

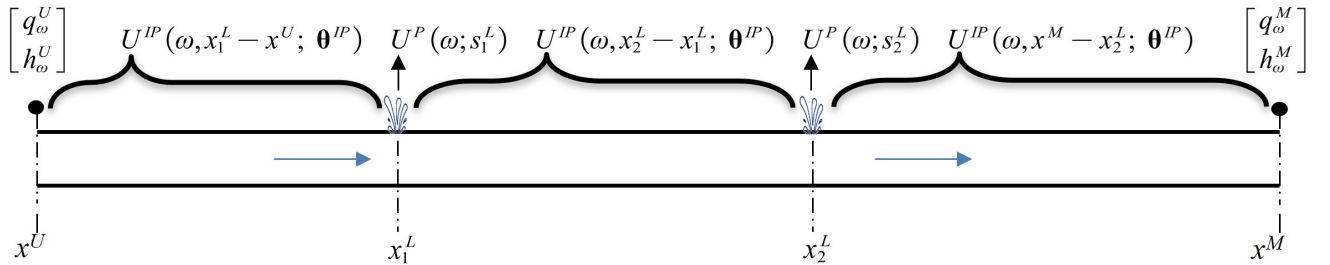


Figure 2.6. Overall transfer matrix for a multi-leak pipe by multiplying the field and point transfer matrices.

We can derive an alternative expression for U^O in the form of a matrix addition by first expanding and rearranging Eq. 2.28 with the substitutions of U^{IP} (Eq. 2.12) and U^P (Eq. 2.20), and then applying the small-leak assumption (i.e., $s_{n_L}^L \ll 1, n_L \in [1, \dots, N_L]$) to neglect the higher-order smaller terms. The detailed derivation can be referred to Appendix A.3. Consequently, as depicted in Fig. 2.7, U^O can be alternatively expressed as a summation of field transfer matrix for the intact pipe and leak-scattering transfer matrices for leaks as:

$$U^O \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}_1^L, \boldsymbol{\theta}_2^L \right) \approx U^{IP} \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP} \right) + U^L \left(\omega, x^M - x^U; \boldsymbol{\theta}_1^L \right) + U^L \left(\omega, x^M - x^U; \boldsymbol{\theta}_2^L \right) \quad (2.29)$$

The detailed expressions for U^{IP} and each U^L in Eq. 2.29 can be referred to Eqs. 2.25 and 2.26 with parameters replacement.

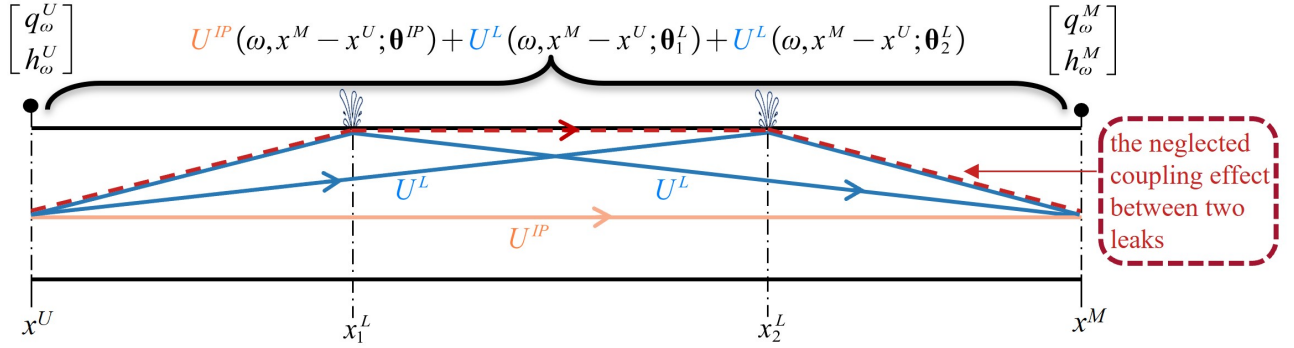


Figure 2.7. Overall transfer matrix for a multi-leak pipe by summing the field transfer matrix of intact pipe and the multiple leak-induced scattering matrices.

Then, for a general N_L -leak case, by following the similar derivation for the 2-leak case, we have:

$$\begin{bmatrix} q_\omega^M \\ h_\omega^M \end{bmatrix} = U^O \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}_1^L, \dots, \boldsymbol{\theta}_{N_L}^L \right) \begin{bmatrix} q_\omega^U \\ h_\omega^U \end{bmatrix} \quad (2.30)$$

where the overall transfer matrix can be expressed in two ways:

$$\begin{aligned}
U^O \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}_1^L, \dots, \boldsymbol{\theta}_{N_L}^L \right) \\
= U^{IP} \left(\omega, x^M - x_{N_L}^L; \boldsymbol{\theta}^{IP} \right) \times U^P \left(s_{N_L}^L \right) \times U^{IP} \left(\omega, x_{N_L}^L - x_{N_L-1}^L; \boldsymbol{\theta}^{IP} \right) \times \dots \\
\times U^{IP} \left(\omega, x_2^L - x_1^L; \boldsymbol{\theta}^{IP} \right) \times U^P \left(s_1^L \right) \times U^{IP} \left(\omega, x_1^L - x^U; \boldsymbol{\theta}^{IP} \right) \quad (2.31)
\end{aligned}$$

or

$$\begin{aligned}
U^O \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}_1^L, \dots, \boldsymbol{\theta}_{N_L}^L \right) \\
\approx U^{IP} \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP} \right) + \sum_{n_L=1}^{N_L} U^L \left(\omega, x^M - x^U; \boldsymbol{\theta}_{n_L}^L \right) \quad (2.32)
\end{aligned}$$

So far, we have solved the hydraulic quantities q and h in frequency domain and express the solutions by using transfer matrix method. In the following section, we further extract the head signal h and analyze it as the wavefield to be utilized in the MFP methodology.

2.4 Pressure Head Response in Frequency Domain

2.4.1 Single-Frequency Response

By combining Eq. 2.30, Eq. 2.32, and Eqs. 2.25 and 2.26, we can solve the pressure head h_ω^M at frequency ω and measurement point x^M for a general N_L -leak pipe, which is given as follows:

$$h_\omega^M \left(\boldsymbol{\theta}^{IP}, \boldsymbol{\theta}_1^L, \dots, \boldsymbol{\theta}_{N_L}^L \right) = h_\omega^M \left(\boldsymbol{\theta}^{IP} \right) + \sum_{n_L=1}^{N_L} h_\omega^M \left(\boldsymbol{\theta}_{n_L}^L \right) \quad (2.33)$$

wherein the measured pressure head consists of a linear superposition of different components. The intact pipe-induced pressure head

$$h_{\omega}^M(\boldsymbol{\theta}^{IP}) = -Z \sinh(\mu(x^M - x^U)) q_{\omega}^U + \cosh(\mu(x^M - x^U)) h_{\omega}^U \quad (2.34)$$

and corresponds to the direct wave propagation along the pipe. The n_L -leak induced pressure head

$$h_{\omega}^M(\boldsymbol{\theta}_{n_L}^L) = -s_{n_L}^L \sqrt{\frac{g}{2H_0^L}} Z \sinh(\mu(x^M - x_{n_L}^L)) \begin{pmatrix} Z \sinh(\mu(x_{n_L}^L - x^U)) q_{\omega}^U \\ -\cosh(\mu(x_{n_L}^L - x^U)) h_{\omega}^U \end{pmatrix} \quad (2.35)$$

and corresponds to the leak-scattered wave. These extracted single-frequency pressure heads form the basis of the expression of the next introduced multi-frequency pressure head vector.

2.4.2 Multi-Frequency Response

For all N_f frequencies and N_m measurement locations, the measured pressure head can be formed as a $N_f \cdot N_m$ dimensional complex vector for a general N_L -leak pipe:

$$\mathbf{h}_{\boldsymbol{\omega}}^{\mathbf{M}}(\boldsymbol{\theta}^{IP}, \boldsymbol{\theta}_1^L, \dots, \boldsymbol{\theta}_{N_L}^L) = \underbrace{\left[h_{\omega_1}^{M_1}, \dots, h_{\omega_{N_f}}^{M_1} \right]}_{N_f}, \dots, \underbrace{\left[h_{\omega_1}^{M_{N_m}}, \dots, h_{\omega_{N_f}}^{M_{N_m}} \right]}_{N_f}^{\top} \quad (2.36)$$

wherein the utilized frequency range $\boldsymbol{\omega} = [\omega_1, \dots, \omega_{N_f}]^{\top}$; the superscript $\mathbf{M} = [M_1, \dots, M_{N_m}]^{\top}$ denotes the measurement locations $\mathbf{x}^{\mathbf{M}} = [x_1^M, \dots, x_{N_m}^M]^{\top}$. By referring to the expansion of h_{ω}^M (Eq.

2.33), $\mathbf{h}_{\omega}^{\mathbf{M}}$ in Eq. 2.36 can be also expanded in a similar way:

$$\begin{aligned}
\mathbf{h}_{\omega}^{\mathbf{M}}(\boldsymbol{\theta}^{IP}, \boldsymbol{\theta}_1^L, \dots, \boldsymbol{\theta}_{N_L}^L) &= \mathbf{h}_{\omega}^{\mathbf{M}}(\boldsymbol{\theta}^{IP}) + \sum_{n_L=1}^{N_L} \mathbf{h}_{\omega}^{\mathbf{M}}(\boldsymbol{\theta}_{n_L}^L) \\
&= \left[\underbrace{h_{\omega_1}^{M_1}(\boldsymbol{\theta}^{IP}), \dots, h_{\omega_{N_f}}^{M_1}(\boldsymbol{\theta}^{IP}), \dots, h_{\omega_1}^{M_{N_m}}(\boldsymbol{\theta}^{IP}), \dots, h_{\omega_{N_f}}^{M_{N_m}}(\boldsymbol{\theta}^{IP})}_{N_f} \right]^{\top} \\
&\quad + \sum_{n_L=1}^{N_L} \left[\underbrace{h_{\omega_1}^{M_1}(\boldsymbol{\theta}_{n_L}^L), \dots, h_{\omega_{N_f}}^{M_1}(\boldsymbol{\theta}_{n_L}^L), \dots, h_{\omega_1}^{M_{N_m}}(\boldsymbol{\theta}_{n_L}^L), \dots, h_{\omega_{N_f}}^{M_{N_m}}(\boldsymbol{\theta}_{n_L}^L)}_{N_f} \right]^{\top} \quad (2.37)
\end{aligned}$$

wherein the components of $h_{\omega}^M(\boldsymbol{\theta}^{IP})$ and $h_{\omega}^M(\boldsymbol{\theta}_{n_L}^L)$ are given in Eqs. 2.34, and 2.35.

2.4.3 Real-System Response

For a real system, the actual measured signal includes noise. Thus, the signal at a single frequency and measurement location can be expressed as the following form

$$h_{\omega}^{M*}(\boldsymbol{\theta}^{IP*}, \boldsymbol{\theta}_1^{L*}, \dots, \boldsymbol{\theta}_{N_L}^{L*}) = h_{\omega}^M(\boldsymbol{\theta}^{IP*}, \boldsymbol{\theta}_1^{L*}, \dots, \boldsymbol{\theta}_{N_L}^{L*}) + n_{\omega}^M \quad (2.38)$$

, and the signal vector for multiple frequencies and measurement locations is

$$\mathbf{h}_{\omega}^{\mathbf{M}*}(\boldsymbol{\theta}^{IP*}, \boldsymbol{\theta}_1^{L*}, \dots, \boldsymbol{\theta}_{N_L}^{L*}) = \mathbf{h}_{\omega}^{\mathbf{M}}(\boldsymbol{\theta}^{IP*}, \boldsymbol{\theta}_1^{L*}, \dots, \boldsymbol{\theta}_{N_L}^{L*}) + \mathbf{n}_{\omega}^{\mathbf{M}} \quad (2.39)$$

where n_{ω}^M and $\mathbf{n}_{\omega}^{\mathbf{M}}$ denote the noise included in the real measured single-frequency and multi-frequency head signals, respectively. The introduction of superscripts $*$ is intended to distinguish between the real measured signal (e.g., h_{ω}^{M*} or $\mathbf{h}_{\omega}^{\mathbf{M}*}$) and that derived from the model (e.g., h_{ω}^M or $\mathbf{h}_{\omega}^{\mathbf{M}}$), or to distinguish between the real parameter values (e.g., $\boldsymbol{\theta}^{IP*}$ or $\boldsymbol{\theta}^{L*}$) and the candidate

parameters set up in model (e.g., $\boldsymbol{\theta}^{IP}$ or $\boldsymbol{\theta}^L$). The real intact-pipeline parameters $\boldsymbol{\theta}^{IP*}$ and the real leak properties $\boldsymbol{\theta}^{L*}$ are unknown and need to be solved ultimately.

The purpose of obtaining the expression of the measured head signals h_{ω}^{M*} or $\mathbf{h}_{\omega}^{M*}$ is to analyze and utilize them in the next proposed MFP techniques for parameter inversion. This inversion process uses the transient wavefields and targets the unknown parameter sets $\boldsymbol{\theta}^{IP*}$ and $\boldsymbol{\theta}_1^{L*}, \dots, \boldsymbol{\theta}_{N_L}^{L*}$, enabling the accurate description of hydraulic dynamics during transients and the prediction of the structural health of polymeric pipes.

2.4.4 Signal-to-Noise Ratio Definition

To include the noise effect into the simulation, the signal-to-noise ratio (SNR) is defined as (in decibel):

$$\text{SNR} = 20 \log_{10} \left(\left| \mathbb{E}(\mathbf{h}_{\omega}^M) \right| / \sigma \right) \quad (2.40)$$

where $|\mathbb{E}(\mathbf{h}_{\omega}^M)|$ is the mean head difference and σ is the standard deviation of the Gaussian noise with zero mean. In our pipeline systems, the types of noise include not only common white noise but also nonwhite noise, such as colored noise and impulse noise [44, 62]. For defining white noise, SNR is set to be a constant in the utilized frequency range ω . For colored noise, in this thesis, we choose blue noise (one type of colored noise) as an example, for which SNR drops by 3 dB per increasing frequency. Therefore, if it is assumed SNR_1 at ω_1 , then SNR uniformly drops to $\text{SNR}_1 - 3(\omega_{N_f} - \omega_1)/2\pi$ at ω_{N_f} . For impulse noise, this thesis assumes a constant SNR over ω , except at certain frequencies, at which SNR suddenly decreases SNR_2 that is much lower than the given SNR. In essence, the impulse noise is introduced as additional perturbations on top of the white noise.

2.5 Summary

This chapter establishes the theoretical framework governing transient wave propagation in viscoelastic pipelines with leaks, focusing on deriving frequency-domain solutions through the transfer matrix method. Starting from the time-domain mass and momentum conservation equations, the governing equations are transformed into the frequency domain via Fourier analysis, explicitly accounting for viscoelastic effects modeled by the Generalized Kelvin-Voigt (K-V) model and leakage as a point mass source. The viscoelastic behavior introduces a frequency-dependent wave speed $a(\omega; \boldsymbol{\theta}^{IP})$, where $\boldsymbol{\theta}^{IP}$ encapsulates the retarded strain parameters of the K-V model.

The transfer matrix method is systematically applied to solve the frequency-domain equations. For intact pipe segments, field transfer matrices U^{IP} relate hydraulic quantities (discharge q and pressure head h) between upstream and downstream nodes. Leaks are modeled using point transfer matrices U^P , derived from linearized orifice equations and mass and momentum conservation laws across the leak, which introduce hydraulic impedance Z^L proportional to leak size s^L . By combining field and point transfer matrices multiplicatively, the overall transfer matrix U^O for systems with multiple leaks is constructed. Crucially, under small-leak assumptions, U^O simplifies to a modular additive form: the intact-pipe transfer matrix U^{IP} plus leak-induced scattering matrices U^L . This modularity is crucial as it enables efficient parameterization of both pipeline properties ($\boldsymbol{\theta}^{IP}$) and leak characteristics ($\boldsymbol{\theta}^L$).

The pressure head response is extracted from the transfer matrix solution and derived as a superposition of contributions from the intact pipe and individual leaks. For multi-leak scenarios, the measured head $\mathbf{h}_\omega^{\mathbf{M}}$ across N_f frequencies and N_m locations forms a complex vector, linearly de-

pendent on the system parameters. Real-system considerations incorporate noise models (white, colored, or impulse), quantified via signal-to-noise ratio (SNR), to reflect practical measurement conditions.

The derived expressions for $\mathbf{h}_\omega^{\mathbf{M}}$ serve as the foundation for the subsequent match field processing (MFP) methodology in Chapter 3. By an inversion on the measured heads, unknown parameters θ^{IP*} and θ^{L*} can be estimated, enabling precise characterization of viscoelastic pipeline dynamics and defect detection. This chapter thus bridges fundamental hydraulic theory with inverse problem-solving techniques, setting the stage for robust, model-based diagnostics in transient wave analysis.

CHAPTER 3

THE GENERALIZED MFP FRAMEWORK FOR PIPELINE PARAMETER INVERSION

Matched Field Processing (MFP) is an inversion technique that estimates unknown parameters by matching measured signal from physical systems to modeled signals (e.g., the measured and modeled transient wavefields in pipelines). In this process, the model parameters are adjusted to optimize the match and infer the correct values of the concerned but unknown parameters. Early studies have successfully applied MFP in pipelines for leak detection. This chapter presents the development of a generalized MFP framework for pipeline parameter inversion. We first introduce the widely used single-frequency linear MFP method and elucidate the mechanisms behind the formation of main lobes and side lobes in the ambiguity space (see Chapter 1 for definitions). Next, we formulate the modified broadband MFP (MB-MFP) methods by averaging the single-frequency MFP outputs across frequencies and integrating innovative algorithms which use the gradient information in the single-frequency MFP outputs. These advanced MB-MFP approaches address critical challenges including: (1) suppressing side lobes to increase the main-to-side lobe ratio; (2) increasing main-lobe resolution; (3) handling non-white noise, and (4) enabling super-resolution defect detection.

3.1 Theoretical foundation

Central to our MFP methodology are the following steps: (i) developing a theoretical frequency-domain model to simulate the generation and propagation of hydraulic transient waves, (ii) adjusting the concerned parameters within a candidate range, (iii) computing the model outputs, and (iv) comparing the modeled fields with the measured field. The optimal parameters are determined by selecting the candidates that best match the modeled and measured fields. According to classical papers on applying the MFP approach to inverse problems [39, 40, 80], the key recommendations for applying MFP to our pipeline environment include:

1. selecting an environment with high temporal stability;
2. selecting an environment which was effectively range independent (thereby minimizing complications and issues relating to environmental data range sampling);
3. determining all non-concerned/non-targeted parameters as accurately as possible, e.g., wave speed, measurement locations, pipe length and diameter, etc.;
4. incorporating an accurate scattering model for the effects of concerned parameters, e.g., the generalized K-V model for the viscoelastic effect, or the leak-induced scattering transfer matrix U^L ;
5. reducing the parameter search, i.e., the parameter dimension, as much as possible
6. using low frequencies to minimize the effects of random perturbations (like internal waves and turbulence)
7. determining the solution parameter values by means of a global search of all possible (discretized) values (thereby avoiding troublesome convergence or search algorithms).

Guided by these recommendations, we proceed to develop the inverse analysis framework based

on MFP methodology, as shown in Figure 3.1.

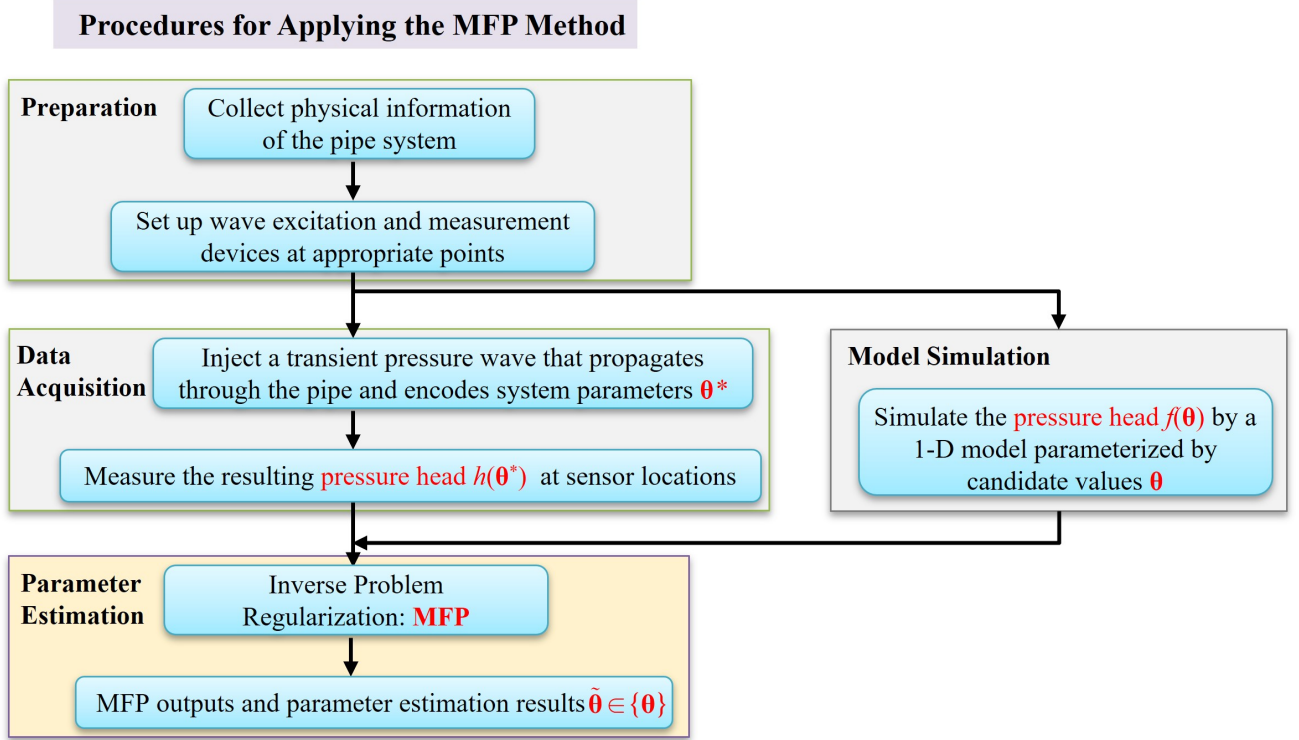


Figure 3.1. Inverse Analysis Framework Based on the MFP Method.

3.2 Single-frequency MFP

We begin by discussing the single-frequency MFP which is defined as a function depending on both frequency ω and candidate parameter set θ :

$$|B(\omega; \theta)| = \left| \left\langle \frac{\mathbf{h}_\omega^{\mathbf{M}*}}{\|\mathbf{h}_\omega^{\mathbf{M}*}\|}, \frac{\mathbf{h}_\omega^{\mathbf{M}}}{\|\mathbf{h}_\omega^{\mathbf{M}}\|} \right\rangle \right| = \frac{|\langle \mathbf{h}_\omega^{\mathbf{M}*}, \mathbf{h}_\omega^{\mathbf{M}} \rangle|}{\|\mathbf{h}_\omega^{\mathbf{M}*}\| \|\mathbf{h}_\omega^{\mathbf{M}}\|} = \frac{\sqrt{(\mathbf{h}_\omega^{\mathbf{M}})^H \mathbf{h}_\omega^{\mathbf{M}*} (\mathbf{h}_\omega^{\mathbf{M}*})^H \mathbf{h}_\omega^{\mathbf{M}}}}{\sqrt{(\mathbf{h}_\omega^{\mathbf{M}*})^H \mathbf{h}_\omega^{\mathbf{M}*}} \sqrt{(\mathbf{h}_\omega^{\mathbf{M}})^H \mathbf{h}_\omega^{\mathbf{M}}}} \quad (3.1)$$

where

- $\mathbf{h}_\omega^{\mathbf{M}*}(\theta^*)$ and $\mathbf{h}_\omega^{\mathbf{M}}(\theta)$ represent the measured and modeled wavefields (i.e., transient pressure

signals). They are both N_m dimensional complex vectors including information from N_m measurement locations at a single frequency ω , and their expressions can be found from Eqs. 2.33 - 2.37;

- $|B|$ is the single-frequency MFP output;
- $\boldsymbol{\theta}^*$ represents the vector of concerned real, yet unknown parameters that need to be estimated; $\boldsymbol{\theta}^*$ can be $\boldsymbol{\theta}^{IP*}$ denoting the vector of viscoelastic coefficients, $\boldsymbol{\theta}^{L*}$ representing the vector of leak properties, or any other concerned parameter set;
- $\boldsymbol{\theta}$ is the vector of candidate parameters set in the model;
- The superscript H signifies the operation of taking the conjugate transpose of a complex vector;
- $\langle \cdots \rangle$ represents taking the dot product of the two complex vectors, that is, multiplying the conjugate transpose of one vector by another vector. The final result is a complex number;
- $|\cdots|$ means taking the absolute value, or modulus, of a complex number, which is equal to the square root of the product of the complex number and its conjugate.
- $\|\cdots\|$ represents taking the L_2 norm of a vector. Note that for a complex vector, as shown in Eq 3.1, its L_2 norm is equal to the square root of the result of its conjugate transpose multiplied by itself, which is also equivalent to the square root of the inner product of the vector with itself;

In Eq 3.1, the normalization of the modeled signal $\mathbf{h}_\omega^{\mathbf{M}}$ is essential. Without it, in the ambiguity space of $|B|$, the region with the highest modeled field intensity will inevitably exhibit the largest output errors. We observe that the MFP output $|B|$ is proportional to the measured signal $\mathbf{h}_\omega^{\mathbf{M}*}$. This implies that even if $\mathbf{h}_\omega^{\mathbf{M}}$ is not normalized, the region corresponding to the predicted parameter set

with the maximum output in ambiguity space remains unchanged (only the maximum value of $|B|$ changes). However, this does not suggest that normalizing $\mathbf{h}_\omega^{\mathbf{M}}$ is unnecessary. On the contrary, this normalization plays a crucial role in enhancing the MFP's robustness against nonwhite noise when using broadband signals, as detailed in the next section. Consequently, MFP incorporates normalization to both measured and modeled fields in its processing.

From the inner product perspective, the output $|B|$ equals to $|\cos \alpha|$, where α is the “angle” between vectors $\mathbf{h}_\omega^{\mathbf{M}*}(\boldsymbol{\theta}^*)$ and $\mathbf{h}_\omega^{\mathbf{M}}(\boldsymbol{\theta})$. In line with the Cauchy-Schwarz theorem, this relationship is defined as

$$|B(\omega; \boldsymbol{\theta})| = \left| \cos \left(\mathbf{h}_\omega^{\mathbf{M}*}, \mathbf{h}_\omega^{\mathbf{M}} \right) \right| = |\cos \alpha| \leq 1 \quad (3.2)$$

Here, $\mathbf{h}_\omega^{\mathbf{M}*}$ has a fixed orientation (determined by the unknown but fixed real parameters $\boldsymbol{\theta}^*$), whereas $\mathbf{h}_\omega^{\mathbf{M}}$ has an orientation varying with candidate parameter set $\boldsymbol{\theta}$. When $\mathbf{h}_\omega^{\mathbf{M}}$ becomes parallel to $\mathbf{h}_\omega^{\mathbf{M}*}$ (indicating a perfect match), $|B|$ reaches its maximum value of one. In all other orientations, $|B|$ is less than one. In other words, only when $\boldsymbol{\theta} = \boldsymbol{\theta}^*$, $|B| = 1$.

In the ambiguity space of $|B|$, the maximum value of one appears as the highest peak (defined as “main-lobe” in Chapter 1.2.2) and corresponds to the optimal parameter set (i.e., closest to the real parameter set). The other high-level peaks are so-called side-lobes, which make the localization of main-lobe becomes ambiguous. For example, as shown in Fig. 3.2, the single-frequency MFP is used for leak detection in pipelines and the results are ambiguous because the same peak not only appears at the true leak location, but also appear periodically at other locations. This observed periodicity occurs due to the decomposition of transient waves into periodic sinusoidal components at each frequency (a comprehensive theoretical discussion is provided in Chapter 4). Additionally, results from single-frequency MFP are susceptible to noise and uncertainties, diminishing

its robustness [81]. These difficulties encountered can be addressed by employing the modified broadband MFP (MB-MFP) methods, which integrate $|B|$ across all frequencies, as discussed next.

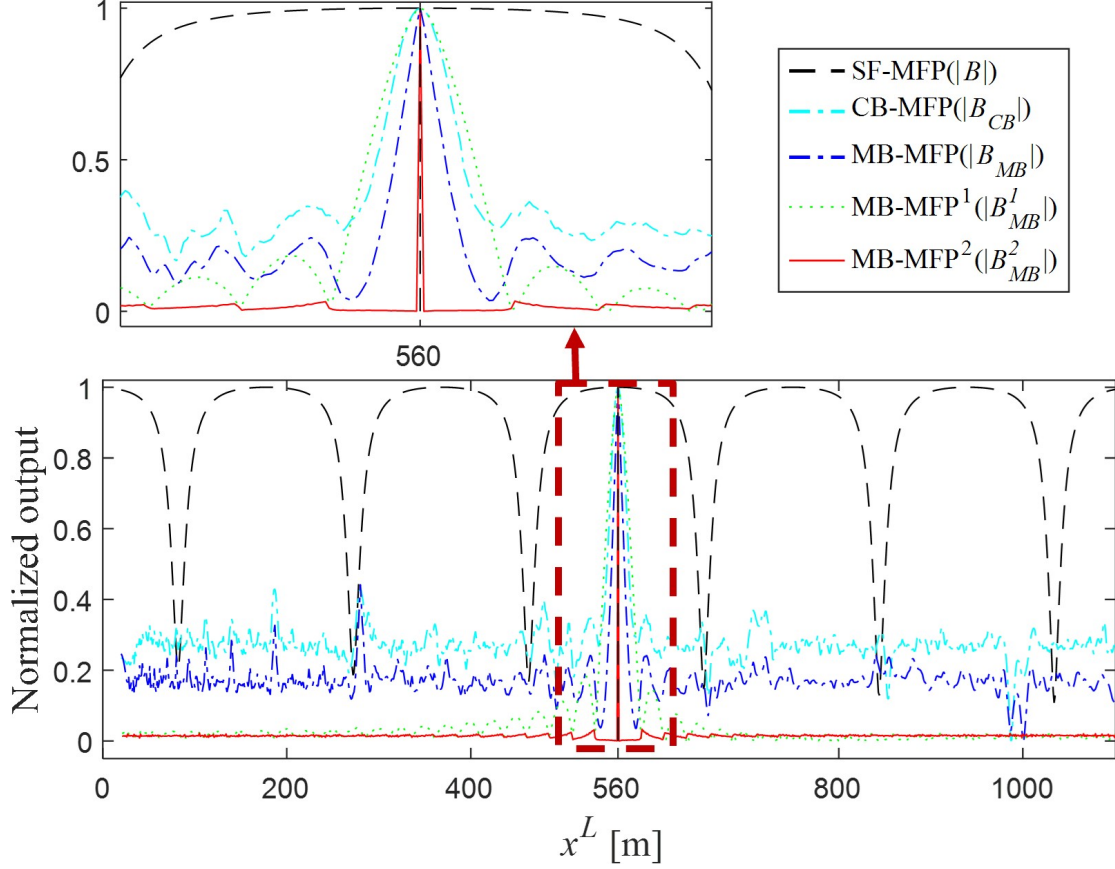


Figure 3.2. The performance comparison between single-frequency MFP (SF-MFP), MB-MFP, MB-MFP¹ and MB-MFP² for an example of single-leak detection, wherein the black vertical dashed line represents the real leak location.

3.3 Modified Broadband MFP

In general, the frequency-sensitive behavior suggests that we should consider the spectral integration to exploit multi-frequency information [39]. Therefore, the modified broadband matched field processing (MB-MFP) technique performs a direct average of the single-frequency MFP outputs

over a given bandwidth is proposed and formulated as:

$$|B_{MB}(\boldsymbol{\theta})| = \frac{1}{\omega_{\max} - \omega_{\min}} \left| \int_{\omega_{\min}}^{\omega_{\max}} B(\omega; \boldsymbol{\theta}) d\omega \right| \quad (3.3)$$

where the utilized frequency range is $(\omega_{\min}, \omega_{\max})$. MB-MFP integrates complex information across the frequency spectrum and has enhanced robustness against white noise, since white noise with its uniform spectrum over the entire frequency range cancels out during the summation process. Consequently, white noise does not introduce a frequency-based bias in the output results. The normalization to the measured field $\mathbf{h}_{\omega}^{\mathbf{M}*}$ in the MFP output function, as expressed in Eq. 3.1, compresses both signal and noise at each frequency so that their combined level equals one. This normalization is particularly effective in reducing the impact of high-amplitude noise at specific frequencies. As a result, MB-MFP, by averaging $|B|$ across frequencies, is also more effective against nonwhite noise types like colored noise and impulse noise.

Compared to single-frequency MFP, MB-MFP exhibits superior ambiguity space characteristics in terms of side-lobes, which display relatively lower peak levels (See the comparison between the performance of SF-MFP, CB-MFP and MB-MFP in Fig. 3.2 for leak-detection example). As discussed in the previous section, under ideal conditions (no mismatch and no noise), the main-lobe appears consistently at the region corresponding to the true parameters $(\boldsymbol{\theta} = \boldsymbol{\theta}^*)$ in each single-frequency MFP ambiguity space. The distribution of side-lobes, however, varies with frequency, and this variability is discussed in more detail in later chapters. After MB-MFP sums the frequency information, the peak level of the main-lobe is enhanced in the final ambiguity space. This is due to the fixed main-lobe region at each frequency. Meanwhile, the peak levels of the side-lobes are attenuated because of their varying distribution across frequencies.

In summary, the newly defined MB-MFP is robust against both white and nonwhite noise. Additionally, the main-lobe to side-lobe ratios are improved compared to single-frequency MFP. However, the issue of side-lobes has not been completely resolved. MB-MFP remains somewhat susceptible to them, although its performance in this regard has improved. Additionally, the initial MB-MFP is constrained by the wave diffraction limit, so its localization resolution is limited to no finer than half the minimum probing wavelength. The next section introduces further treatments for side-lobes and the resolution limit, and proposes improved versions of MB-MFP.

3.4 Improved MB-MFP Versions

To further suppress the produced side-lobes and increase the resolution of estimates in the ambiguity space of MB-MFP, we propose two optimized versions of MB-MFP, which leverage gradient information on the ambiguity space of single-frequency MFP ($|B|$) before averaging $|B|$ across frequencies.

3.4.1 Enhanced MB-MFP¹ with Side-Lobe Canceling Algorithm

For a continuous function $f(x)$, if it possesses a local maximum at x_0 , then $f'(x)$ shifts from positive to negative as it crosses x_0 . Leveraging this property and noting that in the single-frequency MFP ambiguity space, the side-lobes vary with frequency while the main-lobe remains fixed, we refine the MB-MFP method (Eq. 3.3) by incorporating a side-lobe canceling algorithm. The process includes:

1. Calculating the partial derivatives of $\partial|B|/\partial\theta_{n_\theta}$ where $n_\theta = 1, \dots, N_\theta$;

2. Multiplying $|B|$ by the sum of the signs of $\partial|B|/\partial\theta_{n_\theta}$ and then averaging the result across the frequency band to obtain an enhanced MB-MFP method, termed MB-MFP¹.

wherein θ_{n_θ} is the n_θ -th element in the vector $\boldsymbol{\theta} = [\theta_1, \dots, \theta_{N_\theta}]^\top$ composed of N_θ candidate parameters.

Mathematically, MB-MFP¹ is formulated as:

$$|B_{MB}^1(\theta)| = \frac{1}{\omega_{max} - \omega_{min}} \left| \int_{\omega_{min}}^{\omega_{max}} \left(\sum_{n_\theta=1}^{N_\theta} \text{sign} \left(\frac{\partial|B(\omega, \theta_{n_\theta})|}{\partial\theta_{n_\theta}} \right) \right) \cdot |B(\omega; \boldsymbol{\theta})| d\omega \right| \quad (3.4)$$

This inclusion of the derivative's sign helps assign positive values to points at the left edge of $|B|$ peaks and negative values to points at the right edge of $|B|$ peaks, at each frequency ω . Then, when averaging the processed $|B|$ across frequencies, the side-lobes in ambiguity space at one frequency step tend to cancel out these at the former frequency step. Finally, MB-MFP¹ presents reduced side-lobes in its ambiguity space than MB-MFP, while remains the main-lobe. From a physical perspective, the interpretation of Eq. 3.4 is that it utilizes derivative signs as weighting factors to achieve destructive interference of side-lobes in multi-frequency averaging while enhancing the main-lobe region, thereby improving localization resolution and reliability. This embodies the integration of "gradient information" and "frequency diversity" in signal processing, making it suitable for source localization problems in complex environments when using MFP.

3.4.2 Advanced MB-MFP² with Peak Extraction Algorithm

Building on the previous discussion, another algorithm utilizing the sign changes in the partial derivatives of $|B|$ is proposed to extract the peaks from each single-frequency MFP ambiguity space. This peak extraction algorithm helps to formulate another improved MB-MFP method

named MB-MFP² and the steps involve:

1. Calculating the partial derivatives of $\partial|B|/\partial\theta_{n_\theta}$ where $n_\theta = 1, \dots, N_\theta$;
2. Multiplying $|B|$ with the sum of the difference in signs of the derivatives at θ_{n_θ} and $\theta_{n_\theta} + \Delta\theta_{n_\theta}$, then integrating the result over whole frequency range.

These steps are mathematically expressed as:

$$|B_{MB}^2(\theta)| = \frac{\left| \int_{\omega_{min}}^{\omega_{max}} \left(\sum_{n_\theta=1}^{N_\theta} \begin{pmatrix} \text{sign}\left(\frac{\partial|B(\omega, \theta_{n_\theta})|}{\partial\theta_{n_\theta}}\right) \\ -\text{sign}\left(\frac{\partial|B(\omega, \theta_{n_\theta} + \Delta\theta_{n_\theta})|}{\partial\theta_{n_\theta}}\right) \end{pmatrix} \right) \cdot |B(\omega; \boldsymbol{\theta})| d\omega \right|}{\omega_{max} - \omega_{min}} \quad (3.5)$$

The peak extraction algorithm zeroes values at points adjacent to $|B|$ peaks at each frequency, thereby “extracting” the peaks. By integrating this adjusted $|B|$ across frequencies, the extracted main-lobes from consistent region at each frequency are summed to form a final, more distinguished main-lobe in the ambiguity space of MB-MFP². While the extracted side-lobes at each frequency, which are distributed sporadically, sum up to form final side-lobes that are significantly smaller compared to the final main-lobe. Therefore, MB-MFP² achieves side-lobes suppression capability. In addition, the peak extraction operation also enables MB-MFP² to achieve high-resolution parameter estimation, with the resolution limit determined by the predefined increment of model parameters. Fundamentally, the physical essence of this method lies in constructing a spatial difference filter through derivative sign differences between adjacent points, which selectively enhances stable peaks (main-lobes) while suppressing shifting peaks (side-lobes) during multi-frequency integration. Compared to conventional broadband MFP (MB-MFP), it more directly locates peaks and achieve ultra-high resolution, making it particularly suitable for dense multi-source scenarios.

In conclusion, the side-lobe canceling algorithm and the peak extraction algorithm, both utilizing

gradient information from $|B|$, employ distinct strategies for side-lobe suppression, resulting in differences in performance metrics such as side-lobe reduction and parameter estimation resolution. In Fig. 3.2, a simple example clearly illustrates the comparative performance in final parameter estimation of the broadband MFP methods: MB-MFP, MB-MFP¹, and MB-MFP². Further numerical and experimental results are provided in subsequent chapters to demonstrate the effectiveness of these methods in noise resistance, side-lobe suppression, and main-lobe resolution improvement. At the same time, more detailed mechanism is explained.

3.5 Summary

In this chapter, we have explored the fundamental principles of MFP and its application to pipeline parameter inversion. We first established the theoretical framework of single-frequency MFP, including the formulation of its output function as an inner product between measured and modeled signals. While single-frequency MFP provides a baseline approach, we identified its key limitations: sensitivity to noise and solution non-uniqueness due to side-lobes in the ambiguity space. To mitigate these issues, we developed modified broadband MFP (MB-MFP), which integrates single-frequency results across a broad frequency band. This integration enhances robustness against white noise and partially suppresses side-lobes. However, residual side-lobe interference persisted, leading to two proposed improved variants: MB-MFP¹, which incorporates a side-lobe canceling algorithm to actively suppress side-lobes, and MB-MFP², which employs a peak extraction algorithm to sharpen main-lobe resolution while further reducing side-lobes. Both methods demonstrate superior performance in side-lobe suppression and main-lobe clarity, significantly improving inversion accuracy in noisy environments. Subsequent chapters will validate these advancements

through numerical simulations and experimental case studies, highlighting their practical utility in real-world pipeline monitoring.

CHAPTER 4

SINGLE-DEFECT DETECTION BY MATCHED-FIELD PROCESSING

Matched-field processing (MFP) has recently been employed for leak detection in pipelines. The conventional broadband MFP (CB-MFP) method typically produces a main-lobe associated with the actual leak location, along with side-lobes. Such side-lobes have, until now, been considered as a nuisance and source of false identification of defects in pressurized pipelines. In this study, we turn this nuisance into a defect identification advantage. Specifically, we theoretically and experimentally demonstrate that the spatial positioning of the side-lobes of single-frequency MFP varies with the candidate leak location and the probing wave frequency, and that the paths or “trajectories” of the undesirable side-lobes converge to the accurate leak location as the frequency increases. By leveraging this property of the side-lobes, this chapter presents novel modified broadband MFP (MB-MFP) techniques designed to suppress side-lobes and achieve high-resolution leak detection, even in the presence of white, colored, and impulse-type noise. Both numerical simulations and laboratory experiments validate the effectiveness of these proposed MB-MFP techniques.

4.1 Introduction

Measured transient responses are inevitably noisy, influenced by factors like turbulence, traffic, measurement errors, and vibrations from hydraulic devices. To address noisy signals, a transient wave-based leak detection method using the CB-MFP technique, also known as the Bartlett broadband MFP, was recently proposed [43]. The CB-MFP method is conducted in the frequency domain and uses signals over a broad low-frequency band. As stated in the introduction chapter, the proposed method demonstrated better performance compared to existing approaches that depend entirely on system resonance frequencies (e.g., [45]) or inverse modeling techniques like transient analysis inversion [4]. The CB-MFP approach simplifies the search process by separating the two-variable problem (leak position and size) into sequential single-variable searches. First, potential leak position is identified, then this result helps determine leak size through the output function, making calculations more efficient [43]. When plotting CB-MFP results regarding to the possible leak locations, distinct peaks emerge. However, distinguishing between true defect-related peak (main-lobe) from the interference peaks (side-lobes) remains difficult [51–54]. These side-lobes appear because of the measurement limitations: (i) insufficient sensor density (spaced wider than half-wavelength intervals) [60] and (ii) limited frequency range in signal collection [62]. Furthermore, while CB-MFP performs well with white noise and structured nonwhite noise [43], it encounter challenges in real-world applications because nonwhite noise patterns vary unpredictably and prior information is hard to obtain [63].

Previous CB-MFP developments in leak detection [49, 50, 55, 56, 59, 82] have not addressed the side-lobe issues. These false signals create uncertainty in pinpointing leak locations across various fields including pipeline systems [43, 62], underwater acoustics [41], and earth science applications

[61]. In this context, Thode et al. [66] demonstrated that the pattern formed by the surface side-lobe trajectories in the spatial range-frequency plane converges to the true source range. Motivated by their study, this chapter aims to utilize the concealed knowledge within side-lobes to develop more accurate and reliable MFP methods for leak detection in water pipeline systems.

This chapter begins with a problem statement of using CB-MFP for leak detection. The problems include (i) the unknown generation mechanism of side-lobes, (ii) the ambiguity introduced by side-lobes for correct leak localization, and (iii) the lack of robustness of CB-MFP in nonwhite-noise environments [43, 62]. The objective of this study is to investigate the side-lobes and the concealed information pertaining to the leak locations within them, thus addressing the foregoing concerns. First, a derivation on the convergence of single-frequency MFP's surface side-lobe trajectories into the real leak location on the defect location-frequency plane is presented. Next, a modified broadband MFP (MB-MFP) is defined by integrating the MFP output surface across frequencies, which is robust against white and nonwhite noise (e.g., blue noise and impulse noise) due to the normalization of the measured signals. Furthermore, two improved MB-MFP methods, resulting in higher main-lobe to side-lobe ratios, are proposed by using the gradient information in the MFP output surface. Finally, numerical simulations and laboratory experiments are conducted to confirm (i) the convergence of MFP surface side-lobes, and (ii) the robustness of the proposed MB-MFP methods for leak localization and their capability to suppress side-lobes and provide a high-resolution main-lobe.

4.2 Problem Statement

This section begins by briefly describing the CB-MFP method by Wang and Ghidaoui [43] for detecting leaks. It then reviews the challenges with CB-MFP mentioned in the introduction, such as the unclear origins of side-lobes, methods for side-lobes suppression, and its poor performance under nonwhite noise.

4.2.1 Model Description

A reservoir-pipe-valve (RPV) system illustrated in Fig. 4.1, is studied in this chapter. The system consists of a single and horizontal pipe with length L and diameter D , bounded by two reservoirs with pressure heads H_1 and H_2 . A valve, located at the downstream end, is utilized to simulate transient waves via fast ‘closure-open’ maneuvers. The executed discharge perturbation has a pulse shape waveform, such as the sample shown in Fig. 4.2 with a valve closure time of 0.05 s. The introduced transient wave propagates back and forth along the pipe with a certain wave speed a_0 . The system’s fundamental period and frequency are $T_{th} = 4L/a_0$ and $f_{th} = 1/T_{th}$ (or $\omega_{th} = 2\pi/T_{th}$) [31, 32]. A single leak with size s^L and leak location x^L is assumed. The steady-state discharge through the leak is derived from the orifice equation as $Q_0^L = s^L \sqrt{2gH_0^L}$, where H_0^L is the steady-state head at the leak location, determined as $H_0^L = H_1 - (H_1 - H_2)(x^L/L)$. The number and distribution of measurement locations are denoted as N_m and $\mathbf{x}^M = [x_1^M, x_2^M, \dots, x_{N_m}^M]$, respectively. Conventionally, $N_m = 2$ is used for leak detection in simulations and experiments, although additional measurement points can provide more useful information. The distance between the two measurement locations is set to be greater than half a minimum probing wavelength, which is the

spatial-resolution limit of waves [60].

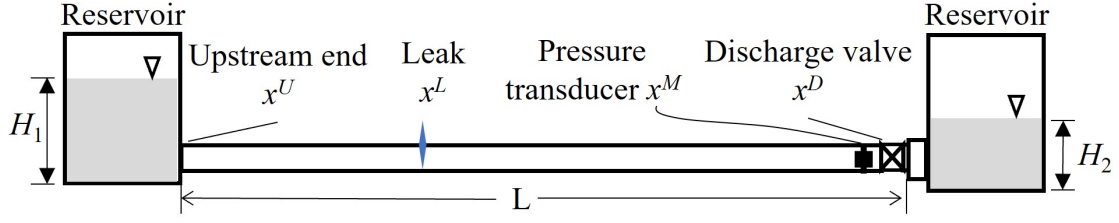


Figure 4.1. Diagram of the leaky RPV model.

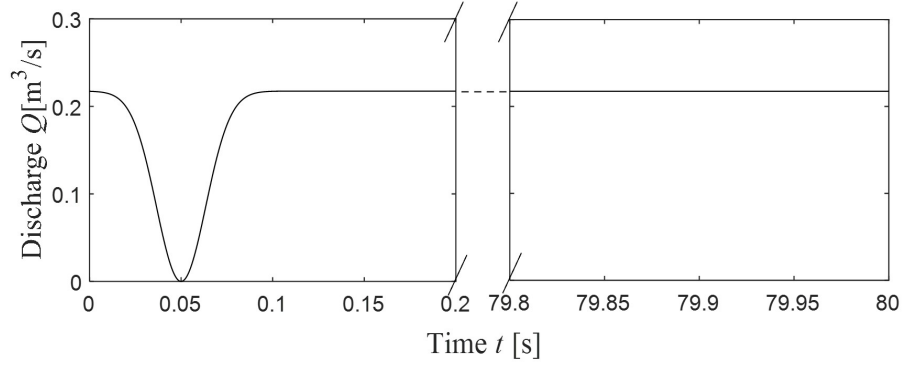


Figure 4.2. Downstream-end discharge perturbation in the form of a pulse-type waveform.

By solving the continuity and momentum equations in the frequency domain, the two hydraulic quantities q and h at a measurement location x^M are connected to q and h at the upstream end x^U with a transfer matrix between them [31]:

$$\begin{bmatrix} q_{\omega}^M \\ h_{\omega}^M \end{bmatrix} = U^O \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}^L \right) \begin{bmatrix} q_{\omega}^U \\ h_{\omega}^U \end{bmatrix} \quad (4.1)$$

where ω is the angular frequency; U^O is a two-by-two overall transfer matrix from x^U to x^M , and

$$U^O \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}^L \right) = U^{IP} \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP} \right) + U^L \left(\omega, x^M - x^U; \boldsymbol{\theta}^L \right) \quad (4.2)$$

In Eq. (4.2), U^{IP} (the superscript IP represents intact pipe) is a field transfer matrix from x^U to x^M and depends on $\boldsymbol{\theta}^{IP}$ (i.e., independent of leak parameters) [31]:

$$U^{IP} \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP} \right) = \begin{bmatrix} \cosh(\mu(x^M - x^U)) & -(1/Z) \sinh(\mu(x^M - x^U)) \\ -Z \sinh(\mu(x^M - x^U)) & \cosh(\mu(x^M - x^U)) \end{bmatrix} \quad (4.3)$$

U^L (the superscript L represents leak) is a scattering transfer matrix derived from the mass and head conservation across the leak and depends on the leak properties $\boldsymbol{\theta}^L = [s^L, x^L]$ [31]:

$$U^L \left(\omega, x^M - x^U; \boldsymbol{\theta}^L \right) = s^L \sqrt{\frac{g}{2H_0^L}} \times \begin{bmatrix} Z \sinh(\mu x^L) \cosh(\mu(x^M - x^L)) & -\cosh(\mu x^L) \cosh(\mu(x^M - x^L)) \\ -Z^2 \sinh(\mu x^L) \sinh(\mu(x^M - x^L)) & Z \cosh(\mu x^L) \sinh(\mu(x^M - x^L)) \end{bmatrix} \quad (4.4)$$

where g is the gravitational acceleration; $Z = (\mu a^2) / (i\omega g A)$ is the characteristic impedance of pipe; A is the cross-sectional area; a is the wave speed; $\mu = a^{-1} (-\omega^2 + i g A \omega R)^{1/2}$ is the propagation function; $R = (f Q_0) / (g D A^2)$ is the linearized steady-state resistance term; f_{DW} is the Darcy-Weisbach friction factor of the pipe; Q_0 is steady-state discharge and $Q_0 = (2g D A^2 (H_1 - H_2) / L f)^{1/2}$.

Combining Eqs. (4.1)-(4.4) gives the real measured pressure head h_ω^{M*} contaminated by noise n_ω^M at the single frequency ω and the measurement location x^M as

$$h_\omega^{M*} \left(\boldsymbol{\theta}^{IP*}, \boldsymbol{\theta}^{L*} \right) = h_\omega^M \left(\boldsymbol{\theta}^{IP*}, \boldsymbol{\theta}^{L*} \right) + n_\omega^M = h_\omega^M \left(\boldsymbol{\theta}^{IP*} \right) + h_\omega^M \left(\boldsymbol{\theta}^{L*} \right) + n_\omega^M \quad (4.5)$$

In this chapter, the goal is to investigate the leak detection performance of using MFP techniques. Therefore, we assume $\boldsymbol{\theta}^{IP*}$ is known and given. $\boldsymbol{\theta}^{L*} = [s^{L*}, x^{L*}]$ are the unknown vector representing the leak properties like leak size and location that need to be predicted. The introduction of superscripts $*$ aims to distinguish the real measured signal from that calculated from model, or to distinguish the real parameters from the candidates ones set in model. From Eqs. 2.34 and 2.35, the intact-pipe pressure head is given as

$$h_{\omega}^M(\boldsymbol{\theta}^{IP*}) = -Z \sinh\left(\mu(x^M - x^U)\right) q_{\omega}^U + \cosh\left(\mu(x^M - x^U)\right) h_{\omega}^U \quad (4.6)$$

and the leak-induced pressure head is expressed as

$$h_{\omega}^M(\boldsymbol{\theta}^{L*}) = s^{L*} G_{\omega}^M(x^{L*}) \quad (4.7)$$

$h_{\omega}^M(\boldsymbol{\theta}^{L*})$ is decomposed into the leak size s^{L*} (i.e. the source amplitude) and the Green's function corresponding to the real leak location x^{L*} :

$$G_{\omega}^M(x^{L*}) = -\sqrt{\frac{g}{2H_0^L}} Z \sinh\left(\mu(x^M - x^{L*})\right) \begin{pmatrix} Z \sinh(\mu(x^{L*} - x^U)) q_{\omega}^U \\ -\cosh(\mu(x^{L*} - x^U)) h_{\omega}^U \end{pmatrix} \quad (4.8)$$

which characterizes the propagation of a simple leak source at frequency ω .

Rewriting Eq. (4.5) defines the pressure head due to scattering from the leak, which is the head difference given as

$$\Delta h_{\omega}^M(\boldsymbol{\theta}^{L*}) = h_{\omega}^{M*}(\boldsymbol{\theta}^{IP*}, \boldsymbol{\theta}^{L*}) - h_{\omega}^M(\boldsymbol{\theta}^{IP*}) = h_{\omega}^M(\boldsymbol{\theta}^{L*}) + n_{\omega}^M = s^{L*} G_{\omega}^M(x^{L*}) + n_{\omega}^M \quad (4.9)$$

The leak-induced head difference Δh at all N_f frequencies and N_m measurement locations can then

be expressed as a vector form:

$$\Delta \mathbf{h}_{\boldsymbol{\omega}}^{\mathbf{M}}(\boldsymbol{\theta}^{L*}) = \mathbf{h}_{\boldsymbol{\omega}}^{\mathbf{M}*}(\boldsymbol{\theta}^{IP*}, \boldsymbol{\theta}^{L*}) - \mathbf{h}_{\boldsymbol{\omega}}^{\mathbf{M}}(\boldsymbol{\theta}^{IP*}) = \mathbf{h}_{\boldsymbol{\omega}}^{\mathbf{M}}(\boldsymbol{\theta}^{L*}) + \mathbf{n}_{\boldsymbol{\omega}}^{\mathbf{M}} = s^{L*} \mathbf{G}_{\boldsymbol{\omega}}^{\mathbf{M}}(x^{L*}) + \mathbf{n}_{\boldsymbol{\omega}}^{\mathbf{M}} \quad (4.10)$$

where $\boldsymbol{\omega}$ is the utilized frequency range and $\boldsymbol{\omega} = [\omega_1, \dots, \omega_{N_f}]$; The superscript $\mathbf{M}$ represents the measurement locations $\mathbf{x}^{\mathbf{M}} = [x_1^{\mathbf{M}}, \dots, x_{N_m}^{\mathbf{M}}]$; $\Delta \mathbf{h}_{\boldsymbol{\omega}}^{\mathbf{M}}$, $\mathbf{h}_{\boldsymbol{\omega}}^{\mathbf{M}*}$, $\mathbf{h}_{\boldsymbol{\omega}}^{\mathbf{M}}$, $\mathbf{G}_{\boldsymbol{\omega}}^{\mathbf{M}}$, and $\mathbf{n}_{\boldsymbol{\omega}}^{\mathbf{M}}$ are all $N_m \cdot N_f$ dimensional complex vectors. For example,

$$\mathbf{G}_{\boldsymbol{\omega}}^{\mathbf{M}}(x^{L*}) = \left[\underbrace{G_{\omega_1}^{M_1}(x^{L*}), \dots, G_{\omega_{N_f}}^{M_1}(x^{L*})}_{N_f}, \dots, \underbrace{G_{\omega_1}^{M_{N_m}}, \dots, G_{\omega_{N_f}}^{M_{N_m}}}_{N_f} \right]^{\top} \quad (4.11)$$

where $\top$ in the superscript represents vector transpose. In the following discussion, we will utilize the simplified notation $\Delta \mathbf{h}$ to represent $\Delta \mathbf{h}_{\boldsymbol{\omega}}^{\mathbf{M}}(\boldsymbol{\theta}^{L*})$ for clarity and conciseness.

The obtained expression of $\Delta \mathbf{h}$ in Eq. (4.10) is used for leak localization. Note that the first equal sign of Eq. (4.10) means that $\Delta \mathbf{h}$ can be computed from the pressure measurements $\mathbf{h}_{\boldsymbol{\omega}}^{\mathbf{M}*}$, and the second and third equal sign indicates the theoretical model of $\Delta \mathbf{h}$ with the real or simulated leak parameters s^{L*} and x^{L*} .

When noise is added in the simulations, the signal-to-noise ratio (SNR) is defined as (in decibel):

$$\text{SNR} = 20 \log_{10} \left(\left| \mathbb{E}(\mathbf{h}_{\boldsymbol{\omega}}^{\mathbf{M}}(x^{L*})) \right| / \sigma \right) = 20 \log_{10} \left(\left| \mathbb{E}(s^{L*} \mathbf{G}_{\boldsymbol{\omega}}^{\mathbf{M}}(x^{L*})) \right| / \sigma \right) \quad (4.12)$$

where $|\mathbb{E}(\mathbf{h}_{\boldsymbol{\omega}}^{\mathbf{M}}(x^{L*}))|$ is the mean head difference, and σ is the standard deviation of the Gaussian noise with zero mean. In pipeline systems, noise types extend beyond white noise to include nonwhite noise like colored noise and impulse noise [44, 62]. In this chapter, we focus on white

noise, blue noise (one type of colored noise) and impulse noise as examples. For white noise, SNR is a constant in the utilized frequency range ω . For blue noise, SNR drops by 3 dB per increasing frequency. Therefore, if it is assumed $\text{SNR} = \text{SNR}_1$ at ω_1 , then SNR uniformly drops to $\text{SNR}_1 - 3(\omega_{N_f} - \omega_1)/2\pi$ at ω_{N_f} . For impulse noise, this chapter assumes that $\text{SNR} = 5$ dB over ω , except at resonant frequencies (i.e., $\omega = z\omega_{th}, z \in \mathbb{Z}^+$), at which SNR suddenly decreases to SNR_2 that is much lower than $\text{SNR} = 5$ dB. In essence, the impulse noise is introduced as additional perturbations on top of the white noise with $\text{SNR} = 5$ dB.

4.2.2 CB-MFP Technique for Leak Localization and Its Challenges

CB-MFP localizes leaks by a one-dimensional search along the entire pipe length. Mathematically, the CB-MFP output function is expressed as:

$$\left| B_{CB}(x^L) \right| \equiv |\langle \Delta \mathbf{h}, \mathbf{w} \rangle| = |\Delta \mathbf{h}^H \mathbf{w}| = \sqrt{\mathbf{w}^H \Delta \mathbf{h} \Delta \mathbf{h}^H \mathbf{w}}; \Delta \mathbf{h} \text{ and } \mathbf{w} \in \mathbb{C} \quad (4.13)$$

where the operator “ $\langle \rangle$ ” means the inner product; the superscript H represents conjugate transpose; the measured data vector $\Delta \mathbf{h}$ is given in Eqs. (4.8), (4.10), and (4.11); the modeled unit data vector $\mathbf{w}$ is defined as

$$\mathbf{w} = \pm \frac{\mathbf{G}_{\omega}^{\mathbf{M}}(x^L)}{\|\mathbf{G}_{\omega}^{\mathbf{M}}(x^L)\|} = \pm \frac{\mathbf{G}_{\omega}^{\mathbf{M}}(x^L)}{\sqrt{\mathbf{G}_{\omega}^{\mathbf{M}}(x^L)^H \mathbf{G}_{\omega}^{\mathbf{M}}(x^L)}} \quad (4.14)$$

where $\mathbf{G}_{\omega}^{\mathbf{M}}(x^L)$ can be obtained from Eqs. (4.8) and (4.11) by replacing x^{L*} with x^L as

$$\mathbf{G}_{\omega}^{\mathbf{M}}(x^L) = \left[\underbrace{G_{\omega_1}^{M_1}(x^L), \dots, G_{\omega_{N_f}}^{M_1}(x^L)}_{N_f}, \dots, \underbrace{G_{\omega_1}^{M_{N_m}}, \dots, G_{\omega_{N_f}}^{M_{N_m}}}_{N_f} \right]^T \quad (4.15)$$

and

$$G_{\omega}^M(x^L) = -\sqrt{\frac{g}{2H_0^L}} Z \sinh(\mu(x^M - x^L)) \begin{pmatrix} Z \sinh(\mu(x^L - x^U)) q_{\omega}^U \\ -\cosh(\mu(x^L - x^U)) h_{\omega}^U \end{pmatrix} \quad (4.16)$$

where x^L is the candidate leak location traversing through the whole pipe length. Equation (4.14) is derived by determining a filter $\mathbf{w}$ which maximizes the signal-to-noise ratio while satisfying the constraint $|\mathbf{w}| = 1$ [43]. The CB-MFP showed in Eq. (4.13) determines how well-matched the measured and modeled signals are. The key idea to obtain the correct estimated leak location $\tilde{x}^L$ is to search for x^L that maximizes the output function $[|B_{CB}|$ in Eq. (4.13)]:

$$\tilde{x}^L = \underbrace{\arg \max_{x^L} |\langle \Delta \mathbf{h}, \mathbf{w} \rangle|}_{x^L} = \underbrace{\arg \max_{x^L} \frac{|\Delta \mathbf{h}^H \mathbf{G}_{\omega}^M(x^L)|}{\sqrt{\mathbf{G}_{\omega}^M(x^L)^H \mathbf{G}_{\omega}^M(x^L)}}}_{x^L} \quad (4.17)$$

Note that Eq. (4.17) can also be derived using the maximum likelihood estimation theory, given that there is only one leak in the system [43].

Some examples of using CB-MFP for leak detection in RPV systems are displayed in Fig. 4.3. The plot (i.e., the ambiguity space) of $|B_{CB}|$ in Fig. 4.3(a) shows a main-lobe at the real leak location (indicated by vertical dashed-line). However, high-level side-lobes also appear at other locations, which can be misconstrued for leaks and, thus, increase the number of false alarms. Wang et al. [55] attempted to address this issue using baseline (defect-free) measured transient signals. However, the applicability of such a method is limited since baseline transient responses are seldom known. Additionally, the examples in Fig. 4.3(b) indicate an resulted degradation of CB-MFP performance, i.e., the main-lobe has a level loss and mislocation under nonwhite noise conditions of blue noise and impulse noise. In other words, CB-MFP fails to identify leaks. Therefore, further research is needed to develop effective ways to suppress the side-lobes and treat nonwhite noise.

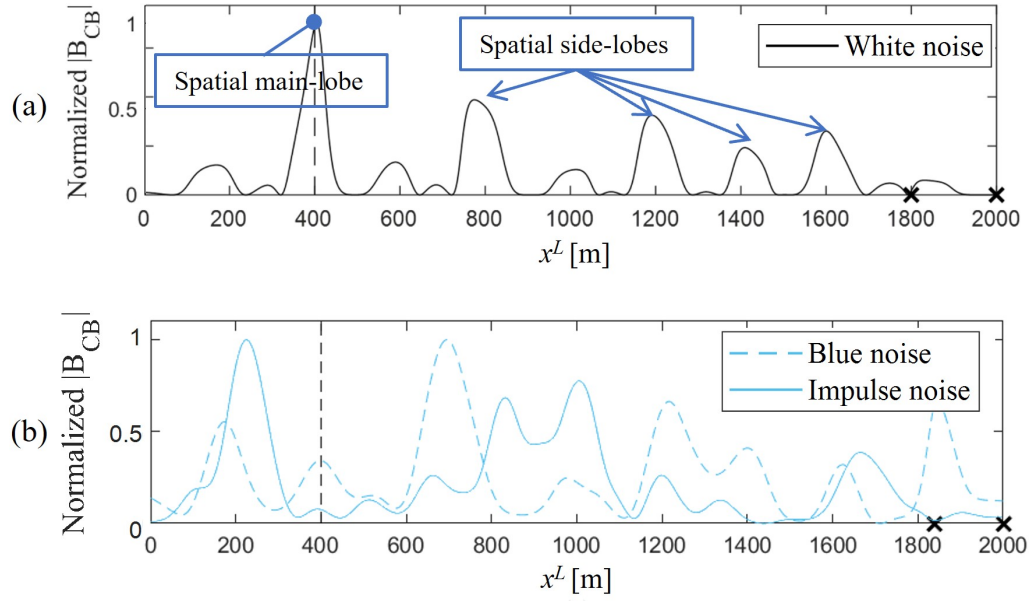


Figure 4.3. Results of using CB-MFP for leak localization under (a) white noise (SNR = 5 dB) and (b) blue noise (SNR₁ = 10 dB), and impulse noise (SNR₂ = -30 dB). The used frequency range is $\omega = [\omega_{th}, \omega_{th} + 0.1\omega_{th}, \dots, 25\omega_{th}]$. The vertical dashed line and the crosses, respectively, represent the real leak location $x^{L*} = 400$ m and the two measurement locations $\mathbf{x}^M = [1800 \text{ m}, 2000 \text{ m}]$.

Thode et al. [66] discovered that in ocean acoustics, the surface side-lobes generated by MFP shift in location across the spatial range-frequency plane as the frequency changes, forming trajectories that converge to the wave source. They successfully used such patterns to determine the wave source range correctly. Additionally, they used the gradient information in the MFP output surface to propose a side-lobe canceling algorithm for the broadband MFP. Inspired by the work of Thode et al. [66], this chapter develops a methodology that addresses the issues of side-lobes as well as nonwhite noise for leak detection application in water supply pipelines, which is discussed next.

4.3 Mechanism behind Single-Frequency MFP Side-Lobes Formation

The previous section introduced CB-MFP for leak localization and the associated issues of side-lobes appearance and non-optimal performance under nonwhite noise. To address these issues, this section explores the properties of the side-lobes in the single-frequency MFP output surface rendered on the defect location-frequency plane, subsequently referred to as “surface side-lobes”. First, the MFP output function from a single-frequency perspective is defined. Second, the relation between side-lobe location and frequency is derived from the MFP output function, which indicates that the side-lobe trajectories form a pattern that converges into the real leak location from low to high frequency on the defect location-frequency plane. Moreover, this convergence property is demonstrated to exist in simulations as well as in experimental data.

4.3.1 Definition of the Single-Frequency MFP Output Function

The MFP output function from a single-frequency perspective is defined as a functional regarding both frequency and candidate leak location:

$$|B(\omega, x^L)| = \left| \left\langle \frac{\Delta \mathbf{h}_\omega^{\mathbf{M}}}{\|\Delta \mathbf{h}_\omega^{\mathbf{M}}\|}, \frac{\mathbf{G}_\omega^{\mathbf{M}}}{\|\mathbf{G}_\omega^{\mathbf{M}}\|} \right\rangle \right| = \frac{\sqrt{(\mathbf{G}_\omega^{\mathbf{M}})^H \Delta \mathbf{h}_\omega^{\mathbf{M}} (\Delta \mathbf{h}_\omega^{\mathbf{M}})^H \mathbf{G}_\omega^{\mathbf{M}}}}{\sqrt{(\Delta \mathbf{h}_\omega^{\mathbf{M}})^H \Delta \mathbf{h}_\omega^{\mathbf{M}}} \sqrt{(\mathbf{G}_\omega^{\mathbf{M}})^H \mathbf{G}_\omega^{\mathbf{M}}}} \quad (4.18)$$

where $\Delta \mathbf{h}_\omega^{\mathbf{M}}$ and $\mathbf{G}_\omega^{\mathbf{M}}$ are both N_m -dimensional vectors, including information from N_m measurement locations at single frequency ω . The MFP output function $[|B|, \text{Eq. (4.18)}]$ differs from the CB-MFP output function $[|B_{CB}|, \text{Eq. (4.13)}]$ in that it is considering a single frequency viewpoint,

and the measured signal has been normalized. This normalization process endows the proposed broadband MFP in the following section with the capability to outperform CB-MFP under non-white noise. From the definition of inner product, $|B|$ is equivalent to $|\cos \alpha|$ where α is the angle between the two data vectors $\Delta \mathbf{h}_\omega^{\mathbf{M}}$ and $\mathbf{G}_\omega^{\mathbf{M}}$. Therefore, as the Cauchy-Schwarz theorem states,

$$\left| B(\omega, x^L) \right| = \left| \cos(\Delta \mathbf{h}_\omega^{\mathbf{M}}, \mathbf{G}_\omega^{\mathbf{M}}) \right| = |\cos \alpha| \leq 1 \quad (4.19)$$

The direction of the measured $\Delta \mathbf{h}_\omega^{\mathbf{M}}$ is fixed due to the unchanged x^{L*} , but the direction of $\mathbf{G}_\omega^{\mathbf{M}}$ changes with x^L along the pipe length. Whenever $\mathbf{G}_\omega^{\mathbf{M}}$ becomes parallel to $\Delta \mathbf{h}_\omega^{\mathbf{M}}$ (i.e., $\Delta \mathbf{h}_\omega^{\mathbf{M}}$ and $\mathbf{G}_\omega^{\mathbf{M}}$ are perfectly matched), $|B|$ reaches the maximum value of one. In other directions, $|B|$ is less than one.

In the RPV system, the pressure head at the upstream-end reservoir can be assumed to be constant. Thus, the head perturbation h at x^U across all frequencies is zero, i.e., $h_\omega^U = 0$. Then, by using the trigonometric relation of $\sinh(\beta) = -i \sin(i\beta)$, $G_\omega^M(x^{L*})$ (Eq. 4.8) becomes

$$G_\omega^M(x^{L*}) = \frac{\sqrt{g}Z^2}{\sqrt{2H_0^L}} \sin(i\mu(x^M - x^{L*})) \sin(i\mu x^{L*}) q(\omega, x^U) \quad (4.20)$$

The influence of damping on source localization has been previously investigated in the literature [60, 83–85]. It has been reported that the presence of damping does not affect the localization characteristics of the matched-filter, given that the TR-parameter ([86]) is less than or close to one and $\text{SNR} \geq 0$ dB. Waqar et al. [84] presented that the influence of various pipeline damping effects (e.g., unsteady friction and pipe wall visco-elasticity) on the peak resolution is negligible in both unbounded and bounded pipe systems. These findings imply that the objective function in Eq. (4.18) would have a well-defined peak (referred to as “main-lobe”) at the true source (i.e.,

real leak) location. This characteristic is explicitly demonstrated by dropping the damping terms (i.e., $f = 0 \Rightarrow \mu = ik = i\omega/a$) and assuming high SNR (i.e., $\mathbf{n}$ is negligible) from Eqs. (4.10), (4.11) and (4.20). Thus, by substituting the expressions of $\Delta \mathbf{h}_\omega^{\mathbf{M}}$ [Eqs. (4.10), (4.11) and (4.20)] and $\mathbf{G}_\omega^{\mathbf{M}}$ [replacing x^{L*} with x^L in Eqs. (4.11) and (4.20)], and using trigonometric identities, the MFP output function given in Eq. (4.18) can be expanded and rearranged as:

$$|B(\omega, x^L)| = \frac{\left| \underbrace{N_m \cos(-k(x^{L*} - x^L))}_{(1)} - \underbrace{\sum_{n=1}^{N_m} \cos(-k(2x^M - x^{L*} - x^L))}_{(2)} \right|}{\sqrt{N_m - \underbrace{\sum_{n=1}^{N_m} \cos(-k(2x^M - 2x^{L*}))}_{(3)}}} \sqrt{N_m - \underbrace{\sum_{n=1}^{N_m} \cos(-k(2x^M - 2x^L))}_{(4)}}} \quad (4.21)$$

The locations of the maxima of $|B|$ in Eq.(4.21) can be derived by finding x^L that satisfies $|B| = 1$.

The solution is

$$x^L = x^{L*} - \frac{m\pi}{k}, \quad m \in \mathbb{Z} \quad (4.22)$$

At these locations, the terms in Eq.(4.21) adhere to the following rules: term (1) = 1 and term (2) = term (3) = term (4), resulting in the numerator and denominator being equal, thus yielding $|B| = 1$.

By introducing the fundamental frequency $\omega_{th} = a\pi/2L$ and the wavenumber $k = \omega/a$, Eq. (4.22) becomes

$$x^L = x^{L*} - \frac{2m}{\omega/\omega_{th}} L, \quad m \in \mathbb{Z} \quad (4.23)$$

where ω/ω_{th} is the dimensionless frequency. Eq. (4.23) implies that the maxima of $|B|$ appear periodically and one of the maxima always appears at $x^L = x^{L*}$ (corresponding to $m = 0$) for all ω . This special maximum at $x^L = x^{L*}$ is the main-lobe of the single-frequency MFP output $|B|$, and the other maxima at $x^L = x^{L*} - 2mL/(\omega/\omega_{th})$ ($m \in \mathbb{Z} \setminus \{0\}$) are the side-lobes. The side-lobes are evenly

distributed on both sides of the main-lobe and appear more and more intensely as the frequency increases. Several examples of $|B(\omega/\omega_{th} = 11, x^L)|$, $|B(\omega/\omega_{th} = 26, x^L)|$, $|B(\omega/\omega_{th} = 41, x^L)|$, and $|B(\omega/\omega_{th} = 56, x^L)|$ are shown in Fig. 4.4(a) to illustrate the distribution of main-lobe and side-lobes.

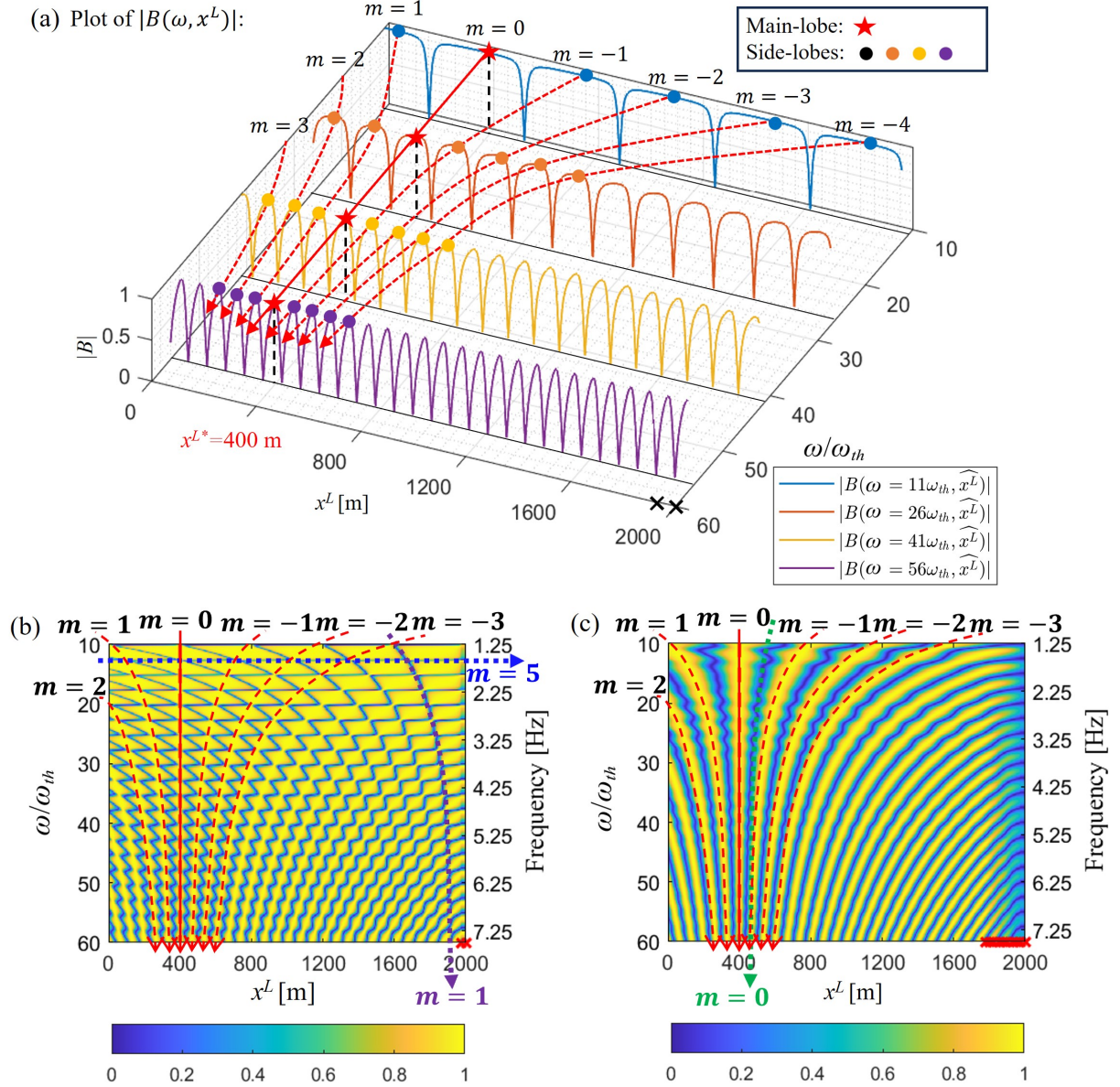


Figure 4.4. (a) Demonstration of $|B(\omega, x^L)|$ at $\omega = 11\omega_{th}, 26\omega_{th}, 41\omega_{th}$ and $56\omega_{th}$ and the trajectories of surface main-lobes and side-lobes. The real leak location $x^{L*} = 400$ m is represented by the vertical dashed lines. The two measurement locations $\mathbf{x}^M = [1975, 2000]$ m are marked with crosses. (b) and (c) Two output surfaces of $|B|$ in the defect location-frequency plane with $\mathbf{x}^M = [1975, 2000]$ m and $\mathbf{x}^M = [1775, 1775+25, \dots, 2000]$ m, respectively.

4.3.2 The Convergence of Surface Side-Lobes Trajectories into the Leak Location on the Defect Location-Frequency Plane

According to Eq. (4.23), the main-lobe (*i.e.*, $m = 0$) always appears at $x^L = x^{L*}$ for all ω . Thus, if rendering the MFP output ($|B|$) in the defect location-frequency plane, these surface main-lobes collectively form a vertical straight line to the real leak location x^{L*} along the frequency axis (see the grey dash line of $x^L = x^{L*}$ in Fig. 4.4(a)). On the other hand, for $m \neq 0$, x^L varies with both m and ω . Eq. (4.23) shows clearly that x^L converges to x^{L*} for all m when ω becomes larger (*i.e.*, when $m \neq 0$, $\tilde{x}^L \rightarrow x^{L*}$ as $\omega \rightarrow \infty$). For consistency in terminology with Thode et al. [66], the relation between x^L and ω given by Eq. (4.23) is hereafter referred to as the trajectories of the surface side-lobes. Figure 4.4(a) shows several of such trajectories. The red dash lines illustrate the trajectories when $m = [-4, -3, \dots, 2, 3]$ in Eq. (4.23).

The observed phenomena – specifically: 1) the spatial periodicity density of $|B|$ maxima increases with frequency, and 2) a local maximum consistently emerges at the true leak location across all frequencies – can be explained through two physical mechanisms:

1. Frequency-Dependent Periodicity (Phenomenon 1)

This behavior originates from the frequency-domain decomposition of transient waves. According to Fourier transform theory, time-domain transient waves can be decomposed into steady-state harmonic components at distinct frequencies. As frequency increases, the corresponding harmonic wavelength decreases, thereby enhancing the spatial resolution of wave-field matching. This physical principle is mathematically characterized by the frequency-domain response function in the MFP algorithm (Eq. 4.21), manifesting as a decrease of

spacing between $|B|$ maxima along operating frequency.

2. Frequency-Independent Maximum at True Leak Location (Phenomenon 2)

The persistent global maximum at the actual leak position stems from the physical foundation of MFP methodology. By matching features of leak-scattered wavefields – which inherently encode source characteristics (e.g., geometric scale, spatial location) – MFP ensures maximum coherence exclusively at the true leak location during wavefield reconstruction. Mathematically, this corresponds to the response output function attaining its global maximum at the authentic leak location. Crucially, this maximum remains frequency-independent, providing robust physical validation for leak localization.

The coupling between these phenomena reveals: frequency selection governs spatial resolution (affecting localization accuracy), while the intrinsic properties of scattered wavefields impose physical constraints for leak identification. This synergy between spectral characteristics and wave propagation principles constitutes the theoretical foundation for MFP's robust leak detection capability.

Two samples of the output surface of $(|B|)$ rendered in the defect location-frequency plane are given in Fig. 4.4(b) and Fig. 4.4(c). The results illustrate the convergence of surface side-lobe trajectories into the real leak location (i.e., $x^L \rightarrow x^{L*}$ as $\omega \rightarrow \infty$). However, Fig. 4.4(b) shows that the trajectories of minima of $|B|$, when only two close sensors are used, exhibit a zigzag pattern. These zigzags become much less pronounced by using more sensors (see Fig. 4.4(c) for the case of ten sensors). These phenomena can be explained as follows. First, for the case of two sensors situated near the downstream end of the pipe (i.e., $N_m = 2$ and $x_2^M \approx x_1^M = L$) in Fig. 4.4(b), $|B|$ in Eq.(4.21) becomes

$$|B(\omega, x^L)| = \frac{|\cos(-k(x^{L*} - x^L)) - \cos(-k(2L - x^{L*} - x^L))|}{\sqrt{1 - \cos(-k(2L - 2x^{L*}))} \sqrt{1 - \cos(-k(2L - 2x^L))}} \quad (4.24)$$

$|B|$ in Eq. (4.24) is minimum when the numerator is zero, i.e., $\cos(-k(x^{L*} - x^L)) = \cos(-k(2L - x^{L*} - x^L))$, which yields

$$-k(x^{L*} - x^L) = -k(2L - x^{L*} - x^L) + 2m\pi \Rightarrow \omega/\omega_{th} = m \frac{2L}{L - x^{L*}}, m \in \mathbb{Z}^+ \quad (4.25)$$

or

$$k(x^{L*} - x^L) = -k(2L - x^{L*} - x^L) + 2m\pi \Rightarrow \omega/\omega_{th} = m \frac{2L}{L - x^L}, m \in \mathbb{Z}^+ \quad (4.26)$$

Eq. (4.25) and Eq. (4.26) respectively describe the horizontal and curved trajectories of minima in Fig. 4.4(b), and their combination forms the zigzag pattern. For example, the black horizontal dot line is one of the horizontal trajectories corresponding to $m = 5$ in Eq. (4.25), and the black curved dash-dot line is one of the curved trajectories corresponding to $m = 1$ in Eq.(4.26).

Next, considering the multi-sensor case in Fig. 4.4(c), where the sensors are widely and uniformly distributed at $\mathbf{x}^M = [x_1^M, x_2^M, \dots, x_{N_m}^M] = [L - (N_m - 1)\Delta x^M, \dots, L - \Delta x^M, L]$ with $N_m = 10$ and $\Delta x^M = 25$ m. In this case, dividing all terms on the right-hand side of Eq. (4.21) by N_m results in the trigonometric summation terms (2), (3) and (4) approaching zero and term (1) becoming $\cos(-k(x^{L*} - x^L))$. Therefore, Eq. (4.21) can be simplified to

$$|B(\omega, x^L)| = |\cos(-k(x^{L*} - x^L))| \quad (4.27)$$

where the locations of minima can be easily obtained from $\cos(-k(x^{L*} - x^L)) = 0$ as

$$x^L = x^{L*} + \left(\frac{1}{2} - m\right) \frac{2L}{\omega/\omega_{th}}, m \in \mathbb{Z} \quad (4.28)$$

which correspond exactly to the curved trajectories of $|B|$ minima in Fig. 4.4(c). For instance, a sample trajectory corresponding to $m = 0$ is shown as the black curved dash line. In a manner

similar to Eq. (4.23), Eq. (4.28) also shows that x^L converges to x^{L*} for all m when ω becomes larger, as seen in Fig. 4.4(c).

4.3.3 Numerical Simulations

This sub-section presents more numerical examples of MFP side-lobe trajectories considering damping ($f = 0.02$) and noise (SNR = 5 dB) effects. The pressure heads at reservoirs are $H_1 = 25$ m and $H_2 = 20$ m. The pipe length $L = 2000$ m and the pipe diameter $D = 0.5$ m. The steady-state discharge is $Q_0 = 0.0153$ m³/s. A transient valve-closure signal, as shown in Fig. 4.2, is inserted. The generated transient wave propagates with a speed $a_0 = 1000$ m/s. This propagation process in the frequency domain is simulated by using the transfer matrix method as expressed in Eq. (4.1). The fundamental frequency is $f_{th} = 1/T_{th} = 0.125$ Hz (or $\omega_{th} = 2\pi/T_{th} = 0.785$ rad/s). The real leak location and leak size are $x^{L*} = 400$ m and $s^{L*} = 1.4 \times 10^{-4}$ m².

The simulation results with $N_m = 2$ and 10 are displayed in Fig. 4.5(a) and 4.5(b), respectively. It is observed that the side-lobe trajectories converge into the real leak location $x^{L*} = 400$ m. The estimated range of the leak is approximately 50 m wide at $f = 7.25$ Hz. The comparison between Fig. 4.4(b, c) and Fig. 4.5 indicates that the presence of damping and noise does not affect the pattern's characteristics but rather decreases the pattern's clarity. By incorporating additional measurement locations, Fig. 4.5(b) displays a more readable set of trajectories than Fig. 4.5(a). The appearance of horizontal interference lines is caused by the Bragg resonance condition of maximum wave transmission at $\omega/\omega_{th} = 2mL/x^{L*} = 10m$ ($m = \mathbb{Z}^+$) [65, 87]. At such frequencies the measured leak-induced data vector $\Delta \mathbf{h}_\omega^{\mathbf{M}}$ contains minimal information about the leak and is predominantly composed of noise. Overall, increasing array length can significantly improve peak resolution, and

reduce side-lobes and mismatch sensitivity.

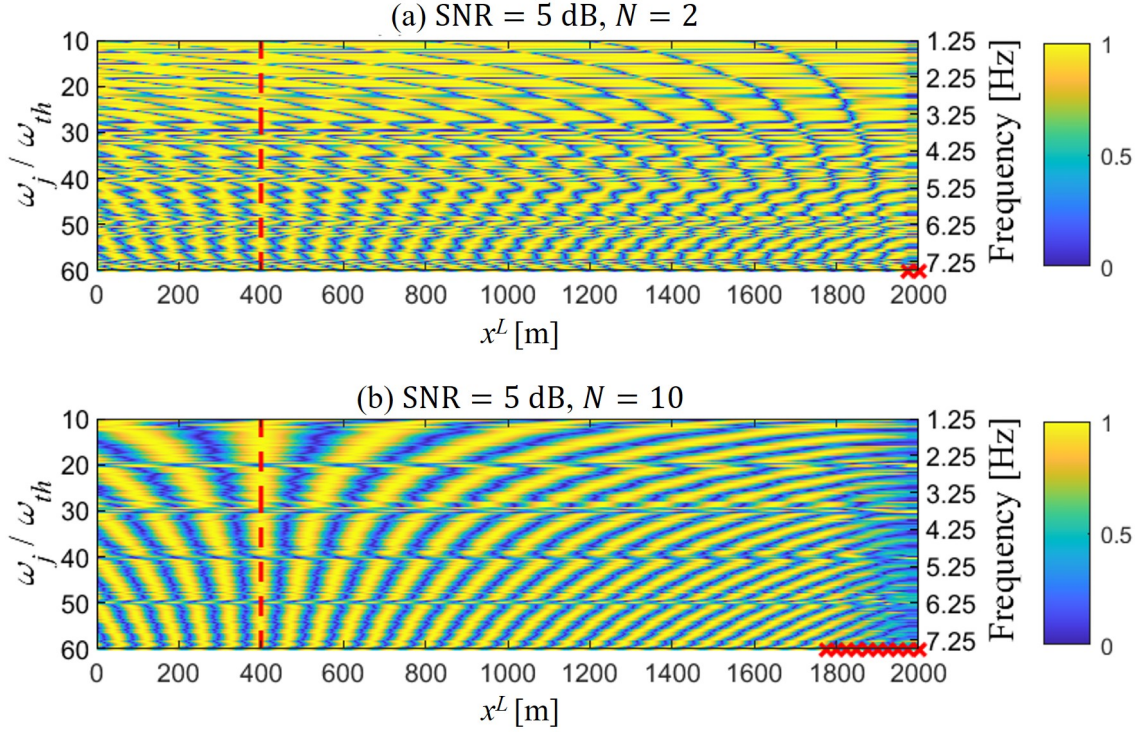


Figure 4.5. The surface side-lobes converge into the real leak location in the defect location-frequency plane for a leak-detection case with (a) SNR = 5 dB, $N_m = 2$ ($\mathbf{x}^M = [1975, 2000]$ m), (b) SNR = 5 dB, $N_m = 10$ ($\mathbf{x}^M = [1775, 1775 + 25, \dots, 2000]$ m). The real leak location is $x^{L*} = 400$ m and represented by the dashed line. The measurement locations are marked with crosses.

4.4 Definition of The Modified Broadband MFP (MB-MFP)

The former section elucidated the concealed leak information in the MFP output surface on the defect location-frequency plane. Building upon this analysis, a modified broadband MFP (MB-MFP) is introduced in this section. MB-MFP involves averaging the MFP output function across frequencies. This newly defined MB-MFP is found to be robust in the presence of both white and

nonwhite noise. However, it is still prone to the presence of side-lobes. In the following section, a treatment for the side-lobes is presented.

CB-MFP [Eq. (4.13)] directly sums the complex information across all frequencies through the inner product operation. As such, the flat spectrum of white noise across the available frequency bandwidth does not have a frequency-dependent bias on the inner product, making CB-MFP robust against white noise. In contrast, CB-MFP is susceptible to nonwhite noise (or colored noise) which has uneven power spectrum density. However, nonwhite noise is often encountered in real-field pipelines due to various factors, including the presence of dynamic devices and turbulence [44, 62]. To overcome this limitation, an MB-MFP method using the frequency average of the MFP output function [Eq. (4.18)] is defined with an integration form:

$$\left| B_{MB}(x^L) \right| = \frac{1}{\omega_{\max} - \omega_{\min}} \left| \int_{\omega_{\min}}^{\omega_{\max}} B(\omega, x^L) d\omega \right| \quad (4.29)$$

The integration is performed through the frequency range $[\omega_{\min}, \omega_{\max}]$ at x^L , i.e., performed along the vertical line at x^L in the defect location-frequency plane (see the example shown in Fig. 4.6). After averaging the MFP output surface along the vertical frequency axis in Fig. 4.6(a), a resulting one-dimensional MB-MFP output plot across all leak candidates is presented in Fig. 4.6(b), where the output is normalized by using min-max feature scaling (e.g., for a given data vector $\mathbf{y}$, $\mathbf{y}_{\text{new}} = (\mathbf{y} - \min(\mathbf{y})) / (\max(\mathbf{y}) - \min(\mathbf{y}))$). The result in Fig. 4.6(b) shows a main-lobe at real leak location x^{L*} , as well as side-lobes with large amplitude at other locations.

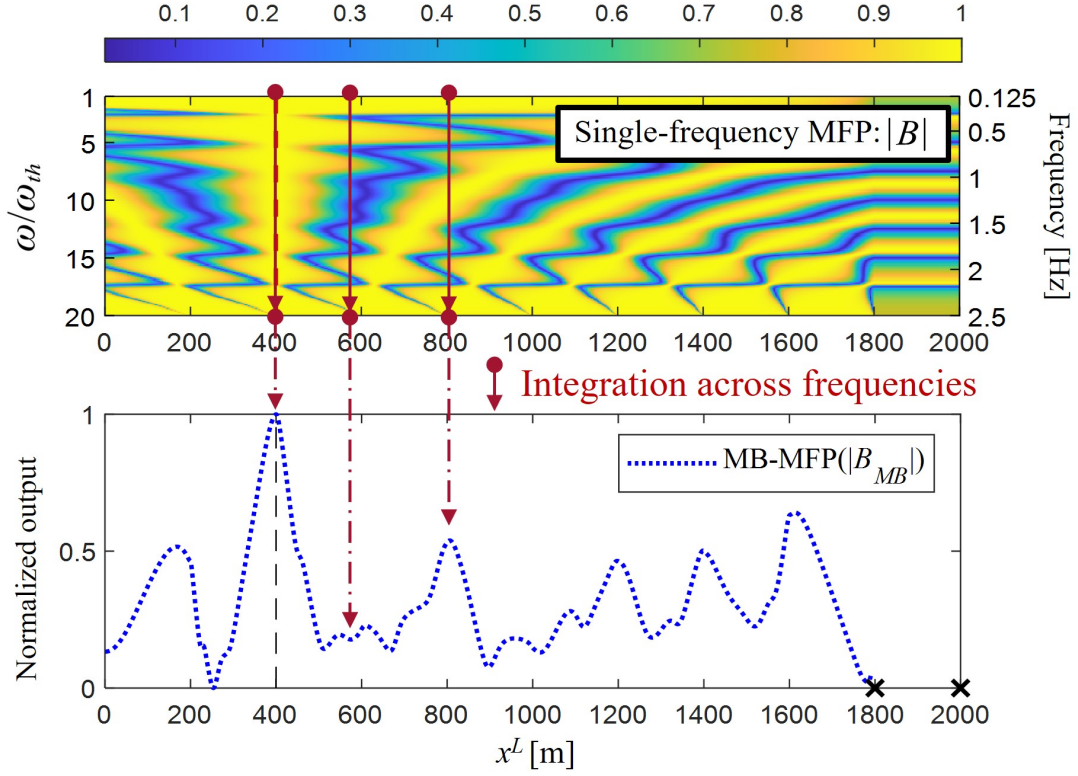


Figure 4.6. (a) An example of MFP output surface ($|B|$) in the defect location-frequency plane. (b) The resulting MB-MFP output plot ($|B_{MB}|$) along x^L by averaging $|B|$ across frequencies. The vertical dashed line and the crosses represent the real leak location $x^{L*} = 400$ m and the measurement locations $\mathbf{x}^M = [1800, 2000]$ m, respectively.

Due to the normalization term $\|\Delta \mathbf{h}_\omega^M\|$ used in the denominator in the MFP output function [$|B|$, Eq. (4.18)], at each ω , the signal and noise are compressed to a level of their sum equal to one. Thus, the noise with relatively high amplitude at certain frequencies is significantly compressed after this normalization process. This makes MB-MFP ($|B_{MB}|$), obtained from the frequency average of $|B|$, suitable against nonwhite noise, such as blue noise and impulse noise. To demonstrate this advantage, three numerical examples are conducted to compare the performance of MB-MFP and CB-MFP under three different noise scenarios: white noise, blue noise, and impulse noise. The results are shown in Fig. 4.7. The utilized frequency range $f = [f_{th}, f_{th} + 0.1f_{th}, \dots, 160f_{th}] =$

$[0.125, 0.125+0.0125, \dots, 20]$ Hz, corresponding to a dimensionless angular frequency range $\omega/\omega_{th} = [1, 1+0.1, \dots, 160]$. Fig. 4.7(a) shows that both CB-MFP and MB-MFP perform well for leak identification in the white-noise case with $\text{SNR} = 5$ dB, but MB-MFP still exhibits a relatively higher main-lobe-to-side-lobes ratio. Fig. 4.7(b) and Fig. 4.7(c) illustrate that CB-MFP fails to localize leak correctly under the blue noise with $\text{SNR}_1 = 10$ dB and the impulse noise with $\text{SNR}_2 = -60$ dB. However, MB-MFP can still successfully identify the real leak location in these cases.

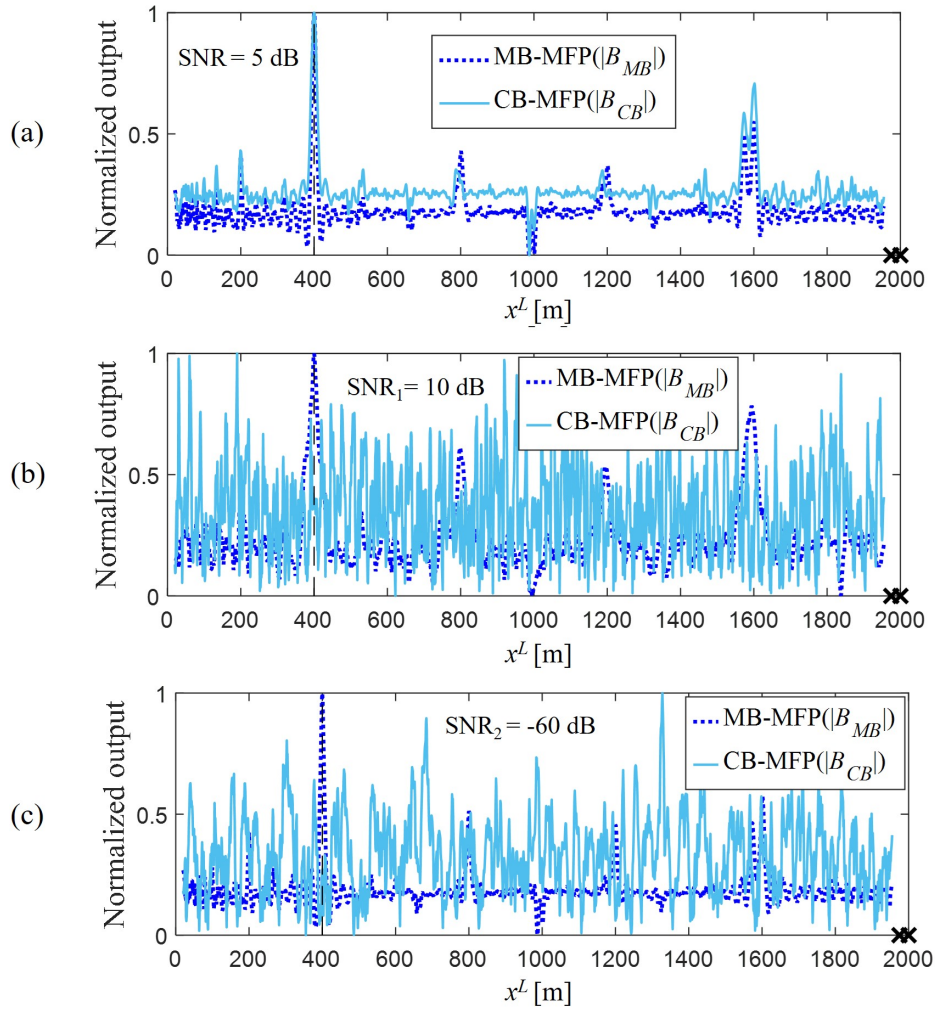


Figure 4.7. Leak localization using CB-MFP ($|B_{CB}|$) and MB-MFP ($|B_{MB}|$) under (a) white noise with $\text{SNR} = 5$ dB, (b) blue noise with $\text{SNR}_1 = 10$ dB, and (c) impulse noise with $\text{SNR}_2 = -60$ dB. The vertical dashed line and the crosses represent real leak location $x^{L*} = 400$ m and the measurement locations $\mathbf{x}^M = [1800, 2000]$ m, respectively.

To extensively assess the accuracy of leak localization while employing CB-MFP and MB-MFP, the absolute localization error $|\tilde{x}^L - x^{L*}|$ regarding various $\text{SNR} = [-8, -8+4, \dots, 12]\text{dB}$, $\text{SNR}_1 = [10, 10+8, \dots, 50]\text{dB}$, and $\text{SNR}_2 = [-70, -70+10, \dots, -20]\text{dB}$ are displayed in Fig. 4.8, where $\tilde{x}^L$ is the predicted leak location. Since the noise generation is random, for each SNR, SNR_1 and SNR_2 , twenty simulations are implemented to average the error. Fig. 4.8(a) demonstrates that both MB-MFP and CB-MFP can accurately localize leaks in the presence of white noise, with a small error less than 1 m. However, under blue noise, as demonstrated in Fig. 4.8(b), MB-MFP exhibits significantly lower leak localization errors compared to CB-MFP. As for impulse noise, Fig. 4.8(c) shows that MB-MFP performs a strong anti-impulse noise ability even at a low $\text{SNR}_2 = -70\text{ dB}$, while CB-MFP fails to perform well since SNR_2 below -40 dB .

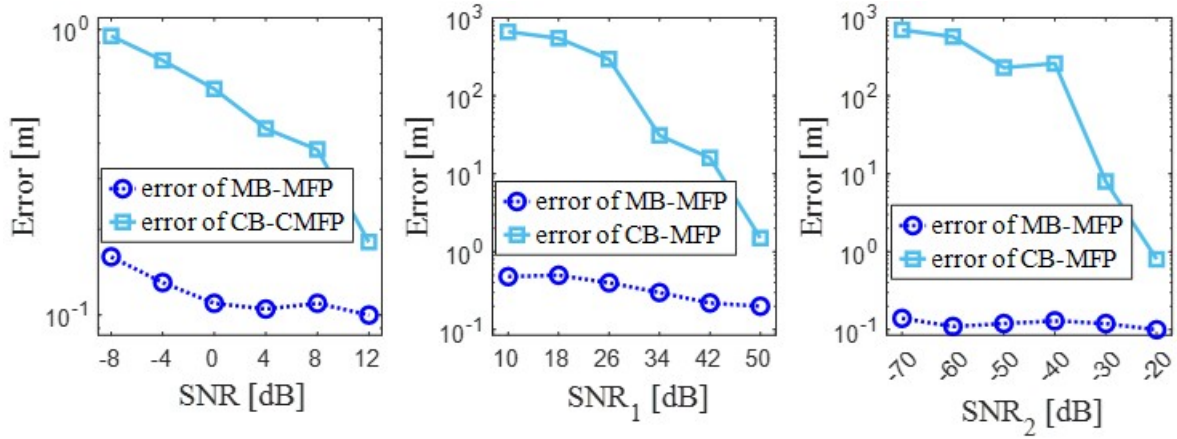


Figure 4.8. Localization errors of using CB-MFP and MB-MFP under (a) white noise with $\text{SNR} = [-8, -8+4, \dots, 12]\text{dB}$, (b) blue noise with $\text{SNR}_1 = [10, 10+8, \dots, 50]\text{dB}$ (the corresponding SNR at $f_{\max} = 20\text{ Hz}$ is $[-50, -50+8, \dots, -10]\text{dB}$), and (c) impulse noise with $\text{SNR}_2 = [-70, -70+10, \dots, -20]\text{dB}$. The real leak location is $x^{L*} = 400\text{ m}$ and the measurement locations are $\mathbf{x}^{\mathbf{M}} = [1975\text{ m}, 2000\text{ m}]$.

4.5 Developed MB-MFP Methods for Side-Lobe Suppression

Despite its accuracy and robustness under white and nonwhite noise, the abovementioned MB-MFP still experiences high-level side-lobes, which introduce ambiguity when deciding on leak detection. To address this issue, this section proposes two additional MB-MFP methods, namely, MB-MFP¹ using a side-lobe canceling algorithm and MB-MFP² using a peak extraction algorithm, to increase the main-lobe to side-lobe ratios. Both methods utilize the gradient information in the MFP output surface. Their effectiveness for suppressing side-lobes is demonstrated and compared in the simulations in this section and the lab tests in the following section.

4.5.1 MB-MFP¹ Using Side-Lobe Canceling Algorithm

For a continuous function $f(x)$, if it has a local maximum at x_0 , then $f'(x)$ changes from positive on the left side to negative on the right side near x_0 . Based on this property, combined with the knowledge that the surface side-lobes are uniformly distributed on both sides of the surface main-lobe at each frequency, we can improve the proposed MB-MFP method [Eq. (4.29)] by applying a side-lobe canceling algorithm on it. To do so, we follow these steps:

- Calculate the partial derivative $\partial|B|/\partial x^L$;
- Multiply $\partial|B|/\partial x^L$ with the sign of $\partial|B|/\partial x^L$, and then average it over the whole frequency band to yield an improved MB-MFP method (named MB-MFP¹). Mathematically, the output function of MB-MFP¹ is expressed as

$$\left|B_{MB}^1(x^L)\right| = \frac{1}{\omega_{\max} - \omega_{\min}} \left| \int_{\omega_{\min}}^{\omega_{\max}} \text{sign}\left(\frac{\partial|B(\omega, x^L)|}{\partial x^L}\right) B(\omega, x^L) d\omega \right| \quad (4.30)$$

Introducing the sign of $\partial|B|/\partial x^L$ assigns positive values to the left-edge points and negative values to the right-edge points of the maxima of $|B|$ at each ω , which can be illustrated by a sample in Fig. 4.9(a), where the blue dot line is the output $|B(\omega = 21\omega_{th}, x^L)|$ and the green solid line is the result after multiplying $|B(\omega = 21\omega_{th}, x^L)|$ with the sign of $\partial|B|/\partial x^L$. The result of multiplying $|B|$ with the sign of $\partial|B|/\partial x^L$ over frequency can be observed in the defect location-frequency plane in Fig. 4.9(b). Then, by integrating the updated $|B|$ over frequency, the surface side-lobes at ω tend to cancel out the surface side-lobes at the former frequency step $\omega - \Delta\omega$. As a result, the side-lobes in the ambiguity space of MB-MFP¹ ($|B_{MB}^1|$) are lower than these of MB-MFP ($|B_{MB}|$), which is demonstrated in Fig. 4.11 and discussed later.

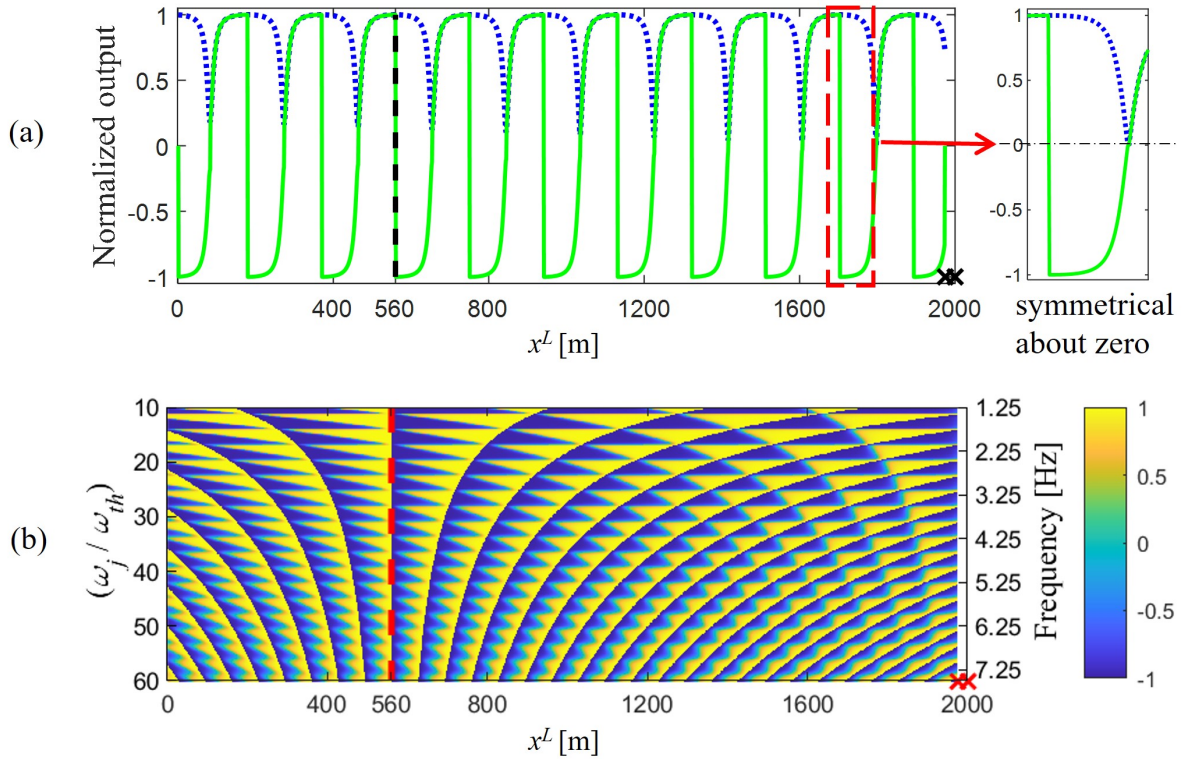


Figure 4.9. Results of multiplying $|B|$ with the sign of $\partial|B|/\partial x^L$ (a) at $\omega = 21\omega_{th}$ and (b) in the defect location-frequency plane. A noiseless environment is considered for clearer observation of the model behavior. The real leak location is $x^{L*} = 560$ m and represented by the vertical dashed line. The measurement locations are $\mathbf{x}^M = [1975 \text{ m}, 2000 \text{ m}]$ and represented by the crosses.

4.5.2 MB-MFP² Using Peak Extraction Algorithm

Building upon the discussion in the previous subsection, we can develop another approach using the derivative information of $|B|$. That is, we can extract the maxima of $|B|$ by using the sign changes of $\partial|B|/\partial x^L$. This peak extraction algorithm follows these steps:

- Calculate the derivative $\partial|B|/\partial x^L$;
- Multiply $|B|$ with the sign of $\partial|B(\omega, x^L)|/\partial x^L$ minus the sign of $\partial|B(\omega, x^L + \Delta x^L)|/\partial x^L$, and then integrate it over the whole frequency band to formulate another improved MB-MFP method (named MB-MFP²) as:

$$\left| B_{MB}^2(x^L) \right| = \frac{\left| \int_{\omega_{\min}}^{\omega_{\max}} \left(\begin{array}{c} \text{sign} \left(\frac{\partial|B(\omega, x^L)|}{\partial x^L} \right) \\ - \text{sign} \left(\frac{\partial|B(\omega, x^L + \Delta x^L)|}{\partial x^L} \right) \end{array} \right) |B(\omega, x^L)| d\omega \right|}{\omega_{\max} - \omega_{\min}} \quad (4.31)$$

The subtraction of the two signs of the derivatives of $|B|$ at x^L and $(x^L + \Delta x^L)$ enables the extraction of peaks of $|B|$ along x^L at each frequency. This is accomplished by assigning zero values to the left-edge and right-edge points close to each peak, which can be illustrated by a sample shown in Fig. 4.10(a). The blue dashed line is $|B(\omega = 21\omega_{th}, x^L)|$. The red solid line is the result after applying the peak extraction algorithm, where the peaks are extracted, and the values on both sides of each peak are suppressed to zero. Fig. 4.10(b) shows the extracted maxima of $|B|$ after using this peak extraction algorithm over frequency, and the traced maxima provide a clear and distinct pattern illustrating the convergence. Then, by integrating the updated $|B|$ across frequencies, the resulting MB-MFP² can have a higher main-lobe-to-side-lobes ratio than MB-MFP. It should be noted that while both peak extraction algorithm [Eq. (4.31)] and side-lobe canceling algorithm [Eq. (4.30)] have utilized the gradient information of $|B|$, they differ in their specific implementation ways. In

the next subsection, numerical examples are implemented to demonstrate the capabilities of MB-MFP¹ and MB-MFP² for suppressing side-lobes.

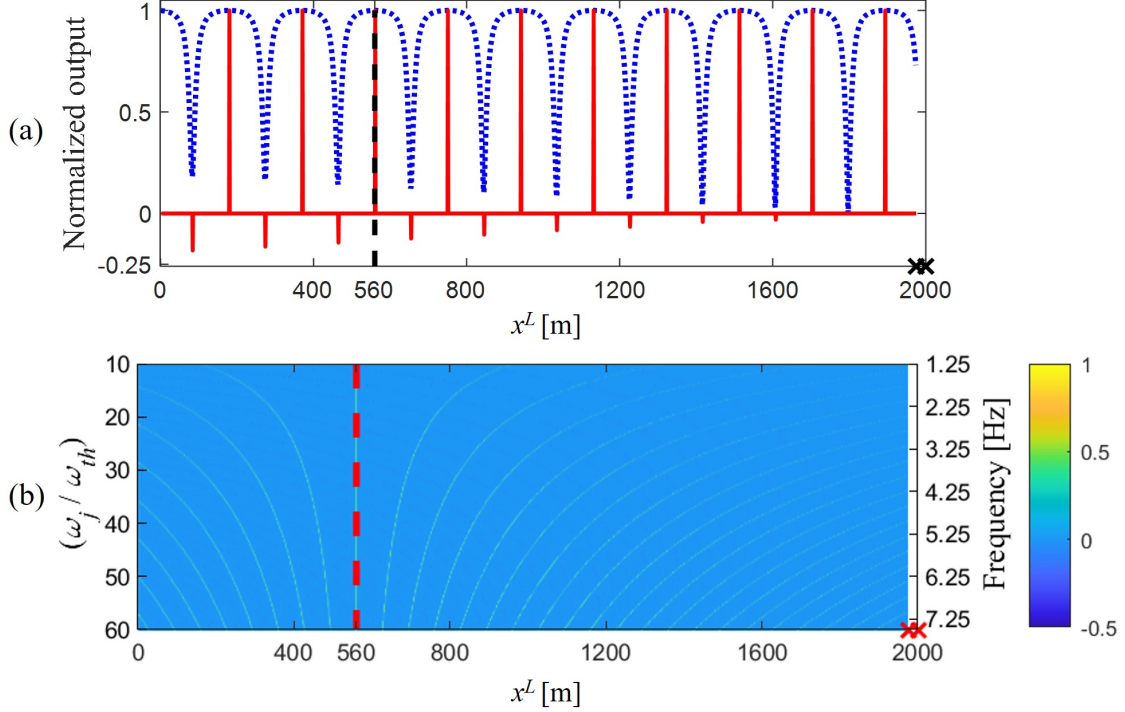


Figure 4.10. Demonstration of the peak extraction algorithm on $|B|$ (a) at $\omega = 21\omega_{th}$ and (b) in the defect location-frequency plane. A noiseless environment is considered for clearer observation of the model behavior. The real leak location is $x^{L*} = 560$ m and represented by the vertical dashed line. The measurement locations are $\mathbf{x}^M = [1975 \text{ m}, 2000 \text{ m}]$ and represented by the crosses.

4.5.3 Numerical Simulations of Using CB-MFP, MB-MFP, MB-MFP¹, and MB-MFP²

The performance of CB-MFP [$|B_{CB}|$, Eq. (4.13)], MB-MFP [$|B_{MB}|$, Eq. (4.29)], MB-MFP¹ [$|B_{MB}^1|$, Eq. (4.30)], and MB-MFP² [$|B_{MB}^2|$, Eq. (4.31)] in terms of leak localization and side-

lobes suppression is well demonstrated in Fig. 4.11. Four cases of (a) no noise, (b) white noise with $\text{SNR} = 5$ dB, (c) blue noise with $\text{SNR}_1 = 30$ dB, and (d) impulse noise with $\text{SNR}_2 = -60$ dB are considered. The utilized frequency range is $f = [f_{th}, f_{th} + 0.1f_{th}, \dots, 160f_{th}] = [0.125, 0.125 + 0.0125, \dots, 20]$ Hz. Fig. 4.11(a) illustrates that in the noise-free case, all methods have successfully localized the real leak. In comparison to CB-MFP and MB-MFP, MB-MFP¹ and MB-MFP² both exhibit significant side-lobes suppression, but MB-MFP² demonstrates stronger suppression capability and higher main-lobe resolution than MB-MFP¹. For the case of white noise with $\text{SNR} = 5$ dB, the results are shown in Fig. 4.11(b) and demonstrate that the performance of all methods for side-lobes suppression is similar to that in the noise-free case in Fig. 4.11(a). However, MB-MFP¹ exhibits a behavior where the initial single peak at x^{L*} in Fig. 4.11(a) splits into two separate peaks in Fig. 4.11(b), which decreases the accuracy of leak identification. Fig. 4.11(c) and Fig. 4.11(d) illustrate that all MB-MFP, MB-MFP¹ and MB-MFP² methods can successfully localize the leak under blue noise with $\text{SNR}_1 = 30$ dB and impulse noise with $\text{SNR}_2 = -60$ dB, while CB-MFP completely deviates from the expected results. In addition, the capability of MB-MFP¹ and MB-MFP² to suppress side-lobes is strong under the impulse noise but it is weaker under the blue noise.

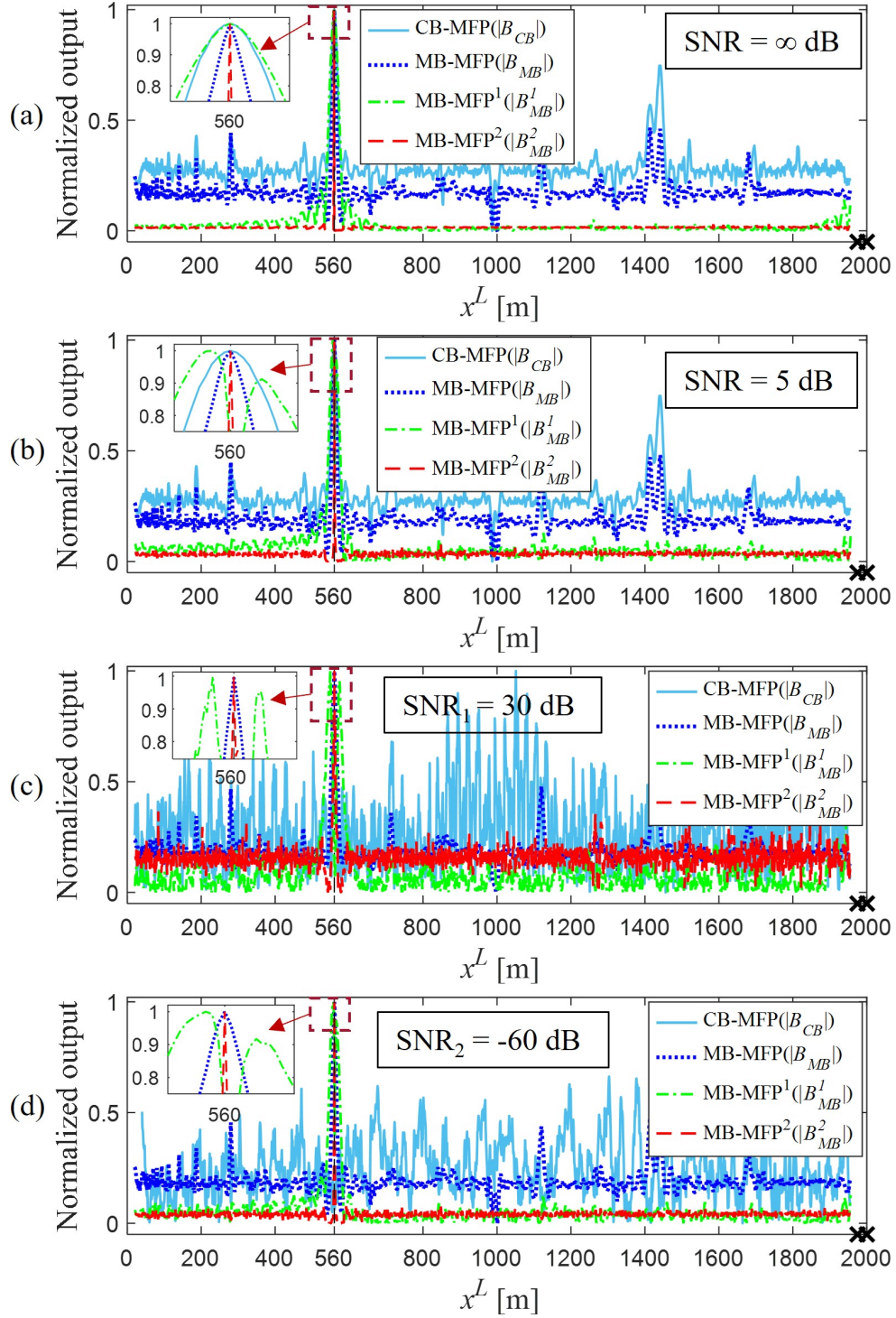


Figure 4.11. Demonstration of using CB-MFP [$|B_{CB}|$, Eq. (4.13)], MB-MFP [$|B_{MB}|$, Eq. (4.29)], MB-MFP¹ [$|B_{MB}^1|$, Eq. (4.30)], and MB-MFP² [$|B_{MB}^2|$, Eq. (4.31)] for leak detection cases with (a) no noise, (b) SNR = 5 dB, (c) SNR₁ = 30 dB, and (d) SNR₂ = -60 dB. The vertical black dashed line represents the real leak location $x^{L*} = 560$ m. The black crosses represent the measurement locations $\mathbf{x}^M = [1975 \text{ m}, 2000 \text{ m}]$.

To systematically evaluate the accuracy for leak localization and the capabilities for side-lobes suppression when using CB-MFP, MB-MFP, MB-MFP¹, and MB-MFP², eighteen cases of white noise with $\text{SNR} = [-8, -8+4, \dots, 12]\text{dB}$, blue noise with $\text{SNR}_1 = [30, 30+5, \dots, 55]\text{dB}$, and impulse noise with $\text{SNR}_2 = [-70, -70+10, \dots, -20]\text{dB}$ are simulated. Each case is repeated twenty times to average the results used for evaluation: (i) the averaged leak localization error, (ii) the ratio of the second highest side-lobe to the main-lobe, and 3) the ratio of the averaged side-lobe to the main-lobe. The findings presented in Fig. 4.12 indicate that overall, MB-MFP² performs the best in terms of accurate leak localization and side-lobes suppression. CB-MFP fails to identify leaks in the presence of blue noise and impulse noise. MB-MFP successfully operates and exhibits strong localization ability in various noise conditions, but it cannot handle the high-level side-lobes. MB-MFP¹ achieves side-lobes suppression in different noise conditions. However, due to its drawback of peak separation near the real leak location in the presence of noise, its localization accuracy is inferior compared to MB-MFP and MB-MFP² methods.

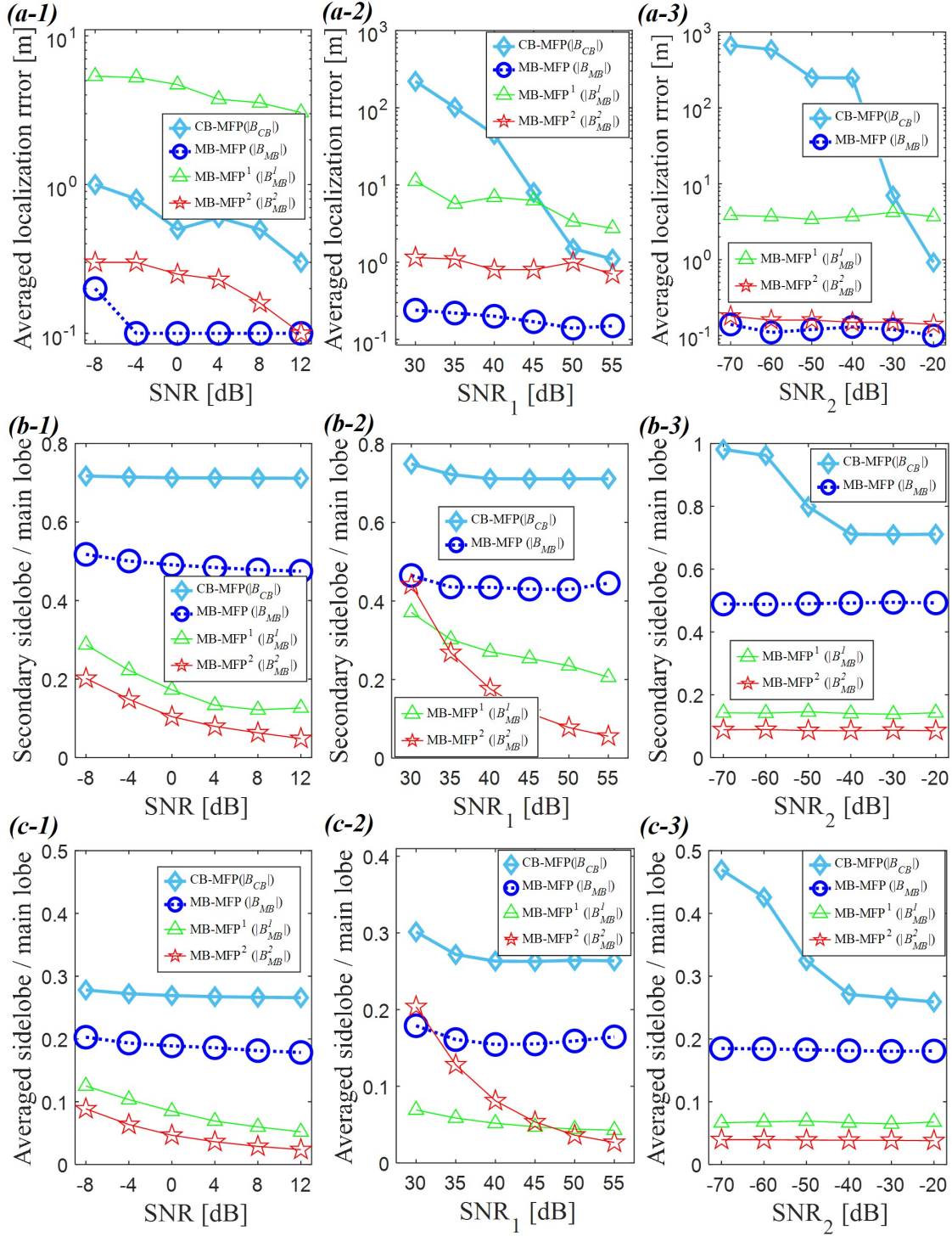


Figure 4.12. Comparisons of the performance of CB-MFP, MB-MFP, MB-MFP¹, and MB-MFP² for leak detection by parameters: (a) averaged localization error, (b) ratio of secondary side-lobe-to-main-lobe, and (c) ratio of averaged side-lobe-to-main-lobe. The noise settings are white noise with SNR = $[-8, -8+4, \dots, 12]$ dB, blue noise with SNR₁ = $[30, 30+5, \dots, 55]$ dB, and impulse noise with SNR₂ = $[-70, -70+10, \dots, -20]$ dB. The real leak location is $x^{L*} = 560$ m. The measurement locations $\mathbf{x}^M = [1975 \text{ m}, 2000 \text{ m}]$.

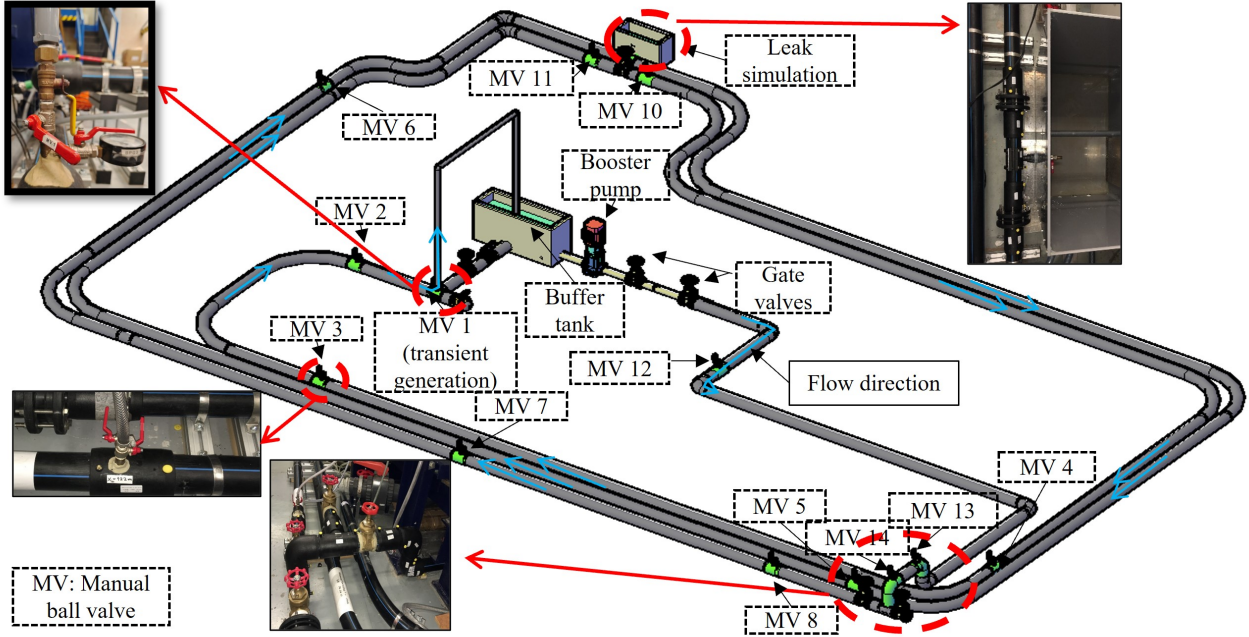
4.6 Experimental Validation

In this section, the facility at the Water Research Resources Laboratory of the Hong Kong University of Science and Technology (HKUST) is used to experimentally validate (i) the convergence of MFP surface side-lobes into the real leak location and (ii) the accuracy of MB-MFP methods for leak localization and their capabilities to suppress side-lobes.

4.6.1 Experimental Setup

The overall 3D schematic and 2D layout of the pipeline system are shown in Fig. 4.13. The upstream and downstream ends are connected to a booster pump and a valve, respectively. A buffer tank is located between the upstream and downstream ends, specifically between the booster pump and the valve, allowing water to flow back into the tank after being pumped through the pipes, enabling it to be recycled. A total of 13 manual ball valves (MV1~8, MV10~14) are installed along the pipe, which can be used to generate transient waves or connected to pressure sensors. For our tests, five of these valves are used (MV1 for generating transient waves, and MV3~4 and MV7~8 for collecting data). The pipe length $L = 142.08$ m. The pipe diameter $D = 0.0792$ m. The pipe wall material is high-density polyethylene (HDPE) with viscoelastic parameters summarized in TABLE 4.6.1 [88], where these parameters are used to calculate the wave speed at different frequencies. The simulated leak location $x^{L*} = 44.09$ m. The transient experiment corresponds to a ratio of steady-state discharge of leak (Q_0^L) and main pipe (Q_0) given as $Q_0^L/Q_0 \approx 36\%$, where $Q_0^L \approx 0.54 \times 10^{-3} \text{ m}^3/\text{s}$ and $Q_0 \approx 1.50 \times 10^{-3} \text{ m}^3/\text{s}$.

(a) 3D schematic:



(b) 2D Layout:

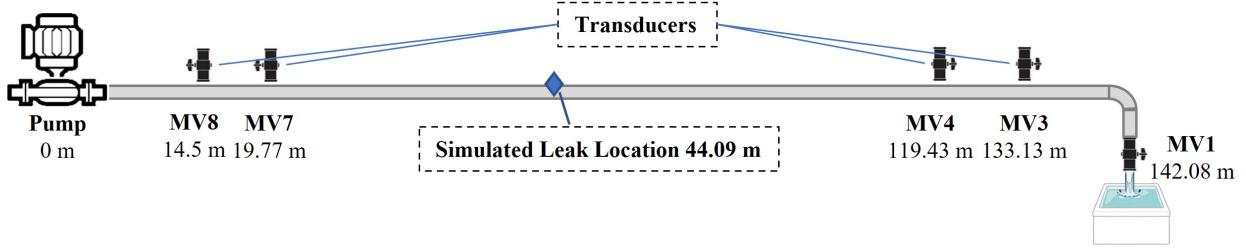


Figure 4.13. The pipeline system in the Water Resources Research Laboratory of HKUST: (a) overall 3D schematic, and (b) simplified 2D layout.

Table 4.1. Coefficients for pipe wall viscosity of the HDPE pipe in the Water Resources Research Laboratory at HKUST

Poisson's ratio $\nu = 0.43$	bulk modulus of water $\kappa = 2.1 \times 10^9$ Pa
density of water $\rho = 10^3$ kg/m ³	Pipe diameter $D = 0.0792$ m
Young's modulus of elasticity of pipe wall $E = 5.4 \times 10^{-3}$ m	
the coefficients of the K-V model: $\tau_1 = 0.038$ s, $\tau_2 = 0.6$ s, $\tau_3 = 1.5$ s	
$C_1 = 1.957 \times 10^{-10}$ Pa ⁻¹ , $C_2 = 1.0954 \times 10^{-10}$ Pa ⁻¹ , $C_3 = 9.05 \times 10^{-12}$ Pa ⁻¹	

4.6.2 Leak Identification

The transient wave is generated by a fast full valve closure at MV1 within 0.05 s. The time-domain pressure signals are measured at MV3~4 and MV7~8 and are further transformed into the frequency-domain signals. The obtained frequency-domain signals at MV7 and MV8 are combined with the transfer matrix to calculate the upstream-end discharge at MV8. The frequency-domain signals at MV3 and MV4 are used to implement broadband MFP methods for leak localization. Tolstoy's work [39, 40] also suggested that the computation of modeled acoustic fields should be at or near the source location, and the space between two sensors equal or larger than $\lambda_{min}/2$. Fig. 4.14 plots the measured time-domain and the transformed frequency-domain signals at MV3 and MV8. Our sampling frequency is 1000 Hz. The frequency interval is $\Delta f = 1/T_m = 0.1$ Hz over the bandwidth $1000/2 = 500$ Hz. T_m is the measurement time. It observes that after 6 Hz, the signals become highly noisy. Therefore, the used frequency range is truncated at 6 Hz.

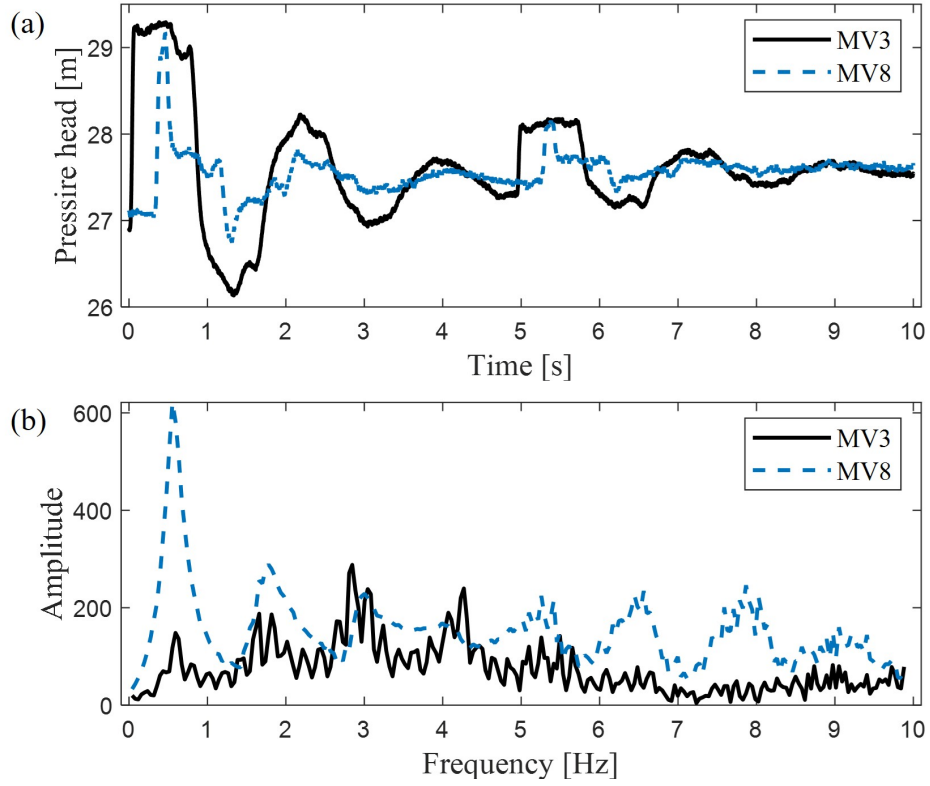


Figure 4.14. (a) Measured time-domain signals and (b) transformed frequency-domain signals at MV3 and MV8

Fig. 4.15(a) shows the output surface of $|B|$ in the defect location-frequency plane, where we observe that the surface side-lobe trajectories converge into the correct leak source range. Fig. 4.15(b) plots the results of using CB-MFP, MB-MFP, and MB-MFP² for leak detection, which indicates that MB-MFP² can significantly reduce the level of side-lobes and have a higher resolution of the main-lobe. However as shown in Fig. 4.15(c), compared to MB-MFP², MB-MFP¹ has not effectively reduced the side-lobe levels and failed to accurately indicate the leak location due to its peak separation behavior.

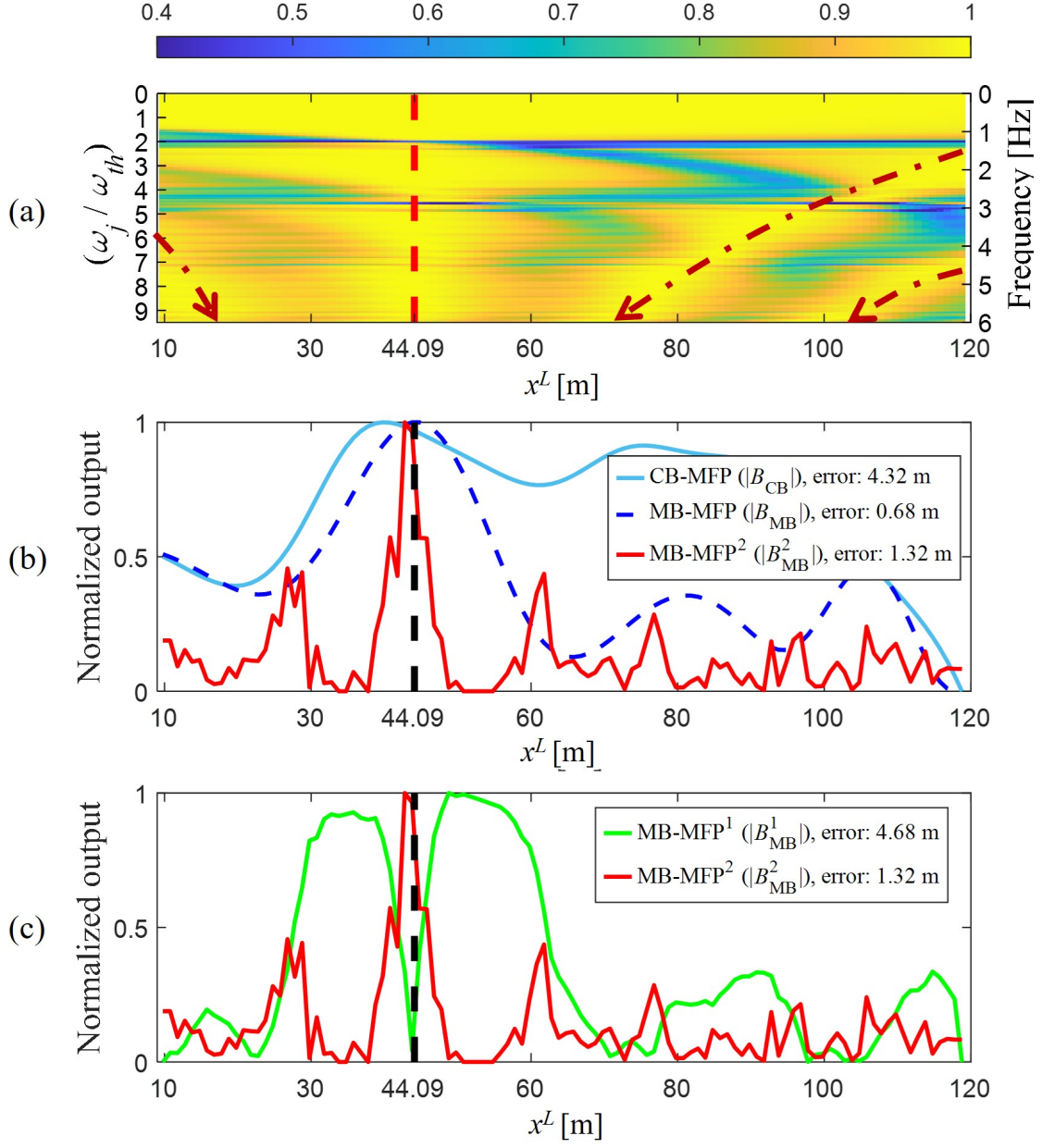


Figure 4.15. (a) Demonstration of $|B|$ on the defect location-frequency plane for lab test. (b-c) Leak localization results by using CB-MFP, MB-MFP, MB-MFP¹, and MB-MFP² methods based on the measured signals at MV3 and MV4. The simulated leak location is $x^{L*} = 44.09$ m and is represented by the black vertical dashed lines.

4.7 Summary

The chapter proposed a novel modified broadband MFP (MB-MFP) technique that increases the robustness for leak localization in the presence of nonwhite noise. Combining the gradient information in the MFP output surface, we further developed two improved modified broadband MFP methods (MB-MFP¹ and MB-MFP²) with lower side-lobe levels and higher localization resolution. The proposed methods addressed three key issues of the used conventional broadband MFP (CB-MFP) method for leak detection: (i) the unclear generation mechanism of side-lobes, (ii) the ambiguity caused by high-level side-lobes for accurate leak localization, and (iii) the poor performance under nonwhite noise. The accuracy and robustness of the proposed MB-MFP methods in localizing leaks while suppressing side-lobes and noise effects is validated experimentally.

CHAPTER 5

MULTI-DEFECT DETECTION BY MATCHED-FIELD PROCESSING

The detection of a cluster of leaks in pressurized water supply pipelines is crucial as it allows for early recognition of the pipeline's poor overall condition, enabling timely interventions. This chapter addresses this problem by developing broadband matched-field processing techniques to identify cluster of leaks in pressurized water mains. The developed methods combine signal processing algorithms with acoustic measurements to achieve super-resolution multi-leak detection, where leaks that are less than half the minimum probing wavelength apart are identified. The effectiveness of the proposed methods is demonstrated through numerical simulations and experimental tests. We also show that the proposed techniques are robust to various noise types, such as white noise with a signal-to-noise ratio (SNR) as low as -8 dB, colored noise and impulse noise.

5.1 Introduction

Identification and localization of leaks in urban water supply systems can prevent water loss, reduce repair costs, and mitigate environmental impacts. Several commercially available technologies exist to detect leaks. Examples include utilizing methods such as acoustic monitoring [89], smart ball [90], Sahara system [91], and LeakfinderRT [92]. The U.S. Environmental Protection Agency

(EPA) conducted a comprehensive assessment of existing leak detection technologies and concluded “*None of the technologies was able to discern two separate leaks in close proximity . . . and appear only to report the larger leak rate for the cluster of leaks. The ability to identify whether a signal is from an isolated leak or multiple leaks in close proximity is helpful in judging the general condition of a pipeline*” [93]. In addition, a pipeline with clusters of defects has typically lower load-bearing capacity than the same pipeline with individual defects [94, 95]. Moreover, [96] notes that the interacting defects are more detrimental towards the structural integrity of a pipeline than single defects. [8] also stated that if small defects appear and are left untreated, they can progressively spread over time, link together, and lead to a severe pipe failure. Therefore, there is a critical need to develop methods that can identify defects that are in close proximity.

In recent years, transient wave-based defect detection methods (TBDDMs) have gained significant attention. These methods utilize wave physics and inverse mathematics, where an intentionally generated transient wave propagates through the pipeline and reflects off defects [23, 35, 97]. By recording and analyzing the transient pressure history within the system, it is possible to localize defects in the pipe system [36, 38]. TBDDMs have proven effective in identifying and characterizing various types of pipeline defects, including leaks, blockages, and structural weaknesses [43, 55–58, 65, 98–101].

The authors have proposed transient-based modified broadband matched-field processing (MB-MFP) techniques for single-leak detection [81]. We demonstrated how the spatial positioning of side-lobes in single-frequency matched-field processing (MFP) output function varies with candidate leak location and probing wave frequency. As frequency increases, the side-lobe paths converge toward the actual leak location. Utilizing this property, we proposed a suite of MB-MFP

strategies that achieved high-resolution leak identification even in low SNR environments with various types of noise while suppressing undesirable side-lobes. In this chapter, we seek to extend these methods to cluster of leaks where the leaks are closely spaced. In the context of wave propagation, “closely spaced” refers to distances between consecutive leaks that are smaller than the diffraction limit, i.e., half the wavelength of the probing wave [60, 84].

The structure of this chapter is as follows: Section 2 presents a frequency-domain transient wave model for pipe systems. It describes how the head difference, caused by leaks during bidirectional transient wave propagation, is extracted and used for leaks detection. Section 3 details the MB-MFP techniques adapted for a multi-leak context, presenting numerical simulations that demonstrate the accuracy and super-resolution of the techniques under different noise conditions and under cases of closely spaced leaks. Section 4 explores the sensitivity of the proposed methods with respect to the number of leaks (N_L) assumed in the model versus the number of real leaks (N_{L*}), showing that the N_L estimated leak locations tend to cluster accurately around the N_{L*} real leak locations when $N_L \geq N_{L*}$. Section 5 validates the MB-MFP techniques through laboratory experiments simulating dual leaks with differing sizes. Finally, the conclusions are drawn in Section 6.

5.2 Model Description

Consider a pipeline system with a cluster of N_{L*} leaks. The measured head response at the measurement point x^M and the frequency ω to an actively generated wave is given as follows:

$$h_{\omega}^{M*}(\boldsymbol{\theta}^{IP*}, \boldsymbol{\theta}_1^{L*}, \dots, \boldsymbol{\theta}_{N_{L*}}^{L*}) = h_{\omega}^M(\boldsymbol{\theta}^{IP*}) + \sum_{n_L=1}^{N_{L*}} h_{\omega}^M(\boldsymbol{\theta}_{n_L}^{L*}) + n_{\omega}^M \quad (5.1)$$

wherein

$$\sum_{n_L=1}^{N_{L^*}} h_{\omega}^M \left(\boldsymbol{\theta}_{n_L}^{L^*} \right) = \underbrace{\sum_{n_L=1}^{N_{L^*}} s_{n_L}^{L^*} G_{\omega}^M \left(x_{n_L}^{L^*} \right)}_{\text{the scatterings from each leak}} \quad (5.2)$$

The derivation to Eqs. 5.1 and 5.2 can be referred to Chapter 1 of Introduction . $h_{\omega}^{M^*}$ is constituted of three distinct components: (i) the baseline head $h_{\omega}^M \left(\boldsymbol{\theta}^{IP^*} \right)$ where superscript IP^* denoting the effect to measured pressure head from the intact pipe, (ii) the scattering $\left(\boldsymbol{\theta}_{n_L}^{L^*} \right)$ arising from n_L -th leak and expressed by multiplying the Green's function G_{ω}^M regarding to the leak location $x_{n_L}^{L^*}$ with the leak size $s_{n_L}^{L^*}$, where the superscript L^* denotes the real leaks, and (iii) the noise n_{ω}^M . In fact, there originally existed a fourth component in Eq. 5.1, which involves the pressure heads resulting from the coupling interaction between multiple leaks. These coupling effects are neglected since they are relatively insignificant compared to the scattering originating from each leak, under the assumption of small leaks (i.e., $s_{n_L}^{L^*} \ll 1 \text{ m}^2, n_L = 1, \dots, N_{L^*}$) which are a subject of interest for transient-based methods [43]. For example, in a two-leak case (as depicted in Fig. 5.1), the transient wave propagation includes three routes: the baseline head $h_{\omega}^M \left(\boldsymbol{\theta}^{IP^*} \right)$, the uncorrelated scatterings with distinct independent sources: $s_1^{L^*} G_{\omega}^M \left(x_1^{L^*} \right)$ and $s_2^{L^*} G_{\omega}^M \left(x_2^{L^*} \right)$, and the negligible coupling effect. In Eq. 5.1, $h_{\omega}^M \left(\boldsymbol{\theta}^{IP^*} \right)$ is expressed as

$$h_{\omega}^M \left(\boldsymbol{\theta}^{IP^*} \right) = -Z \sinh \left(\mu x^M \right) q_{\omega}^U + \cosh \left(\mu x^M \right) h_{\omega}^U \quad (5.3)$$

and G_{ω}^M is given by

$$G_{\omega}^M \left(x_{n_L}^{L^*} \right) = -\frac{\sqrt{g} Z \sinh \left(\mu \left(x^M - x_{n_L}^{L^*} \right) \right)}{\sqrt{2H_0^L}} \begin{pmatrix} Z \sinh \left(\mu x_{n_L}^{L^*} \right) q_{\omega}^U \\ -\cosh \left(\mu x_{n_L}^{L^*} \right) h_{\omega}^U \end{pmatrix} \quad (5.4)$$

Eqs. 5.2-5.4 indicate the relation between $h_\omega^M(\boldsymbol{\theta}^{IP*})$ or $h_\omega^M(\boldsymbol{\theta}_{n_L}^{L*})$ and the boundary conditions q^U and h^U ; $Z = (\mu a^2)/(i\omega g A)$ is the characteristic impedance of pipe, which is determined by the wave speed a , frequency ω , gravitational acceleration g , cross-sectional area A , and propagation function μ defined as $\mu = a^{-1}(-\omega^2 + igA\omega R)^{1/2}$; $R = (f_D Q_0)/(gDA^2)$ is the linearized steady-state resistance term; with Q_0 being the steady-state discharge at the pipe upstream end and f_D as the Darcy-Weisbach friction factor.

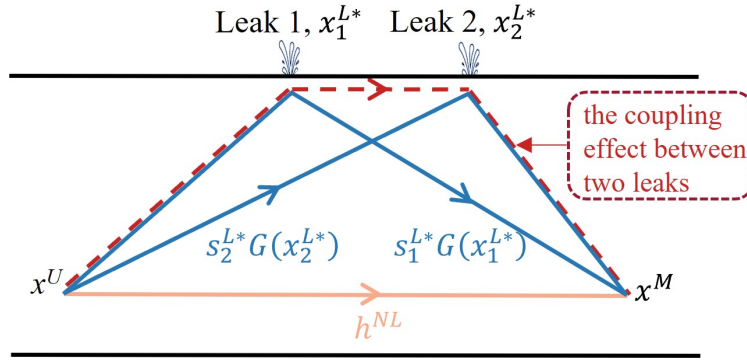


Figure 5.1. Transient wave propagation in a two-leak case.

By taking the difference between h_ω^{M*} and $h_\omega^M(\boldsymbol{\theta}^{IP*})$ in Eq. 5.1, the leak-induced pressure head Δh_ω^M at ω and x^M can be defined and expressed as

$$\Delta h_\omega^M(\boldsymbol{\theta}_1^{L*}, \dots, \boldsymbol{\theta}_{N_L}^{L*}) = h_\omega^{M*}(\boldsymbol{\theta}^{IP*}, \boldsymbol{\theta}_1^{L*}, \dots, \boldsymbol{\theta}_{N_L}^{L*}) - h_\omega^M(\boldsymbol{\theta}^{IP*}) = \sum_{n_L=1}^{N_L} h_\omega^M(\boldsymbol{\theta}_{n_L}^{L*}) + n_\omega^M \quad (5.5)$$

Substituting Eq. 5.2 into Eq. 5.5 yields

$$\Delta h_\omega^M(\boldsymbol{\theta}_1^{L*}, \dots, \boldsymbol{\theta}_{N_L}^{L*}) = \sum_{n_L=1}^{N_L} s_{n_L}^{L*} G_\omega^M(x_{n_L}^{L*}) + n_\omega^M \quad (5.6)$$

which is a linear combination of individual leak contributions. Then, h^L across all N_f frequencies

and N_m measurement locations can be written as:

$$\begin{aligned}\Delta \mathbf{h}_{\omega}^{\mathbf{M}}(\boldsymbol{\theta}_1^{L*}, \dots, \boldsymbol{\theta}_{N_L}^{L*}) &= \mathbf{h}_{\omega}^{\mathbf{M}*}(\boldsymbol{\theta}^{IP*}, \boldsymbol{\theta}_1^{L*}, \dots, \boldsymbol{\theta}_{N_L}^{L*}) - \mathbf{h}_{\omega}^{\mathbf{M}}(\boldsymbol{\theta}^{IP*}) \\ &= \sum_{n_L=1}^{N_{L*}} \mathbf{h}_{\omega}^{\mathbf{M}}(\boldsymbol{\theta}_{n_L}^{L*}) + \mathbf{n}_{\omega}^{\mathbf{M}} = \sum_{n_L=1}^{N_{L*}} s_{n_L}^{L*} \mathbf{G}_{\omega}^{\mathbf{M}}(x_{n_L}^{L*}) + \mathbf{n}_{\omega}^{\mathbf{M}}\end{aligned}\quad (5.7)$$

wherein $\Delta \mathbf{h}_{\omega}^{\mathbf{M}}$, $\mathbf{h}_{\omega}^{\mathbf{M}*}$, $\mathbf{h}_{\omega}^{\mathbf{M}}$, $\mathbf{G}_{\omega}^{\mathbf{M}}$, and $\mathbf{n}_{\omega}^{\mathbf{M}}$ are all $N_m \cdot N_f$ dimensional complex vectors and their forms can be illustrated by considering an example:

$$\mathbf{G}_{\omega}^{\mathbf{M}}(x_{n_L}^{L*}) = \left[\underbrace{G_{\omega_1}^{M_1}(x_{n_L}^{L*}), \dots, G_{\omega_{N_f}}^{M_1}(x_{n_L}^{L*})}_{N_f}, \dots, \underbrace{G_{\omega_1}^{M_{N_m}}, \dots, G_{\omega_{N_f}}^{M_{N_m}}}_{N_f} \right]^{\top} \quad (5.8)$$

The purpose of obtaining the measured head difference vector $\Delta \mathbf{h}_{\omega}^{\mathbf{M}}$ is to facilitate leak localization. Note that the first equal sign in Eq. 5.7 signifies that $\Delta \mathbf{h}_{\omega}^{\mathbf{M}}$ can be determined from the pressure measurement $\mathbf{h}_{\omega}^{\mathbf{M}*}$, and the second equal sign represents the theoretical model of $\Delta \mathbf{h}_{\omega}^{\mathbf{M}}$, incorporating leak parameters $(\boldsymbol{\theta}_1^{L*}, \dots, \boldsymbol{\theta}_{N_L}^{L*})$.

This chapter examines a reservoir-pipe-valve (RPV) system featuring multiple leaks, as illustrated in Fig. 5.2. The system consists of a single pipe, positioned horizontally and extending a length L , which connects two reservoirs at each end. These reservoirs are characterized by pressure heads H_1 and H_2 . A valve is positioned near the downstream end and can be used to generate transient waves by rapid “close-open” maneuvers. An example of such a maneuver is shown in Fig. 5.3, where a discharge perturbation with a pulse shape is introduced. The resulting transient wave propagates back and forth within the pipe at a wave speed of a_0 . The system’s fundamental period and frequency are given by $T_{th} = 4L/a_0$ and $f_{th} = 1/T_{th}$ (or $\omega_{th} = 2\pi/T_{th}$). The utilized angular frequency range is denoted as $\boldsymbol{\omega} = [\omega_1, \dots, \omega_{N_f}]^{\top}$ and the corresponding frequency range is

$\mathbf{f} = [f_1, \dots, f_{N_f}]^\top$, where N_f represents the number of excited frequencies and superscript $\top$ denotes vector transpose. Our study considers N_L leaks with sizes $\mathbf{s}^L = [s_1^L, \dots, s_{N_L}^L]^\top$ and locations $\mathbf{x}^L = [x_1^L, \dots, x_{N_L}^L]^\top$. The steady-state discharge through the n_L -th ($n_L \in [1, \dots, N_L]$) leak is determined using the orifice equation: $Q_{0,n_L}^L = s_{n_L}^L \sqrt{2gH_{0,n_L}^L}$, where H_{0,n_L}^L represents the steady-state head at the leak location, calculated as $H_{0,n_L}^L = H_1 - (H_1 - H_2)(x_{n_L}^L/L)$. The number and distribution of measurement locations are denoted as N_m and $\mathbf{x}^M = [x_1^M, \dots, x_{N_m}^M]^\top$, respectively. Previous experience [81] showed that $N_m = 2$ is sufficient to provide adequate information, although additional measurement locations can yield more valuable insights. The distance between two measurement locations is intentionally chosen to be larger than half of the minimum probing wavelength (i.e., $0.5\lambda_{min} = 0.5a/f_{N_f}$), which represents the spatial resolution limit of the propagating waves due to wave diffraction effect [60, 84].

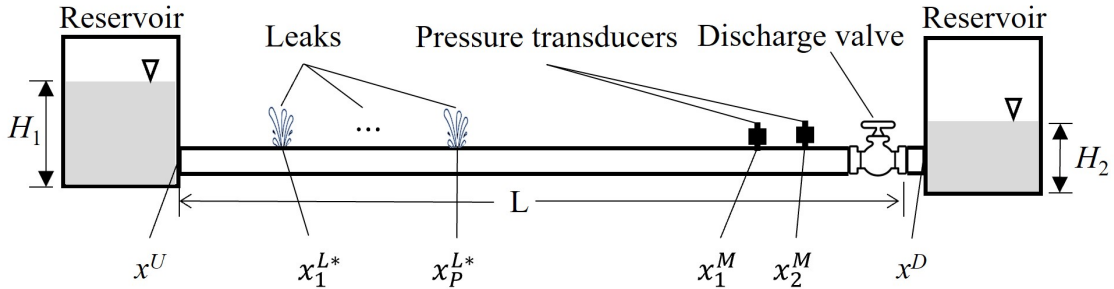


Figure 5.2. Schematic of the RPV model with multiple leaks and measurement locations.

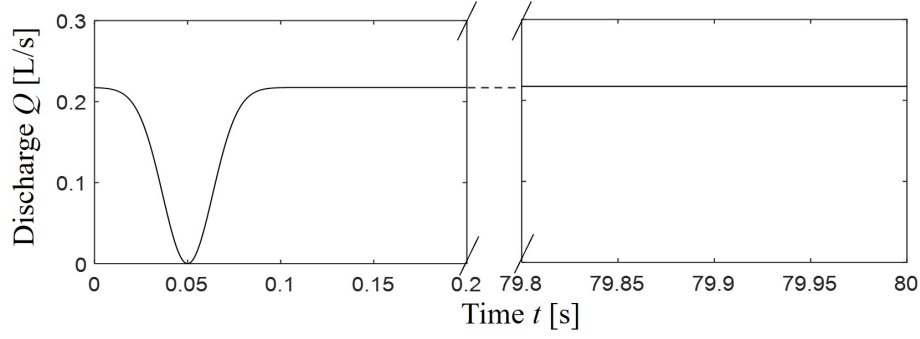


Figure 5.3. Downstream-end discharge undergoes pulse-shaped changes due to valve operations.

To incorporate noise into numerical simulations, the signal-to-noise ratio (SNR) in decibels is defined as

$$\text{SNR} = 20 \log_{10} \left(\left| \mathbb{E} \left(\sum_{n_L=1}^{N_L} s_{n_L}^L \mathbf{G}(\boldsymbol{\omega}, \mathbf{x}^{\mathbf{M}}, x_{n_L}^L) \right) \right| / \sigma \right) \quad (5.9)$$

where $|\mathbb{E}|$ denotes the mean head difference and σ represents the standard deviation of Gaussian noise with zero mean. In pipeline systems, noise can be categorized as white noise, colored noise, or impulse-type noise [44]. For white noise, SNR simply remains a constant across the frequency range $\boldsymbol{\omega}$. For blue noise, SNR decreases by 3 dB per increasing frequency from SNR_1 at ω_1 . Regarding impulse noise, SNR is constant over $\boldsymbol{\omega}$, except for specific frequencies (e.g., $\omega = z \cdot 0.2\omega_{th}$, $z \in \mathbb{Z}^+$), at which the SNR abruptly drops to a much lower value, denoted as SNR_2 .

5.3 Multi-Leak Detection by MB-MFP Techniques

This section extends the scope of using MB-MFP techniques beyond single-leak detection to address the challenge of detecting multiple leaks. This extension aims to achieve super-resolution

localization, even in scenarios with low SNRs (up to -8 dB) where the leaks are in close proximity (i.e., distances less than $0.5\lambda_{\min}$).

5.3.1 MB-MFP Techniques

The concept of MFP with respect to frequency and candidate leak location is first introduced, with a focus on its definition and the underlying mathematical framework. The goal is to provide a better understanding of how the subsequent MB-MFP techniques successfully localize leaks. Similar to the definition given in 4.18, the single-frequency MFP output function is defined as:

$$|B(\omega, \mathbf{x}^L)| = \left| \left\langle \frac{\Delta \mathbf{h}_\omega^M}{\|\Delta \mathbf{h}_\omega^M\|}, \frac{\sum \mathbf{G}_\omega^M}{\|\sum \mathbf{G}_\omega^M\|} \right\rangle \right| = \frac{\sqrt{(\sum \mathbf{G}_\omega^M)^H \Delta \mathbf{h}_\omega^M (\Delta \mathbf{h}_\omega^M)^H \sum \mathbf{G}_\omega^M}}{\sqrt{(\Delta \mathbf{h}_\omega^M)^H \Delta \mathbf{h}_\omega^M} \sqrt{(\sum \mathbf{G}_\omega^M)^H \sum \mathbf{G}_\omega^M}} \quad (5.10)$$

where the measured data vector $\Delta \mathbf{h}_\omega^M$ and the modeled data vector $\sum \mathbf{G}_\omega^M$ are both N_m -dimensional and include information from N_m given measurement locations at n_f -th frequency ω ($n_f \in [1, \dots, N_f]$).

The expressions of $\Delta \mathbf{h}_\omega^M$ and $\sum \mathbf{G}_\omega^M$ can be obtained from Eqs. (5) and (6) as:

$$\Delta \mathbf{h}_\omega^M = \Delta \mathbf{h}_\omega^M(\omega, \mathbf{x}^M, \mathbf{x}^{L*}) = \sum_{n_L=1}^{N_{L*}} s_{n_L}^{L*} \mathbf{G}(\omega, \mathbf{x}^M, x_{n_L}^{L*}) + \mathbf{n} \quad (5.11)$$

and

$$\sum \mathbf{G}_\omega^M = \sum_{n_L=1}^{N_L} \mathbf{G}_\omega^M(x_{n_L}^L) \quad (5.12)$$

where N_L is the unknown and assumed number of candidate leak locations. That is, in Eq. 5.10, a N_L -leak model is used to detect the N_{L*} -leak case. To simplify the problem, this section considers only the case of $N_L = N_{L*}$. The situation where $N_L \neq N_{L*}$ is discussed in the next section.

When $N_L = N_{L*}$ and noise is not considered, $|B|$ reaches the maximum value of one at positions $\mathbf{x}^L = \mathbf{x}^{L*}$ at each single frequency [81]. Consistently, after integrating $|B|$ over all frequencies, the resulting modified broadband MFP (MB-MFP) method exhibits maxima at $\mathbf{x}^L = \mathbf{x}^{L*}$. To suppress the appeared and undesired side-lobes shown in the MB-MFP result and thus enhance the spatial resolution of leak localization, we introduce two optimized broadband MFP methods—MB-MFP¹ and MB-MFP²—leveraging gradient information from the output surface of $|B|$ on the defect location-frequency plane. To be specific, MB-MFP¹ employs a side-lobe canceling algorithm that assigns positive values to the left-edge points and negative values to the right-edge points of the maxima of $|B|$. Conversely, MB-MFP² uses a peak-extraction algorithm, assigning zero values to points adjacent to the maxima of $|B|$ [81]. Then, Integrating the processed $|B|$ over frequencies allows MB-MFP¹ and MB-MFP² to achieve high-resolution leak localization with minimized side-lobes. The robustness of these methods is augmented by the normalization of $\Delta \mathbf{h}_\omega^M$ in $|B|$ [Eq. 5.10], effectively compressing the impact of high amplitude noise $\mathbf{n}$. This makes the proposed broadband MFP methods effective against various noise types, including white, colored (e.g., blue noise), and impulse noise.

The functional expressions for the three broadband MFP methods based on N_L -leak model, i.e., MB-MFP(N_L) [$|B_{MB}(N_L)|$], MB-MFP¹(N_L) [$|B_{MB}^1(N_L)|$] and MB-MFP²(N_L) [$|B_{MB}^2(N_L)|$], are respectively provided in integral form as:

$$\left| B_{MB}(\mathbf{x}^L) \right| = \frac{1}{\omega_{\max} - \omega_{\min}} \left| \int_{\omega_{\min}}^{\omega_{\max}} B(\omega, \mathbf{x}^L) d\omega \right| \quad (5.13)$$

$$\left| B_{MB}^1(\mathbf{x}^L) \right| = \frac{1}{\omega_{\max} - \omega_{\min}} \left| \int_{\omega_{\min}}^{\omega_{\max}} \left(\sum_{n_L=1}^{N_L} \text{sign} \left(\frac{\partial |B(\omega, x_{n_L}^L)|}{\partial x_{n_L}^L} \right) \right) \cdot |B(\omega, \mathbf{x}^L)| d\omega \right| \quad (5.14)$$

and

$$\left| B_{MB}^2(\mathbf{x}^L) \right| = \frac{\int_{\omega_{min}}^{\omega_{max}} \left(\sum_{n_L=1}^{N_L} \begin{pmatrix} \text{sign} \left(\frac{\partial |B(\omega, \mathbf{x}_{n_L}^L)|}{\partial x_{n_L}^L} \right) \\ -\text{sign} \left(\frac{\partial |B(\omega, \mathbf{x}_{n_L}^L + \Delta \mathbf{x}_{n_L}^L)|}{\partial x_{n_L}^L} \right) \end{pmatrix} \right) \cdot |B(\omega, \mathbf{x}^L)| d\omega}{\omega_{max} - \omega_{min}} \quad (5.15)$$

5.3.2 Numerical Simulations

In this sub-section, we demonstrate the anti-noise capabilities and side-lobe suppression performance of the MB-MFP techniques for multi-leak detection through comprehensive numerical simulations. These simulations cover various noise types and levels, and leak intervals. The parameters of the RPV system are listed in the following Table 5.3.2.

Table 5.1. The pipeline parameters used in the simulations

$L = 200 \text{ m}$	$D = 0.08 \text{ m}$	$H_1 = 25 \text{ m}$
$H_2 = 20 \text{ m}$	$a = 1000 \text{ m/s}$	$f = 0.02$
$N_{L^*} = 2$	$\mathbf{x}^{L^*} = [40 \text{ m}, 90 \text{ m}] \text{ or } [40 \text{ m}, 44 \text{ m}]$	$\mathbf{s}^{L^*} = [1.4 \times 10^{-5} \text{ m}^2, 1.4 \times 10^{-5} \text{ m}^2]$
$N_m = 2$	$[x_1^M, x_2^M]^\top = [180 \text{ m}, 200 \text{ m}]$	
$N_f = 1000$	$\boldsymbol{\omega} = [\omega_{th}, \omega_{th} + 0.02\omega_{th}, \dots, 20\omega_{th}]^\top$	or $\mathbf{f} = [f_{th}, f_{th} + 0.02f_{th}, \dots, 20f_{th}]^\top$

The system's fundamental period and frequency are calculated as $T_{th} = 4L/a = 0.8 \text{ s}$ and $f_{th} = 1/T_{th} = 1.25 \text{ Hz}$ (or $\omega_{th} = 2\pi/T_{th} = 7.85 \text{ rad/s}$) [31]. Thus, the shortest (minimum) probing wavelength $\lambda_{min} = a/f_{N_f} = 40 \text{ m}$.

Noise-Free Case

Fig. 5.4 illustrates the results of using $\text{MB-MFP}(N_L)$, $\text{MB-MFP}^1(N_L)$, and $\text{MB-MFP}^2(N_L)$ for the noise-free case, where $N_L = N_{L*} = 2$. Each method successfully identifies two leaks even when the distance between them is below $0.5\lambda_{\min}$, demonstrating super-resolving capabilities beyond the typical spatial resolution limit of propagating waves. Specifically, $\text{MB-MFP}^2(2)$ shows superior main-lobe resolution and side-lobe suppression compared to other techniques.

Various Noise Scenarios

This sub-section presents numerical examples under various noise scenarios: white noise with $\text{SNR} = [-8, -8+4, \dots, 12]$ dB, blue noise with $\text{SNR}_1 = [28, 28+4, \dots, 48]$ dB, and impulse noise with $\text{SNR}_2 = [-60, -60+10, \dots, -10]$ dB. Each scenario is repeated twenty times to compute the average leak localization error, defined as the L_2 -error between estimated and real leak locations:

$$e(\tilde{\mathbf{x}}^L) = \|\tilde{\mathbf{x}}^L - \mathbf{x}^{L*}\|_2 = \sqrt{(\tilde{x}_1^L - x_1^{L*})^2 + (\tilde{x}_2^L - x_2^{L*})^2} \quad (5.16)$$

where $\tilde{\mathbf{x}}^L = [\tilde{x}_1^L, \tilde{x}_2^L]^\top$ is the vector of estimated leak locations.

Fig. 5.5(a) displays the localization errors for the two leaks separated by $1.25 \lambda_{\min}$ across various SNR, SNR_1 and SNR_2 levels. $\text{MB-MFP}(2)$ and $\text{MB-MFP}^2(2)$ consistently demonstrate higher accuracy with localization errors of approximately 1 m, while $\text{MB-MFP}^1(2)$ has errors ranging from 1 to 10 m. Fig. 5.5(b) shows results for two closely placed leaks, only $0.2\lambda_{\min}$ apart. All three methods still successfully identify the leaks, with $\text{MB-MFP}(2)$ and $\text{MB-MFP}^2(2)$ maintaining higher

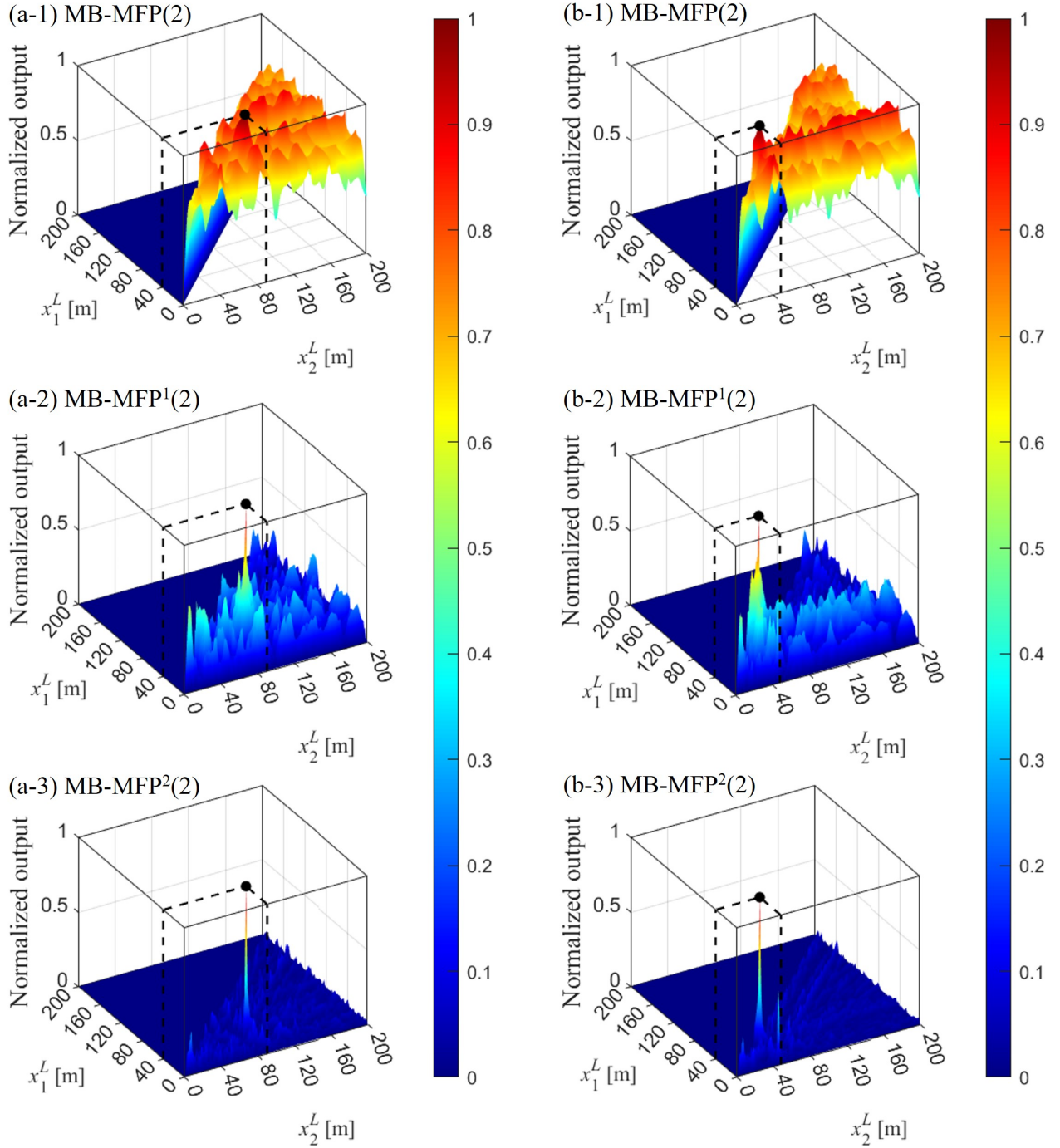


Figure 5.4. Demonstrations of using MB-MFP(2) [Eq. 5.13], MB-MFP¹(2) [Eq. 5.14], and MB-MFP²(2) [Eq. 5.15] for two-leaks detection cases: (a) $\mathbf{x}^{\mathbf{L}*} = [40\text{m}, 90\text{m}]$, and the space between two leaks is $|x_1^{\mathbf{L}*} - x_2^{\mathbf{L}*}| = 50\text{m} = 1.25\lambda_{\min}$ (b) $\mathbf{x}^{\mathbf{L}*} = [40\text{m}, 44\text{m}]$, and the space between two leaks is $|x_1^{\mathbf{L}*} - x_2^{\mathbf{L}*}| = 4\text{m} = 0.2\lambda_{\min}$. In all plots, the black dot represents the real leak locations.

accuracy. The comparison of Fig. 5.5(a) and Fig. 5.5(b) reveals that closer leak proximities result in reduced accuracy for the same method, highlighting the sensitivity of localization performance to the distance between leaks. This indicates that, while these MB-MFP methods are highly effective, the physical proximity of leaks can still impact the precision of the localization results.

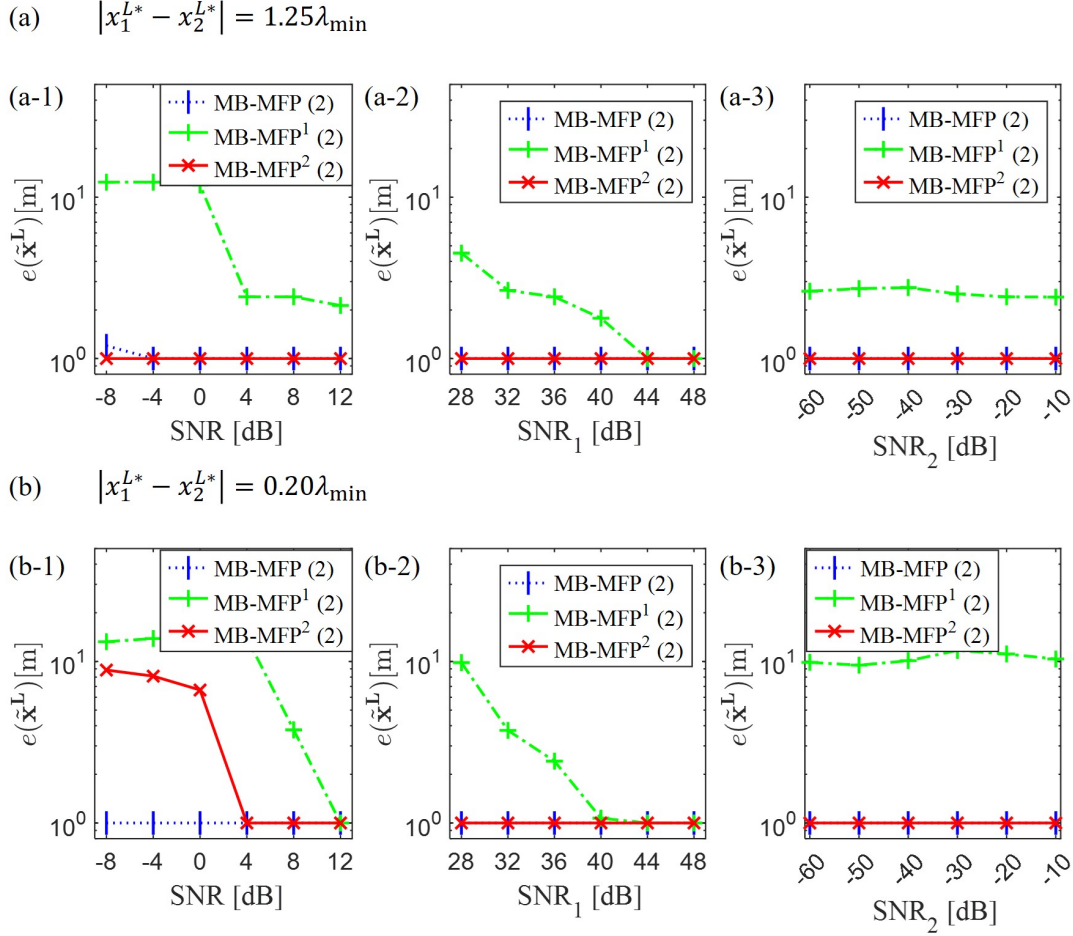


Figure 5.5. Leak localization errors of using MB-MFP(2), MB-MFP¹(2), and MB-MFP²(2) for two-leaks detection cases with (a) $\mathbf{x}^{L*} = [x_1^{L*}, x_2^{L*}] = [40\text{m}, 90\text{m}]$, $|x_1^{L*} - x_2^{L*}| = 50\text{m} = 1.25\lambda_{\min} > 0.5\lambda_{\min}$ and (b) $\mathbf{x}^{L*} = [x_1^{L*}, x_2^{L*}] = [40\text{m}, 44\text{m}]$, $|x_1^{L*} - x_2^{L*}| = 4\text{m} = 0.2\lambda_{\min} < 0.5\lambda_{\min}$ under different noise scenarios.

5.4 A Methodology to Determine the Unknown Leak Number

This section investigates a methodology to determine the unknown number of leaks, building upon the earlier discussion where the assumption $N_L = N_{L*}$ was explored. In real-world applications, accurately knowing the number of leaks (N_{L*}) in advance is often impractical, which prompts the examination of scenarios where $N_L \neq N_{L*}$. Our analysis spans scenarios with $N_L < N_{L*}$, $N_L = N_{L*}$, and $N_L > N_{L*}$. It is found that the N_L estimated leak locations tend to cluster around the N_{L*} actual leak locations as N_L reaches and eventually surpasses N_{L*} . This clustering effect provides a systematic approach to both estimating the number of leaks and achieving precise leak localization.

Fig. 5.6 illustrates leak detection results for $N_{L*} = 1$, $N_{L*} = 2$, and $N_{L*} = 3$ using MB-MFP(N_L), MB-MFP¹(N_L), and MB-MFP²(N_L) with $N_L = 1, 2, 3$, and 4. When $N_L < N_{L*}$, the methods yield inconsistent results. For instance, Fig. 5.6(b-1) shows that MB-MFP(1) successfully locates one leak when $N_{L*} = 2$, while MB-MFP¹(1) and MB-MFP²(1) both incorrectly point out leak locations. Similar inconsistencies are evident in Fig. 5.6(c-1 and c-2). Conversely, when N_L matches N_{L*} , as shown in Fig. 5.6(a-1), (b-2), and (c-3), all three methods accurately pinpoint the N_{L*} simulated leaks. Moreover, when N_L exceeds N_{L*} , as depicted in Fig. 5.6(a-2, a-3, a-4), (b-3, b-4), and (c-4), the estimated N_L leak locations from MB-MFP(N_L), MB-MFP¹(N_L) or MB-MFP²(N_L) converge near the N_{L*} real leak locations. .

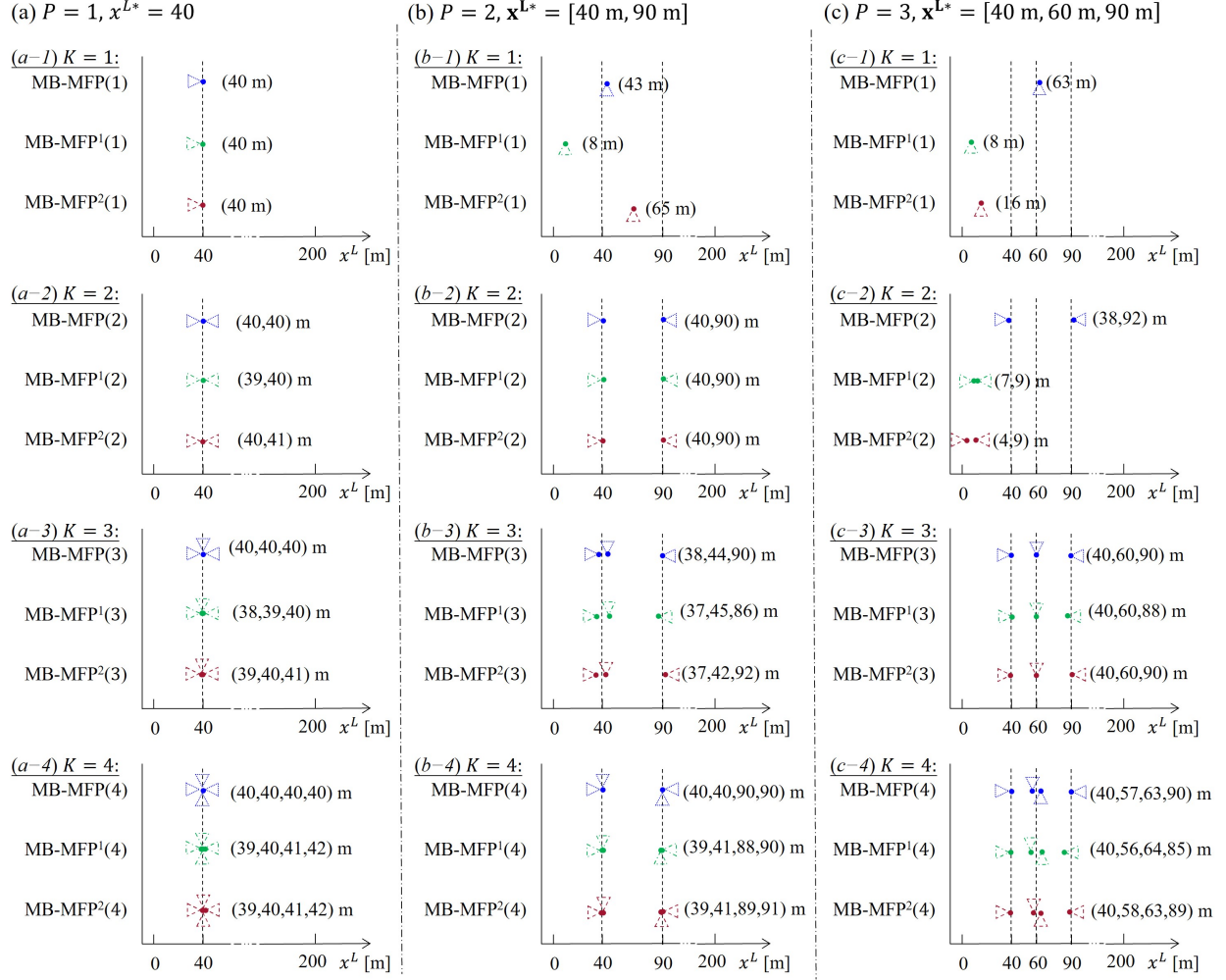


Figure 5.6. Three leak-detection cases corresponding to (a) $N_{L*} = 1, \mathbf{x}^{L*} = 40 \text{ m}$, (b) $N_{L*} = 2, \mathbf{x}^{L*} = [40 \text{ m}, 90 \text{ m}]$, and (c) $N_{L*} = 3, \mathbf{x}^{L*} = [40 \text{ m}, 60 \text{ m}, 90 \text{ m}]$ are tested, by using MB-MFP(N_L), MB-MFP¹(N_L) and MB-MFP²(N_L) with $N_L = 1, 2, 3$ and 4. The vertical dashed black lines and the triangles represent the real and estimated leak locations, respectively.

In fact, the observed clustering phenomena are consistent with the results discussed in Wang and Ghidaoui's work [57]. Following their methodology [57], the number of leakages can then be

determined by calculating the log-likelihood function:

$$\log L \left(N_L, \tilde{\mathbf{x}}^L, \tilde{\mathbf{s}}^L \mid \Delta \mathbf{h}_{\mathbf{w}}^M \right) = - \left\| \Delta \mathbf{h}_{\mathbf{w}}^M - \sum_{n_L=1}^{N_L} \mathbf{G}_{\mathbf{w}}^M \left(\tilde{x}_{n_L}^L \right) \tilde{s}_{n_L}^L \right\|^2 \quad (5.17)$$

The leak sizes can be estimated as:

$$\tilde{\mathbf{s}}^L = \left(M_{\tilde{\mathbf{G}}}^H M_{\tilde{\mathbf{G}}} \right)^{-1} M_{\tilde{\mathbf{G}}}^H \Delta \mathbf{h}_{\mathbf{w}}^M \quad (5.18)$$

and

$$M_{\tilde{\mathbf{G}}} = \left[\mathbf{G}_{\mathbf{w}}^M(\tilde{x}_1^L) \cdots \mathbf{G}_{\mathbf{w}}^M(\tilde{x}_{N_L}^L) \right] \quad (5.19)$$

where $M_{\tilde{\mathbf{G}}}$ is an $N_m \cdot N_f \times N_L$ matrix, $\tilde{\mathbf{x}}^L = [\tilde{x}_1^L, \dots, \tilde{x}_{N_L}^L]$ and $\tilde{\mathbf{s}}^L = [\tilde{s}_1^L, \dots, \tilde{s}_{N_L}^L]$ are N_L -dimensional vectors. As N_L reaches and surpasses N_{L*} , the likelihood initially increases rapidly and then vary slightly. Fig. 5.7 and Fig. 5.8 present the estimated leak locations and sizes and the calculated log-likelihood data. For the case of $N_{L*} = 2$, it can be observed that when using the MB-MFP technique, the log-likelihood converges when $N_L \geq 2$, indicating that the correct number of leaks is two. This aligns with the situation presented in Fig. 5.8, where the predicted $N_L (N_L \geq 2)$ leak locations converge to the vicinity of the correct $N_{L*} (N_{L*} = 2)$ leak locations, confirming the presence of exactly two leaks.

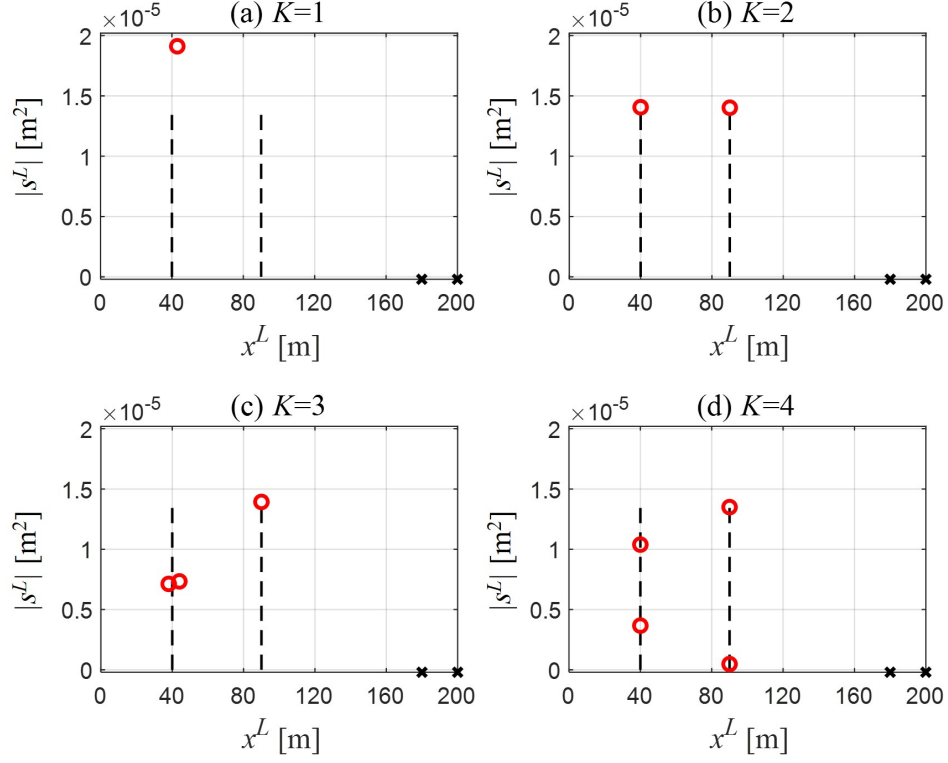


Figure 5.7. Leak detection using MB-MFP with an assumed leak number $N_L = 1, 2, 3$ and 4. The real leak number $N_{L^*} = 2$. The dashed vertical lines represent the real leaks. The real leak locations and sizes are $\mathbf{x}^{L^*} = [40, 90] \text{ m}$ and $\mathbf{s}^{L^*} = [1.4, 1.4] \times 10^{-5} \text{ m}^2$, respectively. The red circles represent the estimated leaks. The estimated leak locations and sizes are (a) $\tilde{\mathbf{x}}^L = [43] \text{ m}$, $\tilde{\mathbf{s}}^L = [1.912] \times 10^{-5} \text{ m}^2$, (b) $\tilde{\mathbf{x}}^L = [40, 90] \text{ m}$, $\tilde{\mathbf{s}}^L = [1.406, 1.402] \times 10^{-5} \text{ m}^2$, (c) $\tilde{\mathbf{x}}^L = [38, 44, 90] \text{ m}$, $\tilde{\mathbf{s}}^L = [0.713, 0.734, 1.393] \times 10^{-5} \text{ m}^2$, (d) $\tilde{\mathbf{x}}^L = [40, 40, 90, 90] \text{ m}$, $\tilde{\mathbf{s}}^L = [0.367, 1.039, 0.048, 1.350] \times 10^{-5} \text{ m}^2$. The black crosses represent the measurement locations.

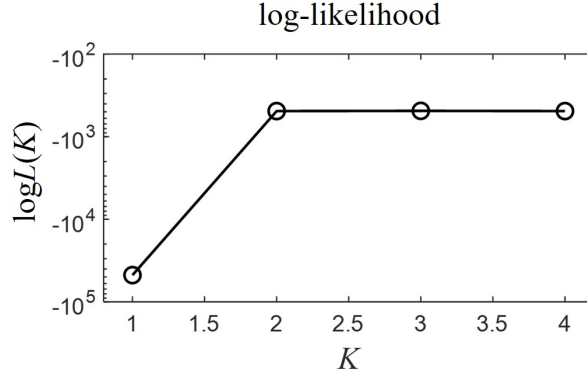


Figure 5.8. The log-likelihood of $N_L = 1, 2, 3$ and 4. The real leak number $N_{L*} = 2$.

5.5 Experimental Validation in a Laboratory Setup

This section discusses the experimental validation conducted in a controlled laboratory setup at the Water Resources Research Laboratory of HKUST, reinforcing the theoretical insights obtained from numerical simulations. The facility comprises a pipeline system, illustrated through a comprehensive 3D schematic and a simplified 2D layout as shown in Figure 5.9. This pipeline system includes essential components such as a booster pump at the upstream end and a valve at the downstream end. Strategically positioned along the pipeline are several manual ball valves (MVs) used either to install pressure transducers or to simulate leaks. In our tests, four MVs are chosen, i.e., MV 1 for generating transient waves by rapid-closure maneuver, and MV 2, MV 4, and MV 8 for data collection purposes.

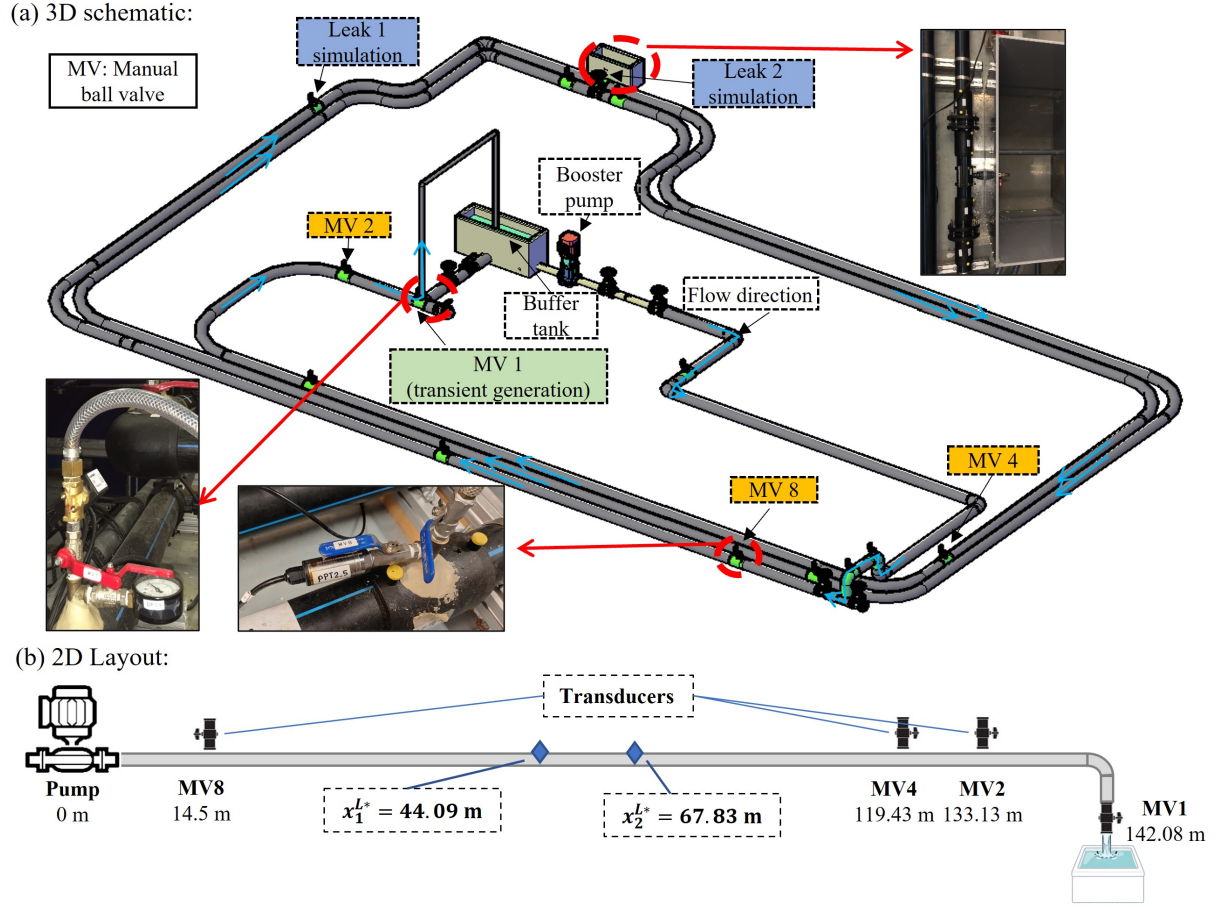


Figure 5.9. The pipeline system in the Water Resources Research Laboratory of HKUST: (a) overall 3D schematic, and (b) simplified 2D layout.

The pipeline has a length of $L = 144\text{ m}$ and a diameter of $D = 0.0792\text{ m}$. The pipe is made of high-density polyethylene, and its detailed viscoelastic parameters are listed in Table 5.5. These parameters are instrumental in calculating the wave speed across various frequencies. The experiment simulated leaks at locations $\mathbf{x}^{L*} = [44.09\text{ m}, 67.83\text{ m}]$. The steady-state flow rates through the main pipe and at the two leaks are measured, being approximately $Q_0 \approx 1.47 \times 10^{-3}\text{ m}^3/\text{s}$, $Q_0^{L1} = 0.40 \times 10^{-3}\text{ m}^3/\text{s}$, and $Q_0^{L2} = 0.63 \times 10^{-3}\text{ m}^3/\text{s}$. The setup of leaks of different sizes is to validate the MB-MFP techniques' ability to simultaneously detect a smaller leak alongside a larger

one.

Table 5.2. Coefficients for pipe wall viscosity of the HDPE pipe in the Water Resources Research Laboratory at HKUST

Water density $\rho = 1000 \text{ kg/m}^3$	Bulk modulus of water $\kappa = 2.1 \times 10^9 \text{ Pa}$
Pipe wall thickness $e = 5.4 \times 10^{-3} \text{ m}$	Poisson's ratio $\nu = 0.43$
Young's modulus of elasticity of pipe wall $E = 1.47 \times 10^9 \text{ Pa}$	
Coefficients of the K-V model: $\tau_1 = 0.038 \text{ s}$, $\tau_2 = 0.6 \text{ s}$, $\tau_3 = 1.5 \text{ s}$	
and $C_1 = 1.957 \times 10^{-10} \text{ Pa}^{-1}$, $C_2 = 1.0954 \times 10^{-10} \text{ Pa}^{-1}$, $C_3 = 9.05 \times 10^{-12} \text{ Pa}^{-1}$	

A rapid full closure of valve MV 1 within 0.05 seconds initiates the transient wave generation. Subsequently, pressure signals are acquired at MV 2, MV 4, and MV 8 in the time domain and transformed into the frequency domain, as depicted in Fig. 5.10. Notably, these signals exhibit significant noise levels beyond the 6 Hz frequency range, leading to the selection of 6 Hz as the upper-frequency limit for subsequent analyses, corresponding to a minimum probing wavelength $\lambda_{\min}$ of around 50 m.

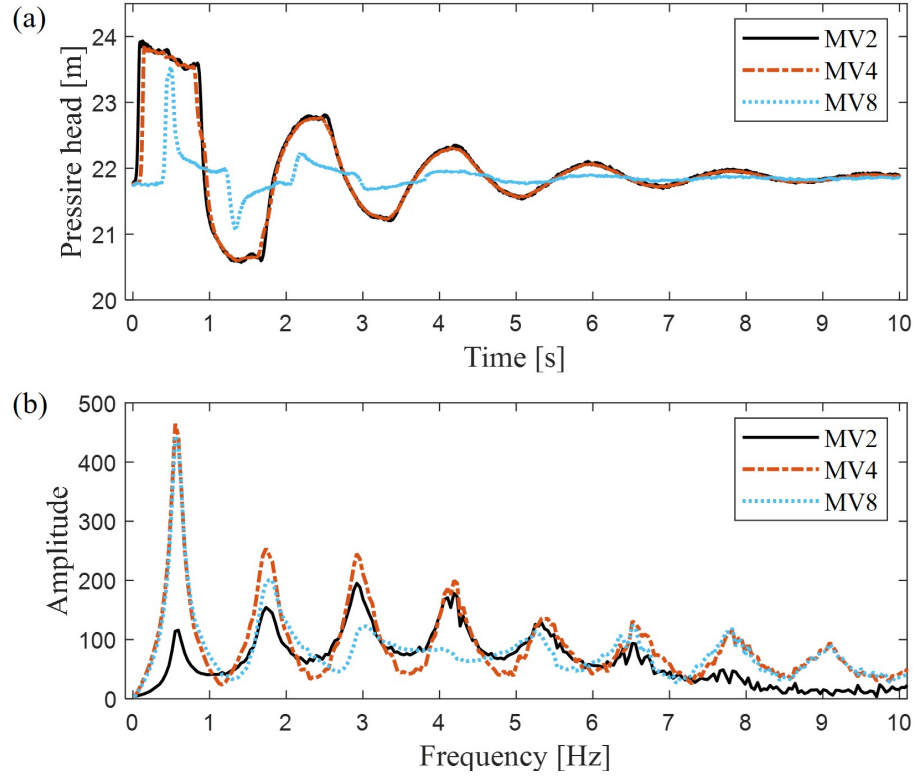


Figure 5.10. (a) Measured time-domain pressure signals and (b) transformed frequency-domain signals at MV 2, MV 4, and MV 8.

The signal at MV8 is employed in conjunction with the transfer matrix to compute the discharge at the upstream end. Meanwhile, the signals at MV3 and MV4 are used for implementing the broadband MFP methods aimed at localizing leaks. Fig. 5.11(A) demonstrates the outcomes of applying MB-MFP(N_L), MB-MFP¹(N_L), and MB-MFP²(N_L) for different values of $N_L = 1, 2, 3, 4$. It is observed that effective leak identification only occurs when $N_L \geq N_{L*} = 2$. Especially, when $N_L > 2$, the estimated leak locations not only converge near the two real leak locations but also correctly indicate the number of leaks. In addition, Fig. 5.11(B) illustrates the results of employing MB-MFP(2), MB-MFP¹(2), and MB-MFP²(2). Here, MB-MFP²(2) demonstrates the highest accuracy and the most effective side-lobe (or ambiguity) suppression, thereby confirming its superiority in

practical leak detection scenarios within the laboratory environment.

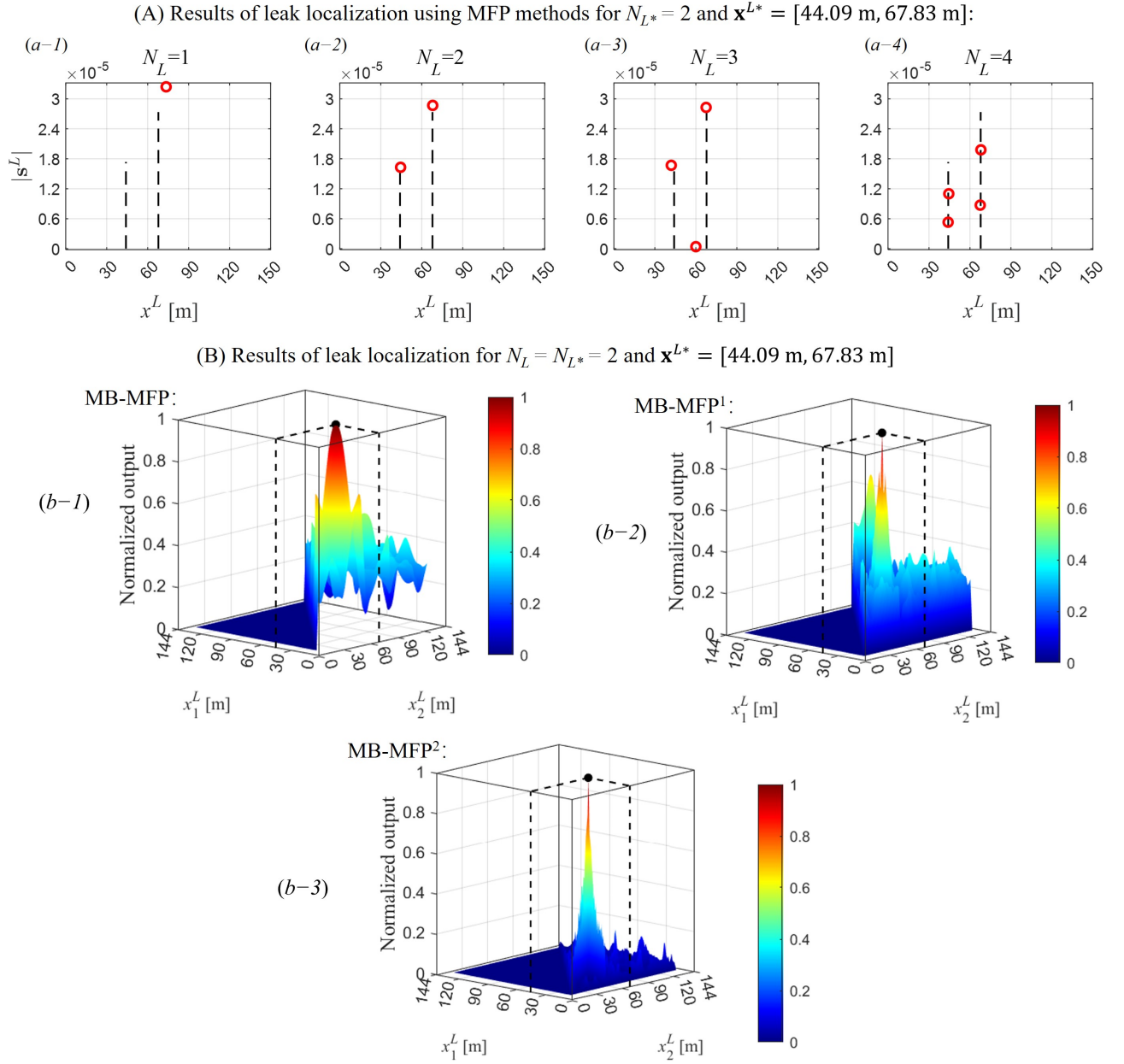


Figure 5.11. Leak localization results of using MB-MFP(N_L), MB-MFP¹(N_L), and MB-MFP²(N_L) with (a-1) $N_L = 1$, (a-2) $N_L = 2$, (a-3) $N_L = 3$, and (a-4) $N_L = 4$. The real leak locations are $\mathbf{x}^{L*} = [44.09 \text{ m}, 67.83 \text{ m}]$ and represented by the black vertical dashed lines. The triangles represent the estimated leak locations. In addition, the results shown in (B) present the capability of MB-MFP(2), MB-MFP¹(2), and MB-MFP²(2) in suppressing the side-lobes.

5.6 Summary

This study has presented a methodology using the modified broadband matched-field processing (MB-MFP) techniques to accurately estimate the locations and numbers of leaks in pressurized water pipelines. The proposed techniques exhibit high efficiency and robustness against various noise types, including white, colored, and impulse noise. This robustness originates from the effective utilization of the full bandwidth of chosen frequencies and the normalization applied to the measured signals. Additionally, these techniques afford super-resolving capability (i.e., enabling the detection of leaks spaced closer than $0.5\lambda_{\min}$) and the ability to simultaneously detect leaks with different sizes. A key finding of our research is that when the number of leaks assumed in the model (N_L) equals or exceeds the actual number of leaks (N_{L*}), the estimated leak locations reliably cluster around the real leak sites. This clustering effect facilitates not only the precise localization of leaks but also the accurate enumeration of them. The proposed methods are analyzed through numerical simulations and validated with experimental data from laboratory settings.

CHAPTER 6

MATERIAL PROPERTY ESTIMATION BY MATCHED-FIELD PROCESSING

This study introduces a novel and non-destructive technique for material property estimation of pressurized pipelines. Utilizing the generalized frequency-domain matched-field processing (MFP) framework, this technique performs an inversion on transient wavefield to determine pipeline parameters. The process involves three straightforward steps: (i) collecting measured data, (ii) computing modeled data for various parameter sets, (iii) identifying the optimal set that best matches the measured and modeled data through MFP framework. Experimental validation on a field-scale high-density polyethylene (HDPE) pipeline network demonstrates the meaningful advancements: (i) first implementation of MFP-based inverse scattering for simultaneous viscoelastic parameter estimation, verified through controlled transient events; (ii) enabling accurate (5% reconstruction signal error) assessment; (iii) computational efficiency achieving 15 ms/parameter evaluation speed; (iv) noise tolerance and scalability to various pipeline systems; and (v) real-time updating capability as evidenced by its successful tracking of a 0.52% creep compliance degradation in transient datasets spanning six years. These factors make MFP a significant breakthrough in pipeline health monitoring technology.

6.1 Introduction

Matched-field processing (MFP) is a premier signal processing technique used in underwater acoustics [40], with many applications as noted by Tolstoy [39]: “... *its successful application to a wide range of problems including the localization in range, depth, and bearing of passive sources, the determination of environmental parameters needed for accurate predictions of acoustic propagation, and the evaluation of model accuracies*”. MFP offers many advantages including (i) noise suppression [102], (ii) insensitivity to the modeling errors [82], (iii) scalability to waveguide/source characteristics [103], and (iv) versatility in handling multi-channel and multi-dimensional data [104].

In the pipeline field, MFP also emerged as a powerful technique to detect defects when combined with hydraulic transients [43, 46, 55]. In particular, researchers have shown that MFP provides an optimal estimate of defect characteristics when noise follows a Gaussian distribution and the estimator is equivalent to the maximum likelihood estimation [43]. The method has already been extended to from a single pipe to complex pipe networks [49, 50, 98, 105]. Moreover, enhanced versions of MFP [81] have been introduced to (i) handle the side-lobes that are commonly found in the conventional MFP or Bartlett MFP, (ii) address the nonwhite noise scenarios, and (iii) achieve super-resolution. Thus far, the scope of MFP in the previous pipeline research has primarily focused on defect detection, with its potential for broader parameter inversion still underexplored.

This chapter extends the application of using MFP to a general method for estimating pipeline system properties by an inversion on hydraulic transient data set. Central to our approach is developing a theoretical frequency-domain model to simulate the generation and propagation of hydraulic transient waves, adjusting the concerned parameters within a candidate range, computing the model

outputs, and comparing the modeled fields with the measured field. The optimal parameters are determined by selecting the candidate ones which best match the modeled and measured data.

To demonstrate MFP as a general parameter inversion technique, we estimate the pipeline viscoelastic (viscoelastic) parameters as an example. These parameters play a significant role in defining the hydraulic transient response in polymeric pipes. Polymeric pipes are either used alone or combined with metallic pipes in hybrid systems. In these systems, even short polymeric segments can significantly affect the transient response of metallic pipelines, underscoring the importance to study polymeric pipes' transient behavior [20]. This study is also crucial for pressure operation safety [21] and pipeline condition assessment[19] during hydraulic transients. However, analyzing transients response in these polymeric pipes is more complex than in elastic ones. This complexity arises from their viscoelastic behavior, which includes both strain and elastic responses, adding a layer of difficulty in understanding the dynamic responses under transient conditions [18, 19]. Moreover, many of the system features like the viscoelastic parameters are impractical to be determined on-site and in real time, mainly because most pipelines are usually buried, making testing not only costly but also time-consuming. This limitation makes MFP—a non-intrusive method to ascertain viscoelastic parameters—highly promising [39]. Furthermore, accurate viscoelastic parameters enhance the precision of models, such as those used in water-hammer studies [22–25] and defect detection methods [36, 43, 46, 106, 107].

This chapter is structured as follows: Section 2 presents the generalized model describing the viscoelastic effect during the transient wave propagation, and details the viscoelastic parameters (i.e., the relevant model coefficients) required to be estimated. Section 3 outlines the methodology for applying MFP techniques to parameter inversion. Section 4 describes the experimental setup at the

Water Resources Research Laboratory at the Hong Kong University of Science and Technology (HKUST) and provides details on transient testing procedures. Section 5 demonstrates the favorable outcomes of applying MFP techniques to lab data, showcasing the highly accurate viscoelastic parameter estimates through minimal error between the reproduced and measured signals. Additionally, the creep compliance rates are calculated from the optimal viscoelastic parameters derived from data collected six years apart, showing physical aging effects on HDPE pipes, i.e., a decreased compliance over time. Finally, Section 6 concludes our findings.

6.2 Preliminaries

6.2.1 Governing Equations in Frequency Domain

By using Kelvin–Voigt (K-V) model to denote ε_r and taking Fourier transform, we obtain the following governing equations in frequency domain [46]:

$$\frac{\partial q}{\partial x} + \frac{gA}{a^2} i\omega h = 0 \quad (6.1)$$

and

$$\frac{\partial h}{\partial x} + \left(\frac{i\omega}{gA} + R \right) q = 0 \quad (6.2)$$

where

$$a(\omega; \boldsymbol{\theta}^{IP}) = \left[\frac{1}{a_0^2} + \frac{\rho D}{e} (1 - \nu^2) \sum_{n_{kv}=1}^{N_{kv}} \frac{C_{n_{kv}}}{1 + i\omega \tau_{n_{kv}}} \right]^{-1/2} \quad (6.3)$$

is the formulated wave speed that depends on the frequency ω and incorporates the viscoelastic effect through K-V models with coefficient sets $\boldsymbol{\theta}^{IP} = [\tau_1, C_1, \dots, \tau_{N_{kv}}, C_{N_{kv}}]^\top$; the superscript $\top$ indicates vector transpose; and $i = \sqrt{-1}$.

6.2.2 Solution of the Frequency-Domain Equations with Boundary Conditions

Solving the frequency-domain governing equations 6.1 and 6.2 with boundary conditions q_ω^U and h_ω^U (i.e., the discharge q and head h at frequency ω and upstream end x^U) yields the flow rate q_ω^M and pressure head h_ω^M (i.e., q and h at frequency ω and measurement location x^M) as follows:

$$\begin{bmatrix} q_\omega^M \\ h_\omega^M \end{bmatrix} = U^{IP}(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}) \begin{bmatrix} q_\omega^U \\ h_\omega^U \end{bmatrix} \quad (6.4)$$

where

$$U^{IP}(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}) = \begin{bmatrix} \cosh(\mu(x^M - x^U)) & -\left(\frac{1}{Z}\right) \sinh(\mu(x^M - x^U)) \\ -Z \sinh(\mu(x^M - x^U)) & \cosh(\mu(x^M - x^U)) \end{bmatrix} \quad (6.5)$$

is the field transfer matrix relating q and h between locations x^M and x^U at frequency ω ; $\boldsymbol{\theta}^{IP} = [\tau_1, C_1, \dots, \tau_{N_{kv}}, C_{N_{kv}}]^\top$ represents the vector of viscoelastic parameters of interest (i.e., K-V coefficients) within the pipeline and influence the transient response; Z is the characteristic impedance of the pipe and $Z = (\mu a^2) / (i\omega g A)$; μ is the propagation function and $\mu = a^{-1} (-\omega^2 + i g A \omega R)^{1/2}$. The relationships depicted in these equations provide insights into how the wave propagates through the pipeline system, as shown in Figure 6.1.

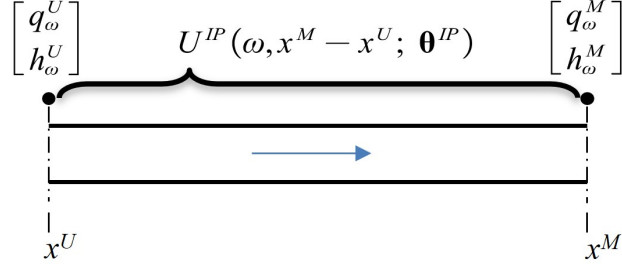


Figure 6.1. The hydraulic quantities (q and h) at the measurement location x^M can be related to q and h at the boundary x^U through a transfer matrix M between them.

6.2.3 Data Model and the Goal of This Study

By rearranging Eqs. 6.4 and 6.5, we can obtain the pressure head h_ω^M at frequency ω and measurement point x^M and as follows [43]:

$$h_\omega^M(\boldsymbol{\theta}^{IP}) = -Z \sinh(\mu(x^M - x^U))q_\omega^U + \cosh(\mu(x^M - x^U))h_\omega^U \quad (6.6)$$

Then, for all N_f frequencies and N_m measurement locations, the measured pressure head can be formed as a $N_f \cdot N_m$ dimensional complex vector:

$$\mathbf{h}_\omega^{\mathbf{M}}(\boldsymbol{\theta}^{IP}) = \left[\underbrace{h_{\omega_1}^{M_1}, \dots, h_{\omega_{N_f}}^{M_1}}_{N_f}, \dots, \underbrace{h_{\omega_1}^{M_{N_m}}, \dots, h_{\omega_{N_f}}^{M_{N_m}}}_{N_f} \right]^\top \quad (6.7)$$

wherein the utilized frequency range $\boldsymbol{\omega} = [\omega_1, \dots, \omega_{N_f}]^\top$; the superscript $\mathbf{M}$ denotes the measurement locations $\mathbf{x}^{\mathbf{M}} = [x_1^M, \dots, x_{N_m}^M]^\top$. For a real system, the actual measured signal includes noise $\mathbf{n}_\omega^{\mathbf{M}}$ and is influenced by the viscoelastic parameters $\boldsymbol{\theta}^{IP*}$ that changes over time and need to be determined to accurately predict the structural health of polymeric pipes. This signal can be expressed

in the following form:

$$\mathbf{h}_\omega^{\mathbf{M}*}(\boldsymbol{\theta}^{IP*}) = \mathbf{h}_\omega^{\mathbf{M}}(\boldsymbol{\theta}^{IP*}) + \mathbf{n}_\omega^{\mathbf{M}} \quad (6.8)$$

The purpose of obtaining the measured head vector $\mathbf{h}_\omega^{\mathbf{M}}(\boldsymbol{\theta}^{IP})$ is to utilize it in next proposed MFP techniques for parameter inversion using transient tests. This inversion targets the unknown K-V coefficients $\boldsymbol{\theta}^{IP*}$, enabling the accurate description of the viscoelastic effect.

Traditionally, $\boldsymbol{\theta}^{IP*}$ are obtained through destructive testing and under specific standards [43]. Worse still, after considerable efforts, these determined coefficients are only valid for specific testing conditions, as they vary with factors such as aging, temperature, and other conditions [108, 109]. Fortunately, MFP is a non-destructive and cost-effective method that can be used to update the relevant viscoelastic parameters in real time.

6.3 Methodology

6.3.1 MB-MFP Techniques

We begin by discussing the single-frequency MFP which is defined as follows:

$$\left| B(\omega; \boldsymbol{\theta}^{IP}) \right| = \left| \left\langle \frac{\mathbf{h}_\omega^{\mathbf{M}*}}{\|\mathbf{h}_\omega^{\mathbf{M}*}\|}, \frac{\mathbf{h}_\omega^{\mathbf{M}}}{\|\mathbf{h}_\omega^{\mathbf{M}}\|} \right\rangle \right| = \frac{\left| (\mathbf{h}_\omega^{\mathbf{M}*})^H \mathbf{h}_\omega^{\mathbf{M}} \right|}{\sqrt{(\mathbf{h}_\omega^{\mathbf{M}*})^H \mathbf{h}_\omega^{\mathbf{M}*}} \sqrt{(\mathbf{h}_\omega^{\mathbf{M}})^H \mathbf{h}_\omega^{\mathbf{M}}}} = \frac{\sqrt{(\mathbf{h}_\omega^{\mathbf{M}})^H \mathbf{h}_\omega^{\mathbf{M}*} (\mathbf{h}_\omega^{\mathbf{M}*})^H \mathbf{h}_\omega^{\mathbf{M}}}}{\sqrt{(\mathbf{h}_\omega^{\mathbf{M}*})^H \mathbf{h}_\omega^{\mathbf{M}*}} \sqrt{(\mathbf{h}_\omega^{\mathbf{M}})^H \mathbf{h}_\omega^{\mathbf{M}}}} \quad (6.9)$$

where the measured signal $\mathbf{h}_\omega^{\mathbf{M}*}(\boldsymbol{\theta}^{IP*})$ and the modeled signal $\mathbf{h}_\omega^{\mathbf{M}}(\boldsymbol{\theta}^{IP})$ are both N_m dimensional vectors at a single frequency ω and their expressions can be found from Eqs. (6.6)–(6.8); $\boldsymbol{\theta}^{IP*}$ represents the vector of actual, yet unknown K-V coefficients that need to be estimated; and $\boldsymbol{\theta}^{IP} =$

$[\tau_1, C_1, \dots, \tau_{N_{kv}}, C_{N_{kv}}]^\top$ denotes the vector of candidate K-V coefficients. This study considers three cases, i.e., K-V model elements with a number of $N_{kv} = 1, 2$, and 3.

Previous research [81] demonstrates that Eq. (6.9) provides substantial insights into the actual parameters $\boldsymbol{\theta}^{IP*}$. In an ideal noise-free environment, $|B|$ reaches its maximum value of one when $\boldsymbol{\theta}^{IP} = \boldsymbol{\theta}^{IP*}$ at each single frequency. We also observe that leak detection (dominated by leak-induced head difference) results in cyclic side-lobe structures at all single-frequencies while viscoelastic parameter estimation (dominated by the viscoelastic effect in transient waves) can result in side-lobe patterns which are non-cyclic, appeared to be randomly located, also varied with frequency. Additionally, results from single-frequency MFP are susceptible to noise and uncertainties, diminishing its robustness [81]. This difficulty encountered can be addressed by employing the modified broadband MFP (MB-MFP) method [81], which integrates $|B|$ across all frequencies as:

$$\left| B_{MB}(\boldsymbol{\theta}^{IP}) \right| = \frac{1}{\omega_{\max} - \omega_{\min}} \left| \int_{\omega_{\min}}^{\omega_{\max}} B(\omega; \boldsymbol{\theta}^{IP}) d\omega \right| \quad (6.10)$$

This method reliably produces a global maximum, or so-called main-lobe, at $\boldsymbol{\theta}^{IP} = \boldsymbol{\theta}^{IP*}$. Moreover, in our previous chapters [81], we show that it is robust under not only white noise, but also colored noise and impulse noise. To further suppress the simultaneously produced local maxima (or side-lobes) while keeping the main-lobe, and enhance the resolution of parameters estimation, we introduce the optimized version— MB-MFP². This version leverages gradient information from $|B|$ on the candidate parameters-frequency space. Specifically, MB-MFP² uses a peak-extraction algorithm that assigns zero values to points adjacent to maxima of $|B|$, by using the gradient infor-

mation of $|B|$ with respect to the candidate parameters $\boldsymbol{\theta}^{IP}$ [81]:

$$\left| B_{MB}^2(\boldsymbol{\theta}^{IP}) \right| = \frac{\left| \int_{\omega_{min}}^{\omega_{max}} \left(\sum_{n_{kv}=1}^{2N_{kv}} \begin{pmatrix} \text{sign} \left(\frac{\partial |B(\omega, \theta_{n_{kv}})|}{\partial \theta_{n_{kv}}} \right) \\ - \text{sign} \left(\frac{\partial |B(\omega, \theta_{n_{kv}} + \Delta \theta_{n_{kv}})|}{\partial \theta_{n_{kv}}} \right) \end{pmatrix} \right) \cdot |B(\omega; \boldsymbol{\theta}^{IP})| d\omega \right|}{\omega_{max} - \omega_{min}} \quad (6.11)$$

wherein $\boldsymbol{\theta}^{IP} = [\theta_1, \theta_2, \dots, \theta_{2N_{kv}-1}, \theta_{2N_{kv}}] = [\tau_1, C_1, \dots, \tau_{N_{kv}}, C_{N_{kv}}]^\top$. The resolution difference between MB-MFP (Eq. 6.10) and MB-MFP² (Eq. 6.11) can be illustrated by an numerical example shown in Fig. 6.2. It is evident that although both methods successfully find the optimal coefficients set, MB-MFP² significantly improves the estimates resolution compared to MB-MFP.

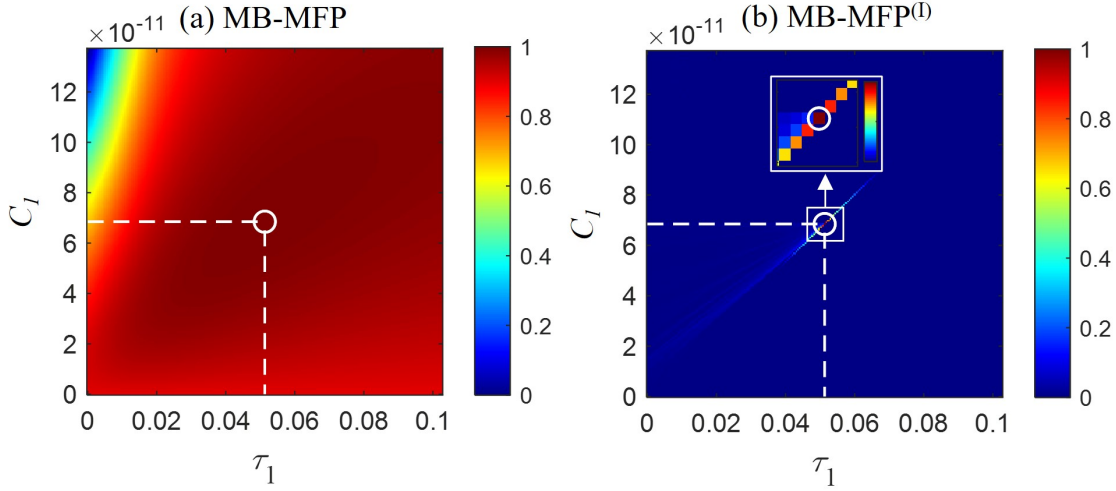


Figure 6.2. Demonstrations of the estimates resolution of using (a) MB-MFP and (b) MB-MFP² for estimating $\boldsymbol{\theta}^{IP*}$ by taking $\boldsymbol{\theta}^{IP} = [\tau_1, C_1]$ (i.e., $N_{kv} = 1$) in a numerical simulation. The white circle and dashed lines indicate the estimated optimal K-V coefficients.

6.3.2 Parameter Inversion Procedures

The viscoelastic parameters (i.e., K-V coefficients) inversion is conducted through a systematic three-step process, utilizing the MB-MFP techniques. The steps are outlined as follows:

Step 1: Selecting frequency range ω and measurement locations x^M

To enhance signal quality, we carefully select a frequency bandwidth with minimal noise, identified through Fourier transformation of the time-domain signal. Measurement locations are strategically chosen near the origin of the transient wave to reduce the impact of unknown pipe characteristics, which might otherwise introduce uncertainty into the inversion process.

Step 2: Conducting a parameters space search using MB-MFP

It is critical to determine not only the key parameters but also the appropriate search interval and increment for each parameter. The choices are usually based on the archival data and prior experience. Then, the MB-MFP techniques as formulated in Eqs. (6.10) and (6.11) are employed to explore the optimal values in the parameters space.

Step 3: Validating and refining results through iterative optimization

MFP involves searching through a limited parameter space, thus sometimes the range of candidate values for the parameters to be varied does not include the true value. This type of error can be avoided by either finer sampling or by iterating to the final parameter value [39]. Therefore, we employ an iterative optimization approach. The process begins with an initial parameter set, and then MFP is used to identify the optimal values within this first set. Subsequently, the optimal

values obtained from the last iteration are used as the intermediate values for the parameter set in the next iteration. This cycle continues until the MFP outcomes converge to stable values, ultimately leading to the identification of the true global optimum.

6.4 Experiments

6.4.1 HKUST Experimental Facility

A field-scale experimental pipeline facility has been designed and constructed in the Water Resources Research Laboratory at HKUST, Hong Kong. Our primary objective is to estimate the unknown K-V coefficients θ^{IP*} , which are crucial for describing the viscoelastic effect within this system by following the proposed MFP methodology. Fig. 6.3 displays both the system's comprehensive 3D schematic and simplified 1D layout.

This system comprises several key components, including a booster pump at the upstream end and a valve at the downstream end. Additionally, multiple manual ball valves (MVs) strategically positioned along the pipeline serve dual purposes: they are not only used to connect pressure transducers but also generate transient waves through rapid discharge adjustments.

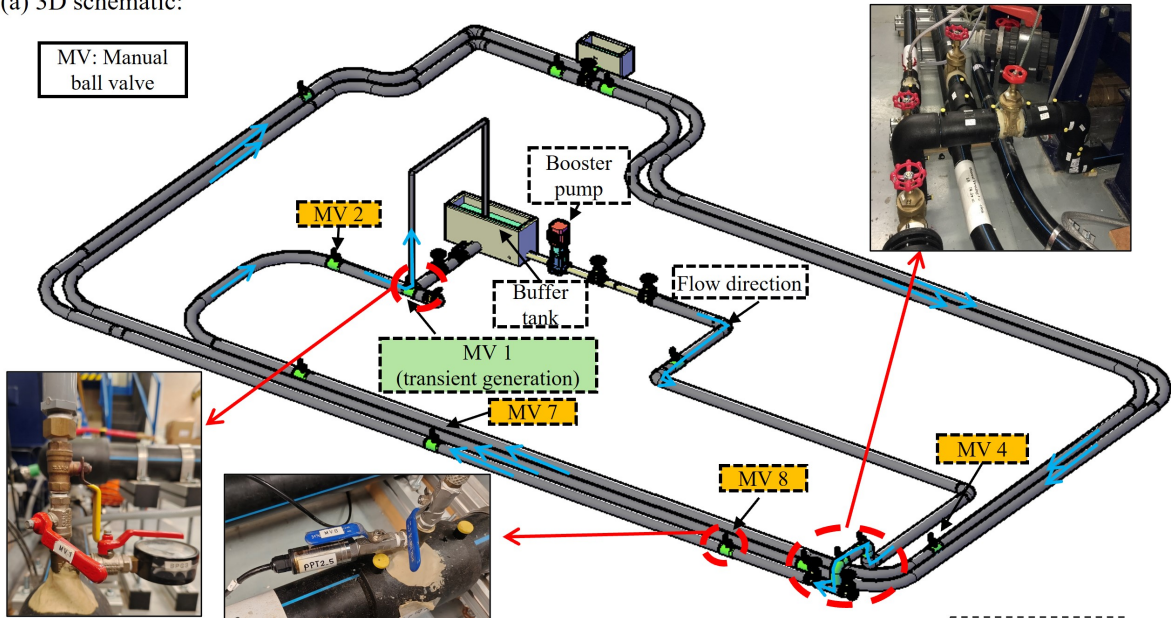
For our transient tests, five MVs are specifically utilized: MV 1 for generating transient waves, MV 7 and MV 8 for connecting signals in conjunction with the transfer matrix to compute the boundary condition at the upstream end, and MV 2 and MV 4 for measuring signals used to implement the MB-MFP methods aiming at parameters estimation. MV 2 and MV 4 are specifically selected

because they are close to where the transient wave is generated, which ensures minimal signal distortion and loss. A summary of the known parameters of the high-density polyethylene (HDPE) pipeline system is given in Table 6.4.1.

Table 6.1. The measured and given parameters of the HDPE pipe system

$\rho = 1 \times 10^3 \text{ kg/m}^3$	$\kappa = 2.1 \times 10^9 \text{ Pa}$	$f_{DW} = 0.02$
$e = 5.4 \times 10^{-3} \text{ m}$	$\nu = 0.43$	$g = 9.81 \text{ m/s}^2$
$L = 144 \text{ m}$	$D = 0.0792 \text{ m}$	$E = 1.47 \times 10^9 \text{ Pa}$

(a) 3D schematic:



(b) 1D layout:

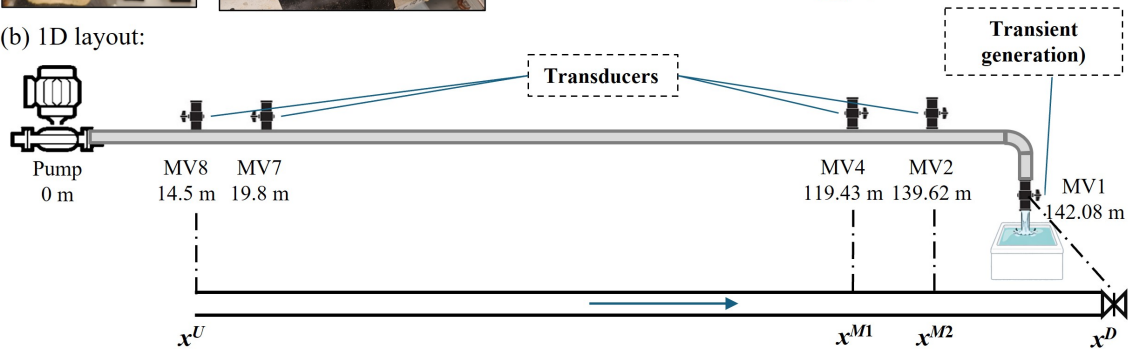


Figure 6.3. The pipeline system in the Water Resources Research Laboratory of HKUST: (a) overall 3D schematic, and (b) simplified 1D layout.

6.4.2 Experimental Procedures

Before initiating the transient tests, we first purge the pipeline of large air bubbles to ensure accuracy in our measurements. This is achieved by running the system at a relatively high pressure (approximately 3 bar), and opening all the manual ball valves (MVs) to allow the entrapped air (if any) to escape, which typically takes about 30 minutes. After this, the pipeline pressure is reduced to around 2 bar by adjusting the upstream pump, thereby setting the stage for our subsequent transient tests.

Once the system is stable, it operates in a steady state with water flowing from the upstream pump to the downstream tank, then recirculating. A hydraulic transient wave is generated by a fast valve operation at MV 1, which is detailed in Fig. 6.4, including valve closure and opening timing and the resulting flow rate oscillations. This wave propagates back and forth within the pipeline and dissipates due to damping effects.

The pressure heads during this transient test are monitored at specific locations—MV 2, MV 4, MV 7, and MV 8—at a sampling frequency of 1,000 Hz for a duration of 10 seconds. To minimize noise and enhance data reliability, 20 experiments are conducted, and then the 20 measured signals are averaged in time domain and transformed into frequency domain, as shown in Fig. 6.5. This averaging approach is also suggested in [39], where it says “*if one has k multiple (statistically independent) sets of data collected over a period of time during which the only effective change in conditions is in the noise, then the simplest estimate is given by the average over the frequency*”.

Due to significant noise in the signals at frequencies above 10 Hz, we truncate the frequency analysis at 10 Hz. In addition, 5 more tests are implemented for the validation of the final estimated

viscoelastic parameters by using MB-MFP techniques.

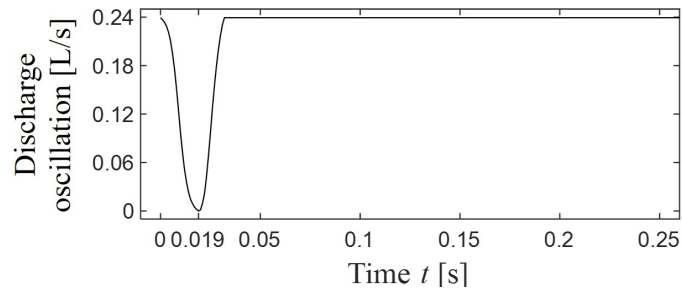


Figure 6.4. Discharge change by manual valve operation.

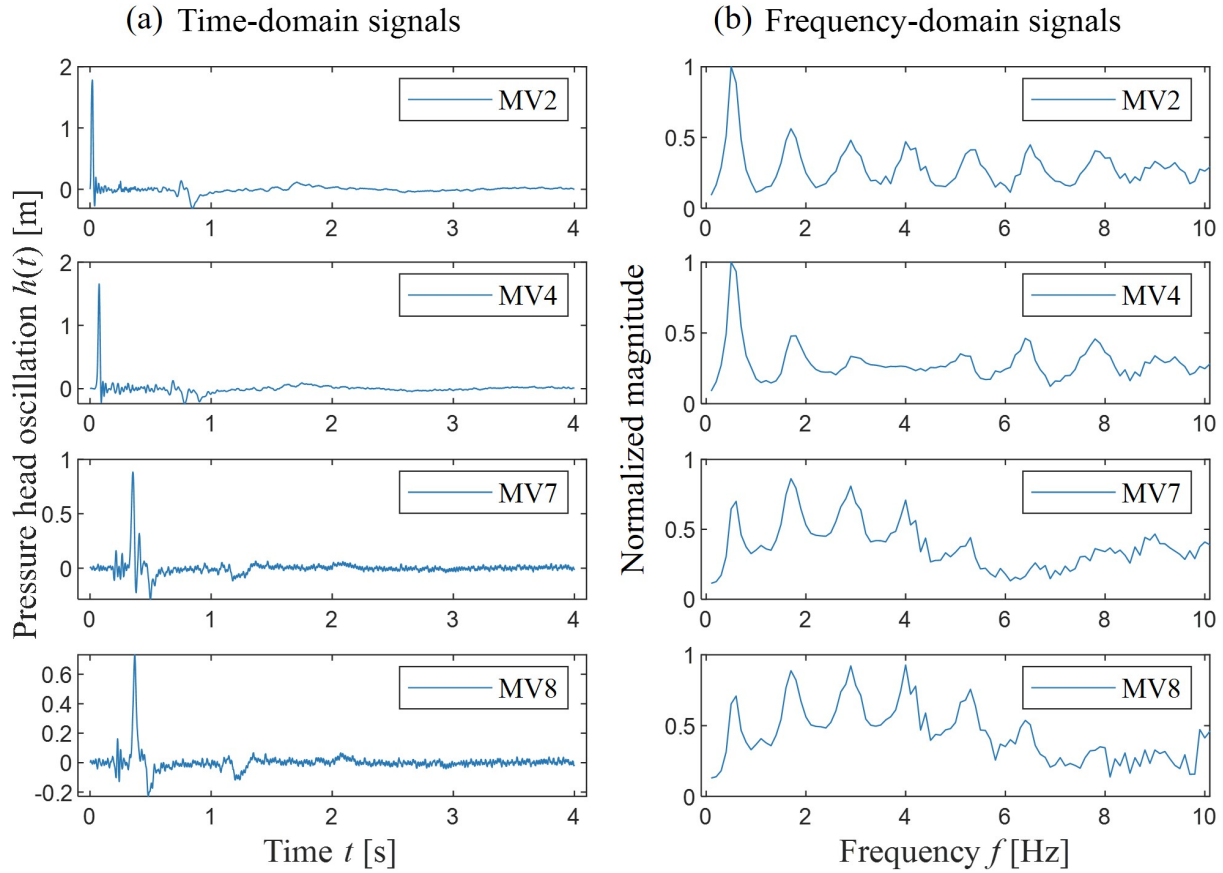


Figure 6.5. The measured time-domain pressure heads and the transformed frequency-domain heads at MV 2, MV 4, MV 7 and MV 8.

6.4.3 Boundary Condition at the Upstream End

Measuring flow rate or pressure head directly at the upstream-end pump poses challenges. As described earlier, we instead measure and use the pressure signals h_{ω}^{MV7} and h_{ω}^{MV8} along with the transfer matrix (Eqs. 6.4 and 6.5) to calculate the flow rate at MV 8, denoted as q_{ω}^{MV8} . We use q_{ω}^{MV8} and h_{ω}^{MV8} as our upstream-end boundary conditions, i.e., let $x^U = x^{\text{MV8}}$. Then, combining q_{ω}^{MV8} and h_{ω}^{MV8} and the transfer matrix allows to compute h_{ω}^{MV4} and h_{ω}^{MV2} , which are matched to the actual measured signals h_{ω}^{MV4} and h_{ω}^{MV2} by using MB-MFP techniques.

6.5 Experimental Results and Analysis

Following the procedures in the former section, transient tests are conducted in the HDPE pipe system at the Water Resources Research Laboratory, HKUST. The measured signals are combined with the MB-MFP methodology to estimate the Kelvin-Voigt (K-V) coefficients.

6.5.1 Estimated K-V Coefficients of Single-Element Model

Inverse problems have traditionally been challenging to solve. One major difficulty arises from computational errors and experimental uncertainties, which mean that the solutions obtained are always approximations of the actual solution. Another challenge is the lack of sufficient data to uniquely determine all the parameters of interest. This raises concerns about the uniqueness of the solution, i.e., multiple values of a parameter could lead to the same observed data. To address this issue, prior information or assumptions are frequently incorporated whenever possible.

From prior experience about this laboratory pipe system [37, 46, 108], the initial values for the single-element model (i.e., $N_{kv} = 1$) are set as $\tau_1 = 0.038$ s, $C_1 = 1.957 \times 10^{-10}$ Pa⁻¹. These values, combined with the search interval and increment given in Table 6.5.1, initiate the first iteration of MB-MFP. The iterations continue until the estimated parameters finally converge to stability. The software used in our study is Matlab R2022b, running on a Windows 11 system equipped with an Intel(R) Core(TM) i7-9700 CPU @ 3.00GHz processor and 24.0GB of memory. For the single-element model, the computational time required for each iteration is approximately 0.015 ~ 0.03 seconds.

Table 6.2. Search parameters for $N_{kv} = 1$, their initial values and search intervals, coarse increments, and number of candidates.

Parameter	Initial value	Interval	Increment	No. values
τ_1	$\tau_1^i = 0.038$ s	$[\tau_1^i - 0.5 \times \tau_1^i, \tau_1^i + 0.5 \times \tau_1^i]$	$0.1 \times \tau_1^i$	11
C_1	$C_1^i = 1.957 \times 10^{-10}$ Pa ⁻¹	$[C_1^i - 0.5 \times C_1^i, C_1^i + 0.5 \times C_1^i]$	$0.1 \times C_1^i$	11

The results of each iteration are displayed in Fig. 6.6. The K-V coefficients start from the initial setting $\tau_1 = 0.038$ s and $C_1 = 1.957 \times 10^{-10}$ Pa⁻¹ and converge to the final estimated $\tau_1 = 0.0513$ s and $C_1 = 6.850 \times 10^{-11}$ Pa⁻¹. To quantitatively show the accuracy of the optimal K-V coefficients in each iteration, we define an error term between the reproduced pressure head $\tilde{\mathbf{h}}_{\omega}^{\mathbf{M}}$ based on the estimated K-V coefficients and the measured head $\mathbf{h}_{\omega}^{\mathbf{M}*}$ as:

$$R_{error} = \|\tilde{\mathbf{h}}_{\omega}^{\mathbf{M}} - \mathbf{h}_{\omega}^{\mathbf{M}*}\| / \|\mathbf{h}_{\omega}^{\mathbf{M}*}\| \quad (6.12)$$

which is the ratio of the Euclidean norm (or modulus) of the difference between $\tilde{\mathbf{h}}_{\omega}^{\mathbf{M}}$ and $\mathbf{h}_{\omega}^{\mathbf{M}*}$ to the

norm of $\mathbf{h}_{\omega}^{\mathbf{M}^*}$. Then, R_{error} for the initial and final parameter sets are respectively calculated and given as 0.7149 and 0.0599 in Table 6.5.1.

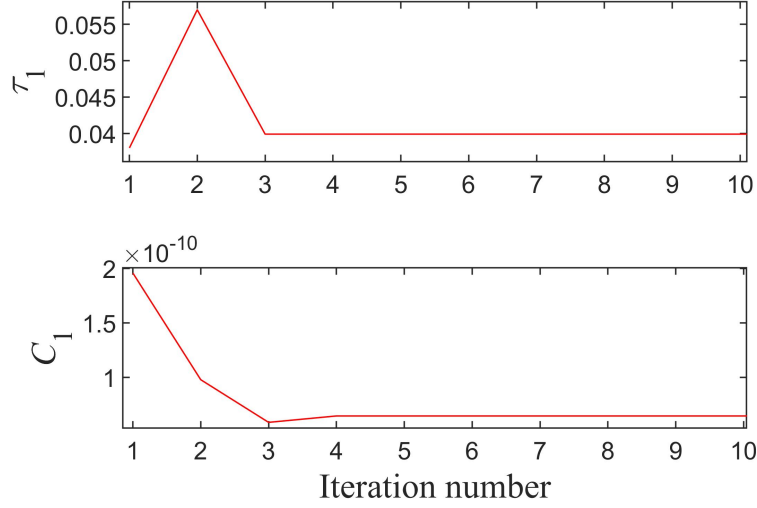


Figure 6.6. Optimization trajectories of parameters (a) τ_1 and (b) C_1 over 10 iterations.

Table 6.3. Optimal K-V parameters for 1-element model, i.e., $N_{kv} = 1$.

Case	Iteration number	Initial assumed values	Initial R_{error}	Final estimated values	Final R_{error}
$N_{kv} = 1$	3	$\tau_1 = 0.038 \text{ s}$	0.6982	$\tau_1 = 0.0399 \text{ s}$	0.0601
		$C_1 = 1.957 \times 10^{-10} \text{ Pa}^{-1}$		$C_1 = 6.458 \times 10^{-11} \text{ Pa}^{-1}$	

By using the same boundary conditions, we have reproduced the pressure signals based on the estimated K-V coefficients and compared these with the five additional measurements in Fig. 6.7. This comparison demonstrates that even using a single-element model, the optimal K-V parameters derived from MB-MFP can closely replicate another tested signal, which indicates the accuracy of

final estimated K-V values. In the next subsection, we extend the methodology to use two-element and three-element K-V models. We anticipate that this expansion will allow for even more precise parameter refinement and further reduction in error margins.

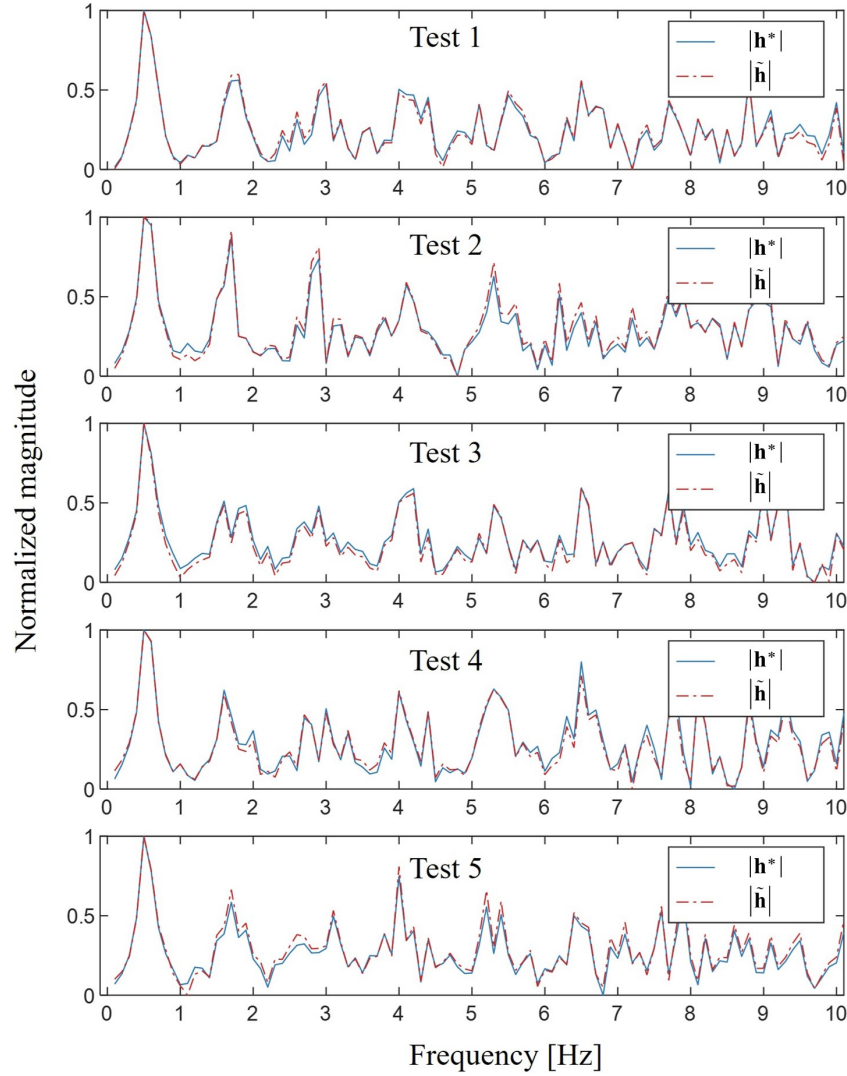


Figure 6.7. The comparisons between the measured and reproduced frequency-domain pressure heads at MV 2 during 5 transient tests, where $|\mathbf{h}^*|$ and $|\tilde{\mathbf{h}}|$ are respectively the measured signal and the reproduced signal based on the final estimated K-V parameters.

6.5.2 Parameter Inversion by Two-Element and Three-Element Models

Building on the successful implementation of MB-MFP methods with a single-element K-V model, we expand our analysis to incorporate two-element and three-element K-V models. The detailed settings and outcomes for these extended models are summarized in Table 6.5.2. The required computational time is approximately 1.8 ~ 2.1 seconds and 1.5 ~ 2 minutes, respectively. Our results demonstrate that the two-element model achieved approximately a 25% improvement in estimation accuracy over the single-element model. This increase in precision is largely due to the additional degrees of freedom available (i.e., K-V coefficients) for modeling the viscoelastic properties of the pipeline system. These additional parameters allow for a more nuanced representation of the system's dynamic behaviors, capturing subtleties that a simpler model might miss. However, the three-element model does not show a significant improvement in accuracy beyond what was achieved with the two-element model. This suggests that when $N_{kv} \geq 2$, our estimation might have reached the maximum accuracy feasible under the experimental conditions and model constraints (i.e., the search intervals) detailed in Table 6.5.1.

Table 6.4. Optimal K-V parameters for 2-element and 3-element models, i.e., $N_{kv} = 2$ and 3.

Case	Iteration number	Initial assumed values	Initial R_{error}	Final estimated values	Final R_{error}
$N_{kv} = 2$	9	$\tau_1 = 0.038 \text{ s}$	0.7529	$\tau_1 = 0.0274 \text{ s}$	0.0446
		$\tau_2 = 0.6 \text{ s}$		$\tau_2 = 0.6048 \text{ s}$	
		$C_1 = 1.957 \times 10^{-10} \text{ Pa}^{-1}$		$C_1 = 0.391 \times 10^{-10} \text{ Pa}^{-1}$	
		$C_2 = 1.095 \times 10^{-10} \text{ Pa}^{-1}$		$C_2 = 2.218 \times 10^{-10} \text{ Pa}^{-1}$	
$N_{kv} = 3$	11	$\tau_1 = 0.038 \text{ s}$	0.7547	$\tau_1 = 0.0274 \text{ s}$	0.0442
		$\tau_2 = 0.6 \text{ s}$		$\tau_2 = 1.8225 \text{ s}$	
		$\tau_3 = 1.5 \text{ s}$		$\tau_3 = 0.2042 \text{ s}$	
		$C_1 = 1.957 \times 10^{-10} \text{ Pa}^{-1}$		$C_1 = 0.391 \times 10^{-10} \text{ Pa}^{-1}$	
		$C_2 = 1.095 \times 10^{-10} \text{ Pa}^{-1}$		$C_2 = 4.710 \times 10^{-10} \text{ Pa}^{-1}$	
		$C_3 = 9.05 \times 10^{-12} \text{ Pa}^{-1}$		$C_3 = 25.65 \times 10^{-12} \text{ Pa}^{-1}$	

6.5.3 Creep Compliance Analysis for Pre- and Post-Six Years

In the Generalized K-V model, the creep compliance function is represented by

$$C(t) = C_0 + \sum_{n_{kv}=1}^{N_{kv}} C_{n_{kv}} \left(1 - \exp \left[\frac{-t}{\tau_{n_{kv}}} \right] \right) \quad (6.13)$$

This function is calculated using the final estimated one-element K-V coefficients from Table 6.5.1, i.e., $\tau_1 = 0.0399 \text{ s}$ and $C_1 = 6.458 \times 10^{-11} \text{ Pa}^{-1}$, for the laboratory data collected in the year of 2024. The result is shown as the solid curve $C(t)_{2024}$ in Fig. 6.8. Additionally, the creep compliance function for an laboratory data set collected from 2018 is also calculated and displayed in Fig. 6.8 as the dashed curve $C(t)_{2018}$, wherein the final estimated and used K-V coefficients are $\tau_1 = 0.0399 \text{ s}$

and $C_1 = 6.850 \times 10^{-11} \text{Pa}^{-1}$.

The difference between the two creep compliance curves in Fig. 6.8 shows a decrease in creep compliance over a period of 6 years (2024 - 2018 = 6 years). This reduction is reported to be due to the physical aging over time in literature, see for example: [108]. The decrease in creep compliance can be explained by free volume theory [109]. As aging time increases, the free volume fraction in HDPE decreases, reducing the activation energy. This causes the molecules to move closer together, increasing the viscosity and enhancing the material's rigidity (or stiffness). Consequently, this results in a higher elastic modulus and lower creep compliance. However, this also increases brittleness, leading to premature failure (earlier than the designed fracture stress) over time [110]. According to the Plastics Pipe Institute [111] in North America, when HDPE pipes are subjected to lower stresses for longer duration (similar to our laboratory conditions, where they have been under approximately 2 bar of pressure for up to six years), they transit from ductile to brittle with time and their failures are more brittle in nature.

Furthermore, for both data sets, the frequency-dependent wave speed a [Eq. 6.3] is calculated and displayed in Figure 6.9. Here, the initial value of a_{2024} is greater than that of a_{2018} , but as frequency increases, both values gradually converge. This consequence can be viewed in terms of a [Eq. 6.3], where $a_0 = 360.5 \text{m/s}$ is set in both a_{2024} and a_{2018} . As the frequency increases, the second term in Eq. 6.3 gradually approaches zero, causing both a_{2024} and a_{2018} to converge to a_0 . In the lower frequency range, the gap between a_{2024} and a_{2018} is primarily due to the variations in the K-V coefficients from the two data sets.

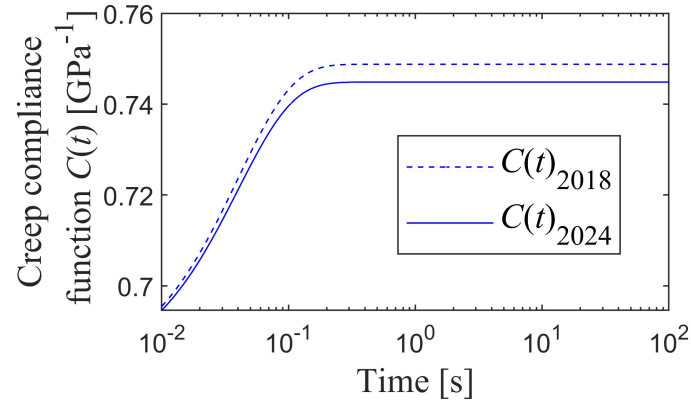


Figure 6.8. Two momentary creep-compliance curves for HDPE pipeline with a six-year aging time. The curves are calculated by using the optimal 1-element K-V coefficients which are respectively obtained from the laboratory data sets at the year 2018 and the year 2024.

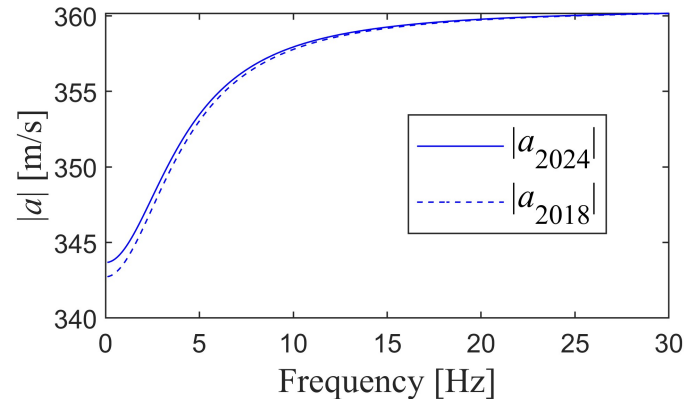


Figure 6.9. The predicted frequency-dependent wave speed a based on the final estimated K-V coefficients.

6.6 Conclusions

This study establishes a novel and general framework for pipeline structural/hydraulic parameter estimation through transient-based matched-field processing (MFP), demonstrating some key scientific advancements. First and foremost, we pioneer the application of MFP methodology to parameter inversion, such as viscoelastic properties, in field-scale pipeline systems. The predicted results are highly accurate as evidenced by the around 5% signal reconstruction error whose definition is given as Eq. 6.12. Additionally, the results also represent a significant improvement in computational efficiency (e.g., the evaluation speed for 1-element K-V model is 15 ms/parameter), noise tolerance, and estimates resolution.

The second innovation lies in establishing an optimal complexity framework through systematic model element analysis. Our quantitative demonstration that a two-element K-V model achieves 25% accuracy gain over single-element configurations — while maintaining computational efficiency — provides critical guidance for balancing model fidelity and practicality. The marginal 0.09% improvement observed when introducing a third element establishes a clear threshold for complexity optimization in viscoelastic modeling.

From a broader perspective, by cross-validating creep compliance rates from six-year-interval laboratory data, we not only confirm the physical aging mechanism (0.52% compliance reduction in our laboratory environment) but also establish MFP as a promising technique enabling real-time parameter updating for aging infrastructure. This dual capability of capturing both instantaneous viscoelastic responses and long-term material evolution establishes our approach as a self-calibrating monitoring framework for pipeline structural health.

CHAPTER 7

CONCLUSIONS AND FUTURE DIRECTIONS

7.1 Overall Conclusions

This thesis presents a comprehensive study on the integration of hydraulic transient wave analysis and matched-field processing (MFP) techniques for parameter inversion in pressurized pipelines. The proposed inverse analysis framework is broadly applicable and capable of providing accurate estimated solutions. The effectiveness of the MFP framework is validated through its applications to solving a range of inverse problems in pressurized pipelines, including single-defect detection, clustered-defect identification, and viscoelastic parameter estimation. The research objectives established in Chapter 1 (Introduction) are successfully achieved, and the key findings and scientific contributions (theoretical, methodological, and practical) can be summarized as follows:

7.1.1 Generalized MFP Inversion Technique for Parameter Estimation

The thesis establishes a modular, adaptable, and flexible MFP framework capable of addressing a wide range of parameter estimation tasks in hydraulic pipeline systems. By formulating the governing equations in the frequency domain and solving them using the transfer matrix method, the pressure head response can be derived as the combined effect of contributions from both the intact

pipe and defects. This theoretical foundation enables the separation of parameters associated with the intact pipe (such as material properties) from those related to defects (such as defect location and size). Such parameter decoupling facilitates targeted MFP analysis by enabling the selective input of parameter-specific signal components corresponding to the parameters of interest. The resulting MFP framework demonstrates robust versatility in handling various parameter inversion tasks across different pipeline systems, offering a unified approach to parameter estimation that integrates multiple tasks and techniques.

7.1.2 Enhanced Robustness Against Noise

A significant advancement in this work is the formulation of modified broadband MFP (MB-MFP) techniques, which enhance system robustness against various noise types, including white noise, colored noise, and impulse noise. This improvement is achieved by normalizing the measured signals at each frequency and then averaging the MFP outputs across frequencies to obtain the MB-MFP methods. MB-MFP methods remain effective even under very low signal-to-noise ratios (SNRs), maintaining high detection accuracy. For example, the experimental validation confirms that the MB-MFP techniques successfully detect single defects under the following SNR conditions: $\text{SNR} = -8$ dB for white noise, $\text{SNR}_1 = 30$ dB for blue noise, $\text{SNR}_2 = -70$ dB for impulse noise.

7.1.3 Enhanced Side-Lobe Suppression and Main-Lobe Resolution

A fundamental contribution of this work is the theoretical analysis of side-lobe behavior in single-frequency MFP outputs, providing a mechanistic understanding of MFP output patterns in the

parameter-frequency space. That is, the side-lobes in single-frequency MFP are inherent to the method. As for broadband MFP, side-lobes arise from constructive interference of single-frequency peak distributions across spectral components. These insights guide the design of gradient-based algorithms to develop improved versions of MB-MFP, namely MB-MFP¹ and MB-MFP², which enhance the side-lobe suppression and the main-lobe resolution. For instance, experimental results show that the secondary side-lobe-to-main-lobe level ratio decreases from approximately 0.5 in the original MB-MFP to about 0.2 in MB-MFP¹ and roughly 0.04 in MB-MFP². Similarly, the average side-lobe-to-main-lobe level ratio drops from around 0.2 to approximately 0.05 in MB-MFP¹ and 0.02 in MB-MFP². Furthermore, MB-MFP² can significantly increase the main-lobe resolution, where the resolution limit at this stage is determined by the choice of parameter increment. In the experimental validation, MB-MFP² achieves sub-meter accuracy in detecting leaks, with a 0.5 m error in a 144-meter-long pipe, under realistic noise conditions, outperforming conventional methods.

7.1.4 Achieved Super-Resolution Defect Detection

The proposed MB-MFP methods demonstrate super-resolving capability for clustered defects through employing a multi-defect model with $N_L \geq N_{L*}$, where N_L represents the number of defects assumed in the model, and N_{L*} represents the actual number of defects in the real pipeline system. For example, defects as close as $0.2\lambda_{\min}$ are resolved, surpassing the traditional resolution limit of $0.5\lambda_{\min}$ and demonstrating the capability of MFP framework to handle complex defect scenario. When $N_L \geq N_{L*}$, the results show that the N_L estimated leak locations tend to cluster around the N_{L*} actual leak locations. This spatial clustering phenomenon enables a dual-purpose framework that not only achieves accurate leak localization but also helps in estimating the number of leaks.

7.1.5 Accurate Material Property Estimation

The effectiveness of the proposed MFP methodologies in estimating material properties is validated through both numerical simulations and laboratory experiments. Specifically, the approach is demonstrated on an HDPE pipe system, where it achieves accurate estimation of viscoelastic parameters. We achieve 5% reconstruction error (by using the predicted viscoelastic parameters) with 15 ms/parameter computational speed. The optimized two-element K-V model shows 25% accuracy gain over single-element versions, while a third element provides only 0.09% improvement, which sets a clear threshold for complexity optimization in viscoelastic modeling. Furthermore, validation against six-year-interval creep compliance data confirms material aging mechanisms, showing a 0.52% reduction in compliance, and demonstrating MFP's dual capability for real-time monitoring and long-term structural health assessment.

In conclusion, this thesis bridges theoretical wave physics and practical pipeline parameter estimation by developing innovative MFP inversion techniques. The proposed MFP framework systematically addresses critical challenges, including noise resilience, spatial resolution limits, side-lobe interference, and complexity in handling multiple parameters, through comprehensive theoretical analysis and methodological advancements. Numerical simulations and experimental validations confirm the effectiveness of the proposed MFP framework for achieving accurate parameter estimation. These improvements contribute directly to maintaining hydraulic infrastructure integrity, enhancing operational safety, and promoting long-term sustainability for WSSs.

7.2 Future Directions

While this thesis advances the state of the art in pipeline parameter inversion by using MFP, the following future research directions, driven by interdisciplinary collaboration in hydraulics, advanced signal processing architectures, and data science, hold potential to further enhance the applicability of MFP methods.

1. Involvement of Multi-Data Sources and Alternative Method for Multi-Defect Detection

Future research in this field is likely to focus on integrating multiple data sources, such as noise signals, flow rates, and vibration measurements, to improve the accuracy and reliability of parameter estimation

An alternative strategy for identifying multiple sources through matched field processing (MFP) involves iteratively removing the strongest detected signals from the localization results. The procedure begins by creating an initial ambiguity space under the assumption of a single source in the environment. The region showing main-lobe in this first analysis is identified as the primary target. Subsequently, a simulated data set containing only this dominant source is created to produce a secondary localization ambiguity space. By subtracting this simulated ambiguity space from the original ambiguity space, the main-lobe caused by the strongest source and its side-lobes can be effectively eliminated. The remaining ambiguity space then reveals weaker sources combined with noise. This cycle continues until only noise remains – though this determination may not easily be made.

2. Extension to Complex Pipeline Networks and High-Frequency Wave Application:

Current methodologies focus on single-pipe systems, but real-world water supply networks often include branches, loops, and junctions. Future work should investigate the network topology effects like how wave reflections and transmissions at junctions influence MFP performance, or adapting MFP to distinguish defects in interconnected pipelines with overlapping transient responses. Another promising application of transient waves is to determine the layout of pipe networks.

While the used MFP focuses on low-frequency regimes (<100 Hz), preliminary results suggest viability in high-frequency scenarios (1-100 kHz). Key challenges include developing a theoretical models for high-frequency wave propagation in pipelines, especially when considering turbulence and damping effect.

3. Integration with Machine Learning

Further exploration of ML and deep learning techniques holds significant promise for advancing the state-of-the-art in pipeline condition assessment. They can offer potential synergies with MFP. For example, in noise classification, trained neural networks may identify and filter non-stationary noise (e.g., traffic-induced vibrations). Additionally, for parameter space reduction, using ML to optimize candidate parameter ranges can decrease the computational costs in high-dimensional searches, especially for our case it is usually necessary to calculate thousands of replica fields. Moreover, machine learning approaches using physics-informed neural networks may be used to predict the global optimum regions corresponding to the interested parameters.

4. Field-Scale Implementation Challenges

Translating laboratory success to field applications requires addressing practical constraints such as the sensor placement optimization, because the sensor locations, which we can control, can have significant effects on the MFP performance. We may need to develop guidelines for minimal yet effective sensor arrays in large-scale systems. At the same time, environmental variability need to be accounted, like the temperature fluctuations, soil-pipe interactions, and seasonal hydraulic load variations.

5. Uncertainty Quantification and Probabilistic Frameworks

Current methods provide deterministic estimates, but probabilistic approaches could improve reliability. Bayesian inference can be used to quantify parameter uncertainties and incorporating prior knowledge (e.g., historical defect data). To have a deep understanding and a standard for model mismatch, we can evaluate performance under incomplete or inaccurate pipeline specifications.

It is important to mention that, up until now, MFP has been used on data with the assumption that a deterministic signal has already been identified. In other words, MFP is currently an estimation method rather than a detection method. In practice, however, MFP is frequently applied in cases where the detection probability $p(d) \neq 1$. Therefore, some statistical algorithms like Bootstrap can be further investigated for this purpose.

6. Theoretical Analysis of Parameter Increment Selection and Frequency Optimization:

The current empirical approach for selecting range increments and source frequencies raises fundamental theoretical questions. A three-pronged investigation is proposed: (i) Establishing quantitative criteria for optimal space increment sizing as there is no a prior analytic

approach presently available, (ii) Developing frequency selection frameworks that account for parameter sensitivity, and (iii) Systematic characterization of frequency-dependent side-lobe interference patterns. This theoretical foundation should address the observed mismatch phenomenon where certain frequencies generate misleading localization peaks, potentially through developing weighted frequency combination schemes that suppress spurious side-lobe contributions.

7. Computational Acceleration through Optimization and Adaptive Refinement:

MFP pays for its improved performance with intensive computer processing since it is usually necessary to calculate thousands of replica fields. To address the computational bottleneck, implementing optimization methods (e.g., genetic algorithms, particle swarm optimization, etc.) with adaptive stopping criteria to eliminate redundant replica field computations. This requires developing probabilistic models of peak distribution in pipeline environments.

An alternative method involves using a numerical approach that refines local maxima and ensures they converge to their accurate values. This method could rely on techniques such as Newton's method, gradient methods, or by progressively sampling the regions near the initial rough estimate of the maxima, gradually narrowing the search until the peak is identified within a set tolerance.

APPENDIX A

SUPPLEMENTARY MATERIAL FOR GOVERNING EQUATIONS

A.1 Solution to the Frequency Response of An Intact Pipe

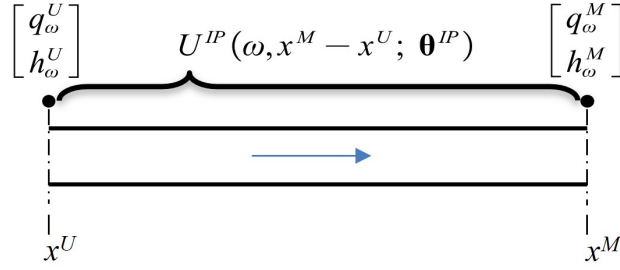


Figure A.1. Field transfer matrix for an intact pipe segment.

For intact pipe shown in Fig. A.1 (same to Fig. 2.2), the right-hand-side term relating to leak in Eq. 2.6 is neglected. Then, the frequency-domain mass and momentum conservation equations are given as

$$\frac{\partial q}{\partial x} + \frac{gA}{a^2} i\omega h = 0 \Rightarrow \frac{\partial q}{\partial x} = -\frac{gA}{a^2} i\omega h \quad (\text{A.1})$$

and

$$\frac{\partial h}{\partial x} + \left(\frac{i\omega}{gA} + R \right) q = 0 \Rightarrow \frac{\partial h}{\partial x} + \frac{i\omega + RgA}{gA} q = 0 \quad (\text{A.2})$$

Differentiating Eq. A.2 with respect to x and substituting Eq. A.1 into it yields

$$\frac{\partial^2 h}{\partial x^2} - \frac{-\omega^2 + i\omega g A R}{a^2} h = 0 \Rightarrow \frac{\partial^2 h}{\partial x^2} - \mu^2 h = 0 \quad (\text{A.3})$$

where μ is the propagation function and $\mu = \sqrt{-\omega^2 + i\omega g A R}/a$. Due to the derivative rules to the hyperbolic functions: $(\sinh \mu x)' = \cosh \mu x$, $(\cosh \mu x)' = \sinh \mu x$, the homogeneous solution for Eq. A.3 is derived as

$$h = \alpha_1 \sinh \mu x + \alpha_2 \cosh \mu x \quad (\text{A.4})$$

where α_1 and α_2 are the coefficients need to be determined. This expression of h (Eq. A.4) can be substituted back into Eq. A.1 and it obtains

$$\frac{\partial q}{\partial x} = -\frac{i\omega g A}{a^2} (\alpha_1 \sinh \mu x + \alpha_2 \cosh \mu x) \quad (\text{A.5})$$

Similarly, based on the derivative rules to the hyperbolic functions, the homogeneous solution to Eq. A.5 is

$$q = -\frac{1}{Z} (\alpha_1 \cosh \mu x + \alpha_2 \sinh \mu x) \quad (\text{A.6})$$

wherein Z is the characteristic impedance of the pipe and $Z = \mu a^2 / i\omega g A$.

Now consider the denotations of q and h in Fig. A.1. From Eqs. A.4 and A.6, we can obtain the following relationship:

$$q_\omega^M = -\frac{1}{Z} \left(\alpha_1 \cosh \left(\mu (x^M - x^U) \right) + \alpha_2 \sinh \left(\mu (x^M - x^U) \right) \right) \quad (\text{A.7})$$

and

$$h_\omega^M = \alpha_1 \sinh \left(\mu (x^M - x^U) \right) + \alpha_2 \cosh \left(\mu (x^M - x^U) \right) \quad (\text{A.8})$$

By giving the boundary conditions q_ω^U and h_ω^U at $x^M = x^U$, then we obtain

$$q_\omega^U = -\frac{1}{Z} \left(\alpha_1 \cosh \left(\mu \left(x^U - x^U \right) \right) + \alpha_2 \sinh \left(\mu \left(x^U - x^U \right) \right) \right) = -\frac{1}{Z} \alpha_1 \Rightarrow \alpha_1 = -Z q_\omega^U \quad (\text{A.9})$$

and

$$h_\omega^U = \alpha_1 \sinh \left(\mu \left(x^U - x^U \right) \right) + \alpha_2 \cosh \left(\mu \left(x^U - x^U \right) \right) = \alpha_2 \Rightarrow \alpha_2 = h_\omega^U \quad (\text{A.10})$$

Substituting the expressions of the coefficients α_1 (Eq. A.9) and α_2 (Eq. A.10) into Eqs. A.7 and A.8 yields the final solutions for q_ω^M and h_ω^M :

$$q_\omega^M = \cosh \left(\mu \left(x^M - x^U \right) \right) q_\omega^U - \frac{1}{Z} \sinh \left(\mu \left(x^M - x^U \right) \right) h_\omega^U \quad (\text{A.11})$$

and

$$h_\omega^M = -Z \sinh \left(\mu \left(x^M - x^U \right) \right) q_\omega^U + \cosh \left(\mu \left(x^M - x^U \right) \right) h_\omega^U \quad (\text{A.12})$$

A.2 Derivation to the Linearized Orifice Equation

As given in Eq. 2.13, the orifice equation relates Q_0^L (the steady-state discharge extracted through leak) and H_0^L (the steady-state pressure head at the leak location) as

$$Q_0^L = s^L \sqrt{2gH_0^L} \quad (\text{A.13})$$

where s^L is the effective cross-sectional area of the leak. For the unsteady state during a transient, we have

$$Q_0^L + q^L = \left(s^L + \Delta s^L \right) \sqrt{2g \left(H_0^L + h^L \right)} \quad (\text{A.14})$$

where q^L and h^L are respectively the discharge and pressure head perturbations across the leak; Δs^L is the leak size change during unsteady state, and it is often small and negligible comparing to s^L ; For a relatively small pressure perturbation,

$$\sqrt{1 + \frac{h^L}{H_0^L}} \approx \left(1 + \frac{1}{2} \frac{h^L}{H_0^L}\right) \quad (\text{A.15})$$

Hence, Eq. A.14 can be rewritten as

$$Q_0^L + q^L \approx s^L \sqrt{2gH_0^L} \left(1 + \frac{1}{2} \frac{h^L}{H_0^L}\right) \quad (\text{A.16})$$

By substituting $Q_0^L = s^L \sqrt{2gH_0^L}$ (Eq. A.13) into Eq. A.16, we can yield the expression of linearized orifice equation during the unsteady state as

$$q^L = s^L \sqrt{\frac{g}{2H_0^L}} h^L = \frac{Q_0^L}{2H_0^L} h^L = \frac{1}{Z^L} h_\omega^L \quad (\text{A.17})$$

where

$$Z^L = 2H_0^L / Q_0^L = \left(s^L \sqrt{\frac{g}{2H_0^L}}\right)^{-1} \quad (\text{A.18})$$

is the hydraulic impedance of leak;

A.3 Overall Transfer Matrix for a Multi-Leak Pipe

By substituting U^{IP} (Eq. 2.12) and U^P (Eq. 2.20) into the expression of U^O (Eq. 2.28) for the two-leak pipe system, we have

$$\begin{aligned}
& U^O \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}_1^L, \boldsymbol{\theta}_2^L \right) \\
&= U^{IP} \left(\omega, x^M - x_2^L; \boldsymbol{\theta}^{IP} \right) \times U^P \left(s_2^L \right) \times U^{IP} \left(\omega, x_2^L - x_1^L; \boldsymbol{\theta}^{IP} \right) \\
&\quad \times U^P \left(s_1^L \right) \times U^{IP} \left(\omega, x_1^L - x^U; \boldsymbol{\theta}^{IP} \right) \\
&= \begin{bmatrix} \cosh \left(\mu \left(x^M - x_2^L \right) \right) \cosh \left(\mu \left(x_2^L - x_1^L \right) \right) & -\frac{1}{Z} \cosh \left(\mu \left(x^M - x_2^L \right) \right) \sinh \left(\mu \left(x_2^L - x_1^L \right) \right) \\ +\frac{Z}{Z_2^L} \cosh \left(\mu \left(x^M - x_2^L \right) \right) \sinh \left(\mu \left(x_2^L - x_1^L \right) \right) & -\frac{1}{Z_2^L} \cosh \left(\mu \left(x^M - x_2^L \right) \right) \cosh \left(\mu \left(x_2^L - x_1^L \right) \right) \\ +\sinh \left(\mu \left(x^M - x_2^L \right) \right) \sinh \left(\mu \left(x_2^L - x_1^L \right) \right) & -\frac{1}{Z} \sinh \left(\mu \left(x^M - x_2^L \right) \right) \cosh \left(\mu \left(x_2^L - x_1^L \right) \right) \\ -Z \sinh \left(\mu \left(x^M - x_2^L \right) \right) \cosh \left(\mu \left(x_2^L - x_1^L \right) \right) & \sinh \left(\mu \left(x^M - x_2^L \right) \right) \sinh \left(\mu \left(x_2^L - x_1^L \right) \right) \\ +-\frac{Z^2}{Z_2^L} \sinh \left(\mu \left(x^M - x_2^L \right) \right) \sinh \left(\mu \left(x_2^L - x_1^L \right) \right) & +\frac{Z}{Z_2^L} \sinh \left(\mu \left(x^M - x_2^L \right) \right) \cosh \left(\mu \left(x_2^L - x_1^L \right) \right) \\ -Z \cosh \left(\mu \left(x^M - x_2^L \right) \right) \sinh \left(\mu \left(x_2^L - x_1^L \right) \right) & +\cosh \left(\mu \left(x^M - x_2^L \right) \right) \cosh \left(\mu \left(x_2^L - x_1^L \right) \right) \end{bmatrix} \\
&\quad \times \begin{bmatrix} 1 & -\frac{1}{Z_1^L} \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} \cosh \left(\mu \left(x_1^L - x^U \right) \right) & -\frac{1}{Z} \sinh \left(\mu \left(x_1^L - x^U \right) \right) \\ -Z \sinh \left(\mu \left(x_1^L - x^U \right) \right) & \cosh \left(\mu \left(x_1^L - x^U \right) \right) \end{bmatrix} \tag{A.19}
\end{aligned}$$

According to trigonometric functions, we can obtain the following results:

$$\left\{ \begin{aligned} & \cosh(\mu(x^M - x_2^L)) \cosh(\mu(x_2^L - x_1^L)) + \sinh(\mu(x^M - x_2^L)) \sinh(\mu(x_2^L - x_1^L)) \\ &= \cosh(\mu(x^M - x_1^L)) \\ & -\frac{1}{Z} \cosh(\mu(x^M - x_2^L)) \sinh(\mu(x_2^L - x_1^L)) - \frac{1}{Z} \sinh(\mu(x^M - x_2^L)) \cosh(\mu(x_2^L - x_1^L)) \\ &= -\frac{1}{Z} \sinh(\mu(x^M - x_1^L)) \\ & -Z \sinh(\mu(x^M - x_2^L)) \cosh(\mu(x_2^L - x_1^L)) - Z \cosh(\mu(x^M - x_2^L)) \sinh(\mu(x_2^L - x_1^L)) \\ &= -Z \sinh(\mu(x^M - x_1^L)) \end{aligned} \right. \quad (\text{A.20})$$

The above result can be substituted into the overall transfer matrix U^O (Eq. A.19), and we obtain:

$$\begin{aligned} & U^O(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}_1^L, \boldsymbol{\theta}_2^L) \\ &= \begin{bmatrix} \cosh(\mu(x^M - x_1^L)) & -\frac{1}{Z} \sinh(\mu(x^M - x_1^L)) \\ +\frac{Z}{Z_2^L} \cosh(\mu(x^M - x_2^L)) \sinh(\mu(x_2^L - x_1^L)) & -\frac{1}{Z_2^L} \cosh(\mu(x^M - x_2^L)) \cosh(\mu(x_2^L - x_1^L)) \\ -Z \sinh(\mu(x^M - x_1^L)) & \cosh(\mu(x^M - x_1^L)) \\ +-\frac{Z^2}{Z_2^L} \sinh(\mu(x^M - x_2^L)) \sinh(\mu(x_2^L - x_1^L)) & +\frac{Z}{Z_2^L} \sinh(\mu(x^M - x_2^L)) \cosh(\mu(x_2^L - x_1^L)) \end{bmatrix} \\ & \times \begin{bmatrix} 1 & -\frac{1}{Z_1^L} \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} \cosh(\mu(x_1^L - x^U)) & -\frac{1}{Z} \sinh(\mu(x_1^L - x^U)) \\ -Z \sinh(\mu(x_1^L - x^U)) & \cosh(\mu(x_1^L - x^U)) \end{bmatrix} \quad (\text{A.21}) \end{aligned}$$

Through the small-leak assumption (i.e., $s_{n_L}^L \ll 1, n_L \in [1, \dots, N_L]$), we can conclude that, compared to $\frac{1}{Z_1^L} = s_1^L \sqrt{\frac{g}{2H_{0,1}^L}}$ and $\frac{1}{Z_2^L} = s_2^L \sqrt{\frac{g}{2H_{0,2}^L}}$, the term $\frac{1}{Z_1^L} \frac{1}{Z_2^L}$ can be neglected as it is a much smaller term. Then, by continuing to expand and rearrange the above Eq. A.21, substituting the small-leak assumption, and continuing to use the trigonometric relations (Eq. A.20), we obtain:

$$\begin{aligned}
& U^O \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP}, \boldsymbol{\theta}_1^L, \boldsymbol{\theta}_2^L \right) \\
& \approx \begin{bmatrix} \cosh(\mu(x^M - x^U)) & -\frac{1}{Z} \sinh(\mu(x^M - x^U)) \\ +\frac{Z}{Z_2^L} \cosh(\mu(x^M - x_2^L)) \sinh(\mu(x_2^L - x^U)) & -\frac{1}{Z_2^L} \cosh(\mu(x^M - x_2^L)) \cosh(\mu(x_2^L - x^U)) \\ +\frac{Z}{Z_1^L} \cosh(\mu(x^M - x_1^L)) \sinh(\mu(x_1^L - x^U)) & -\frac{1}{Z_1^L} \cosh(\mu(x^M - x_1^L)) \cosh(\mu(x_1^L - x^U)) \\ -Z \sinh(\mu(x^M - x^U)) & \cosh(\mu(x^M - x^U)) \\ -\frac{Z^2}{Z_2^L} \sinh(\mu(x^M - x_2^L)) \sinh(\mu(x_2^L - x^U)) & +\frac{Z}{Z_2^L} \sinh(\mu(x^M - x_2^L)) \cosh(\mu(x_2^L - x^U)) \\ -\frac{Z^2}{Z_1^L} \sinh(\mu(x^M - x_1^L)) \sinh(\mu(x_1^L - x^U)) & +\frac{Z}{Z_1^L} \sinh(\mu(x^M - x_1^L)) \cosh(\mu(x_1^L - x^U)) \end{bmatrix} \\
& = \begin{bmatrix} \cosh(\mu(x^M - x^U)) & -\frac{1}{Z} \sinh(\mu(x^M - x^U)) \\ -Z \sinh(\mu(x^M - x^U)) & \cosh(\mu(x^M - x^U)) \end{bmatrix} \\
& + s_1^L \sqrt{\frac{g}{2H_0^L}} \begin{bmatrix} Z \cosh(\mu(x^M - x_1^L)) \sinh(\mu(x_1^L - x^U)) & -\cosh(\mu(x^M - x_1^L)) \cosh(\mu(x_1^L - x^U)) \\ -Z^2 \sinh(\mu(x^M - x_1^L)) \sinh(\mu(x_1^L - x^U)) & Z \sinh(\mu(x^M - x_1^L)) \cosh(\mu(x_1^L - x^U)) \end{bmatrix} \\
& + s_2^L \sqrt{\frac{g}{2H_0^L}} \begin{bmatrix} Z \cosh(\mu(x^M - x_2^L)) \sinh(\mu(x_2^L - x^U)) & -\cosh(\mu(x^M - x_2^L)) \cosh(\mu(x_2^L - x^U)) \\ -Z^2 \sinh(\mu(x^M - x_2^L)) \sinh(\mu(x_2^L - x^U)) & Z \sinh(\mu(x^M - x_2^L)) \cosh(\mu(x_2^L - x^U)) \end{bmatrix} \\
& = U^{IP} \left(\omega, x^M - x^U; \boldsymbol{\theta}^{IP} \right) + U^L \left(\omega, x^M - x^U; \boldsymbol{\theta}_1^L \right) + U^L \left(\omega, x^M - x^U; \boldsymbol{\theta}_2^L \right) \tag{A.22}
\end{aligned}$$

At this point, we have derived Eq. 2.29, which indicates that U^O for a two-leak pipe can alternatively be expressed as the summation of U^{IP} for the intact pipe and U^L for the two leaks. For the case of a more general N_L -Leaks pipe, similarly, we can derive Eq. 2.32.

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